NEWSLETTER AND JOURNAL

THE GAUGE ONE MODEL RAILWAY ASSOCIATION

GAUGE '1' MODEL RAILWAY ASSOCIATION

ISSUE 240

WINTER 2013/14

THE OFFICIAL JOURNAL OF THE GAUGE ONE MODEL RAILWAY ASSOCIATION
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The Gauge One Model Railway Association

Newsletter and Journal

No.240 Winter 2013/14

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On the Cover

Rodney Maennling's scratchbuilt CNR Caboose. Photo Rodney Maennling.

FROM THE EDITOR

Many thanks to all of you who responded to my request in NL&J 239 for articles. Such was the influx that I have had to carry several over to the next issue. I look forward to seeing articles on the models you have completed between family duties over the Christmas period!

Photographs are generally being submitted satisfactorily, with JPEG file sizes over 1Mb each, which ensures that they will print well. Computer drawings are causing me some concern, however, especially if the line thickness is very thin. As with photographs, just because a drawing looks great on the computer screen doesn't mean that it will print well and if a line is too thin, it might not appear on the printed page. I'm more than happy to receive 'proper' technical drawings or sketches - I think they have a character of their own - as long as they are clear.

I tend to try to edit any articles as soon as I receive them. That way, any queries I may have can be dealt with without the added pressure of an approaching deadline, but it does mean that if you send in a revised version of your article, I might have to redo all my editing, so may I suggest contributors put their articles to one side for a week before submitting them, to give yourselves the chance for an extra read through before submission?

We usually see the head of a train on the front cover of the NL&J, but this time we have a photo of the tail-end charlie, a caboose. You wait ages for a photo of a brake van and then two turn up at once! Rodney Maennling sent in some thoughts on scratchbuilding, together with photos of some of his models, which included the model caboose on the cover and Phillip Taylor sent in an article about the caboose he made to run a fair distance behind his Berkshire.

Markus Neeser finally concludes his long running series "I Could Never Do That" in this issue. Hopefully many of you will have found inspiration in his articles and found time to talk to Markus about his models when he has visited the UK. Roger Thornber has provided an article about setting up the valve gear on a Project. With the new Fifth Edition of the Project book having just been released, this is timely, as is an article by Harry Wade who, incidentally, made the long journey from Kentucky to the Fosse for the Expo. I have heard Harry referred to as the 'Sage of Nashville', so it will almost certainly be worth your while reading his ideas.

Happy New Year!

Bill Read

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GAUGE ONE EXPO 2014 CANCELLED: REPLACED BY SPRING MEETING ON 14 JUNE

The popularity of Gauge One Expo 2013, among visitors and traders, led your Committee to plan a repeat performance in 2014, although only for one day. The date set was Saturday 14 June 2014.

A couple of weeks prior to the AGM at Woking on 16 November 2013, we received notice from the Yorkshire Group that certain of their members wished to raise the subject of the very poor financial results of the Expo at the AGM. They described the losses, contained in the Financial Statement for the year, as 'huge', and queried the expenses incurred. They were concerned about possible further losses in 2014. Their written statements were widely distributed and created much consternation within the Committee, as well as fertile ground for rumours about the nature of the expenses.

At the AGM, we circulated a written breakdown of the expenses. The Chairman made it clear that no member of the Committee had received any expenses for travel to or from The Fosse, for accommodation or for food. All costs were strictly related to the event itself, and it was clear that financial control had been weak, leading to escalation of unforeseen costs. In fact, the anxiousness to make the event a success (which, for visitors and traders, it was) led to over-provision in some areas. The cost of the venue itself was also a major element, although much of that was foreseen.

Along with the financial breakdown, an outline projection of figures was provided for Gauge One Expo 2014. The Chairman stated that the Committee had already decided that, if the feeling in the room was that the Expo should not be proceeded with, and there being no obligation to do so, then the Committee would accept that as its course of action. After a sometimes-heated discussion, a vote was taken and it was effectively decided that Gauge One Expo 2014 would be cancelled.

Since then, we have received a number of messages of support for the Expo idea, both from members and from traders (who are mostly one and the same thing) and the Committee will be considering the future of Expo. It is worth remembering that the Committee's policy, prior to the Expo, for bringing Gauge One to the attention of non-members, was to increase and improve G1MRA's attendance at local shows. This approach will be further promoted, and investment made in display materials. Involvement of the Groups will be sought in this process.

There has been some comment that the Association should not mount further 'commercial' (not our word) shows, open to non-members, with their attendant risks. Given the financial results of Gauge One Expo 2013, this is understandable, but your Committee thinks that, if the positive results (see NLJ 239) could be achieved without unplanned losses, then further Expos should not be ruled out. Comment has also been heard to the effect that such exhibitions are more in the interests of traders than of members, but your Committee cannot accept such a blinkered view of the facts, when it is clear that members benefit from what the trade offers, and that the scope of trade offerings is an influence on deciding whether to join the Association. Indeed, our Association is in need of new blood and new ideas, and the Expo of 2013 provided both of these. In effect, the results have provided us with the opportunity to discuss the wider subject of our policy related to events and venues, and your Committee will be doing so.

In the meantime, we have decided to organise a Spring Meeting to replace the 2014 Expo. This will take place at Shepshed High school, near Loughborough, LE12 9DA, on 14 June 2014.

Chris Ludlow, Chairman

Gauge One Model Railway Association Present

Spring Meeting 2014

To be held on Saturday 14th June

at Shepshed High School • Forest Street • Shepshed

Near Loughborough • Leicestershire • LE12 9DA

10.30AM – 5.00PM

Running track • Demonstrations • Trade Stands

Refreshments

Members and their invited guests welcome

5 minutes from M1 Junction 23

Half-hourly bus service (No.4 'Shepshed Sprinter') from Loughborough Railway Station to the venue. East Midlands Airport – Castle Donnington

Further information in NL&J 241 - April 2014

ANNOUNCEMENTS

2014 SUBSCRIPTIONS DUE NOW

The Association Membership Year runs from 1st January to 31st December

Subscriptions for 2014 are due now.

If you do not have a standing order in place or you pay via an overseas co-ordinator, please pay now using the enclosed form.

If you pay by Standing Order please check that the amount is correct with your bank.

Annual Subscription is £20.00

Airmail supplement for members outside the EEA is £20.00

Should your subscription (including any sum outstanding since the subscription was increased on 1 January 2013) not have been received by 31st March then Article 9 will apply. A Joining Fee of £8 for the UK and £10 for overseas members will apply to reactivate your membership.

Please help the Association by paying the correct amount in a timely manner.

Many thanks – *Andrew Barnes – Membership Secretary*

BAKEWELL
GAUGE I NORTH
2014

Saturday 12th April 2014
Bakewell Agricultural Business Centre,
Haddon Road, Bakewell, Derbyshire, DE45 1HR

The G1MRA Yorkshire Group is pleased to be sponsoring this Exhibition once again. The main attraction this year will be Midsomer Norton, by kind permission of the Barrett family.

Over twenty traders are already booked, including for the first time Tenmille Products and Northern Finescale. For a full up-to-date list of trade stands and exhibitors, please look at the website www.gauge1north.org.uk

The popular 'I Built it Myself' display will be repeated, with £50 prize vouchers for the two categories, locomotives and rolling stock. Entry forms are available from the secretary: secretary@gauge1north.org.uk

Invitations for runners will be distributed at the end of January, but if you wish to register an interest in either running or helping, please contact the secretary. Stewards and exhibitors get free entry to the exhibition. Runners only will have to pay the entrance fee. Advance booking tickets will be available this year and a limited number of free parking passes will be available to G1MRA members applying for tickets. Again, contact secretary@gauge1north.org.uk for details.

BRING AND BUY STAND

The first run of the new arrangements for the G1MRA Bring and Buy worked well at Woking. We provided seller's forms and labels to the very few who arrived without them, no one with more than ten items was turned away and the tables were filled with a good selection of items. The 'common cash till' helped us to serve buyers quickly. When it came to paying out to sellers or returning unsold items, of which there were not many, the forms and labels made the task clear and easy.

In many ways we were sorry to see the old paperless system go after so many years of good service but the huge volume of transactions and goods was overwhelming us.

Thanks to the sellers for following the new arrangements, to the buyers, and especially to the volunteer helpers for giving up their time on the day.

Roger Olding

STOP PRESS: CHRIS LUDLOW RESIGNS

Our Chairman Chris Ludlow has submitted his resignation to take effect from the end of January. His reasons are essentially personal, but he also feels that he is too distant from the centre to lead G1MRA through the period of self-searching that he feels is essential for its future wellbeing. He also cites the amount of diplomatic negotiation and some unpleasantness between a small number of members, which have made the stresses of office for him unbearable. The Committee has formulated a plan to deal with his absence, and this will be announced soon. Chris has told us of his intention to write an article about his experiences, for the interest of members and hopefully the good of the Association.

MISSING OR DAMAGED NEWSLETTER?

If you receive a damaged NL&J or you believe that your copy has gone missing in the post, please contact the Membership Secretary for a replacement.

PROJECT BOOK EDITION 5

by Martin Hulse (584)

The Project book has already been in print for many years. The fourth edition aimed to create a base for further corrections. Unfortunately, the computer files have been lost, which has meant that for some time copies have had to be reproduced from ageing paper masters, and corrections were not practical. I decided to revisit this aim, and have recreated a fifth edition of the Project book using today's technology.

A fresh approach to the layout has been taken with:

- Words from the fourth edition, revised where necessary.
- The drawings redrawn by Roger Thornber, with some design changes.
- Exploded views and parts lists have been added for each section. The table of contents shows where part numbers can be found. The exploded views show how parts come together.
- Some of the photographs have been replaced by 3D views where these provided more clarity.
- Colour has been added to the drawings, especially to distinguish dimensions from outlines.
- The modifications suggested by Ron Poulter to the lubricator steam feed and boiler manufacture have been used to modify these sections .
- A new cover picture.
- General arrangement drawings have been added.

The fifth edition aims to keep to the original design, except where a change is useful. The most obvious visible change is moving the lubricator to the right side, to give a neater backhead. The changes from previous editions are:

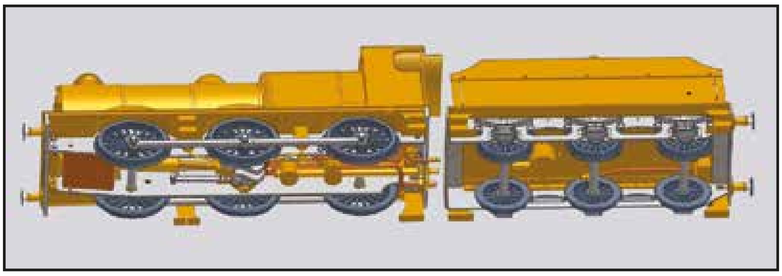
- Slide valve rod and nut instead of filed rod. Easier to construct, get dimensions right, and adjust.
- Lubricator now on right. Tidier backhead.
- Tender has axle bushes. Means they wear, not the holes in the frames.
- Alternative way of setting valve gear included.
- Sheet metal in metric thicknesses or fractions instead of SWG because that is what can be bought today.

We now have an edition that preserves the original work whilst adding those modifications that have proved themselves. We also have a base which can be built on to incorporate future changes and corrections as these are identified.

Equally important, the printing technology that is best for G1MRA can be used each time more copies are needed, and we can start to consider other ways of using this asset, such as electronic copies.

If readers find errors then let me know, please, so we can correct them. If builders explore variations then please write these up for the G1MRA Newsletter & Journal - we can build up a library of variations to consider via the G1MRA website.

Gauge 1 has changed since the original Project book was written. We used to assume the builder would make everything except the pressure gauge and wheel castings. As the membership of G1MRA has grown to over 2500 members,



commercial suppliers have provided many more choices to the builder of this Project design. You will find details in the G1MRA suppliers list. So if you would like to buy some of the Project parts rather than make them, you will find laser cut frames, ready made boilers, pre-turned wheels, complete buffer sets, etc, are all available from suppliers. But do make sure they will work together - it is difficult, for example, to fit wheels with round holes to axles with square end shafts!

Many hands have worked on the Project book over the years, too many to name them all. We have Ron Poulter & Bob Hines to thank for the original design; Barry Applegate for technical advice and the axlepump; and new drawings for this edition from Roger Thornber that make it come alive. To everyone else who is not mentioned by name we apologise, but rest assured that we are very grateful for all your help.

As Editor, I hope to have kept the Project design tradition whilst improving the layout and readability. The changes and additions are my decisions, as are any mistakes. Enjoy this 21st century Edition 5 of what is already a G1MRA classic book.

Michael Wrottesley built this model of a Pickersgill 4-4-0 engine ten years ago, using some Keith Cousin parts, a Project boiler from Brian Beck and a twin cylinder set and slip eccentric motion by the late Harold Denyer (HB Models). The chimney and safety valve cover were made by Dave Brutnell. It is a model of a Caledonian Railway locomotive constructed in April 1916 as CR No 115, LMS 14463 and the last of the class to be withdrawn in December 1962 as BR 54663. The fledgling Scottish Locomotive Preservation Fund made a big effort to preserve one but could raise only enough money for one ex-CR engine and opted for the McIntosh 0-6-0 No 828 instead. Michael saw the last Pickersgill 4-4-0 in service on the former Highland Railway and with his father's help located it at Wick Shed on 31 August 1961 and that inspired him to model one of the class.
Photo Alan Leslie.

PHOTOGRAPHIC COMPETITION RESULTS

Last year your Committee decided to promote a Photographic Competition. The aim was to encourage the membership to send in photographs (taken between AGMs) that could be used as a library 'stockpile' for the NL&J, or used in G1MRA promotional material.

Just three people responded to the announcement of the competition with one, Elizabeth Hodgkins, sending in several photographs of her layout. Ultimately, the decision as to which photo would be declared the winner came down to your Editor, who decided that Mike Palmer's photograph of his GWR Saddletank outshone the other entries. So congratulations to Mike for being the winner of the first G1MRA Photographic Competition, winning a book offered by Graham Colover as the prize and thank you to Elizabeth Hodgkins and Harvey Smith for sending in entries.

Mike Palmer's winning photo of his completed GWR Saddle Tank on his layout.


Elizabeth Hodgkins took this photo on her layout.


This photo by Harvey Smith was taken at a typical small GTG.

I COULD NEVER DO THAT

Part 19: Final Bits and Pieces
by Markus Neeser (2476)

Dear Newsletter readers, quite some time has passed by since the appearance of the last part of ICNDT and the engine has been mostly finished since then, including final bits and pieces.

First, before I start with Part 19, a brief review: shortly after Part 18 appeared in the NL&J, the late Tony Hall-Patch rang me one evening to ask me whether I was using a machine reamer with a short lead-in to ream the pump barrel, which I duly confirmed (I don't think that I have any hand reamers at all in my workshop). He pointed me to the fact that using a machine reamer makes a big difference, as the use of a hand reamer with a long lead-in would inevitably produce a conical hole since the reamer could not be run through completely. I had to agree with this. But don't worry: simply drill the hole all through the barrel and ream it, then close the barrel with a suitably shaped plug. The result will be the same and you will not need to get a machine reamer.

The discussion then moved towards quality tools and their availability. Speaking for Switzerland, I can state that it is fortunately no problem to get tools of excellent industrial quality here. Not necessarily at the sometimes dead cheap prices as Chinese tools, but at reasonable cost. 'You get what you pay for' and the result is usually not of a better quality than the tools used. While speaking of Chinese tools, I believe to have noticed that their quality level has not improved over the last 10 to 12 years. At least the ones offered in our hobby environment. I am observing the development with great interest since my first visits to the Sinsheim Hallendampftreffen in 1998 or 1999. I sometimes rather got the impression that the quality has even declined and wonder what you think about this.

But now back to the topic of this article. This most recent part of ICNDT features some details about mechanical completion of the engine and the tender by attaching some missing details before transfer to the paint shop after final tests on the track.

To speed completion up a little bit, I decided to prepare some patterns to have castings produced by my preferred foundry. Whenever it comes to producing fairly exposed items like lamp irons, I prefer to have them cast to avoid repetitive work (too many hobbies, you know) and especially because the cast items are significantly more robust than the patterns. A cast lamp iron will last longer and will most probably never have to be replaced.

So I turned towards pattern production for a while and by the end of August 2012 an order for about five different castings was placed with the local foundry. The castings were returned some two weeks later in the usual excellent quality.

Anybody interested in obtaining some of the casting (all in 10mm scale) is kindly invited to contact me (see the Membership CD for details).

The almost finished MR Johnson 2601 Class single wheeler locomotive.

Brake pipes

Unlike most other Midland locomotives, the front end brake pipe of the 2601 class 'Spinners' runs through the running plate and then down behind the buffer beam. It is not attached to the outside of the buffer beam. So the removable lid covering the oiler was drilled just behind the buffer beam and the brake hose casting was soldered to the lid, using a little flanged bush on the underside of the lid to increase the joint area and stability. A side effect of this attachment is that the lid can now be easily lifted off using the brake hose as a handle. The rear end brake hose was fitted in a similar way by drilling a hole through the drag beam. The brake hose is simply inserted in the hole and held in place with a tiny set screw.

Guard irons

While waiting for the castings to be returned, I worked on the guard irons. Both the rear and the front guard irons were cut from 1mm thick semi-hard brass sheet by transferring the shape of the guard irons from a side elevation drawing I had. The irons were then cut out using a fretsaw and filed to shape.

The front guard irons were quite easily attached by bolting them to the inner frames and bending the guard irons to shape afterwards.
To attach the rear guard irons, I had to mill T-shaped adapter pieces as there were no inner frames. The guard irons were soft soldered into slots in the adapters before the assemblies were bolted underneath the drag beam.

In case of a severe derailment with damage to the guard irons, both

Castings for the MR single wheeler, still with the sprues on.

the rear and the front end guards can be easily exchanged. This was, by the way, successfully tested on one of the tracks visited in September 2013 and worked to the designer's satisfaction…

Driving wheel brake hangers

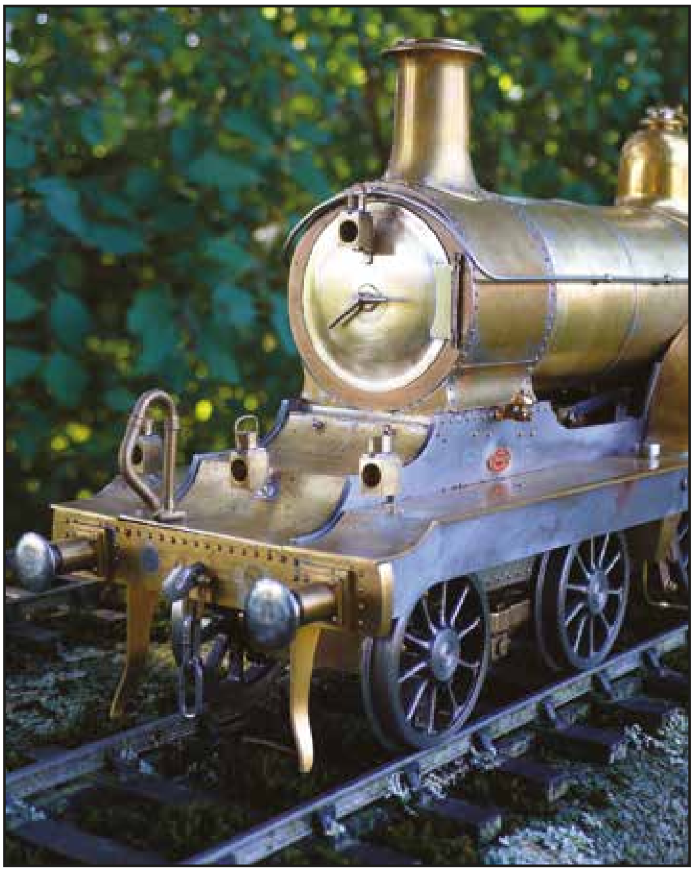
The driving wheel brake hangers are to my eyes a very prominent detail and had to be added. Although all detail work below the footplate is at permanent risk of being broken off during accidents, it is out of the question to me that it may be omitted. So I rather focus on "How can these details be quickly replaced if they should get lost" rather than leaving them off.

To produce the pattern for the foundry, I started as usual with the production of the brake block. In the case of the Midland Johnson single wheeler, the brake block has contoured sides, which makes it almost impossible to mill the block in one go from a solid piece of brass without using a numerically controlled mill. So I produced the block from two pieces of brass which could be profiled with a file before soldering them together. The top end of the brake hanger received a spigot on both sides which allowed me to bolt the brake hangers between the sandwich frames. A fixing bolt runs through the outer frame, then through the brake hanger before it is grabbed by a threaded hole cut into the inner frame. Thereby, the brake hangers also act as spacers and stiffen the sandwich frame. Finally, the lower ends of the hangers were connected via the brake gear rigging. The rear one needed to have a curved shape to allow some space for the meths burner.

Sanding gear

The sanding gear was, like the brake hangers, a detail I didn't want to leave out. The only snag was that I just couldn't make out the correct geometry of the sander control valves placed between the sandboxes and the sander pipes. Although the drawings I had at hand gave an idea of the side elevation, I was puzzled by the shape of the front elevation. Especially the lid that I presumed to be a lid for a cleaning opening gave me some headaches. Although I looked and asked around, I couldn't find any photos of this detail so I finally decided to postpone it until the next visit to the NRM. In the meantime, simple sanding pipes made from brass wire were fitted.

During our September 2013 G1MRA trip to the UK, we started with a visit to the NRM at York. Right after our visit to the archive, I moved on to the MR single on display in the hall where the Royal Trains are parked. But much to my dismay, some safety responsible staff had decided to completely bridge the gap between platform and engine with wooden boards all along the engine and tender. Well, I did not return to the archive to bother the staff with a minor request and will install a simple 'might be' sander control gear.



Brake hose casting, guard irons and lamps on the engine's front end.

Lamp irons

Right after completing the guard irons I tackled the lamp irons of the locomotive. I decided to have them cast as the strength and toughness of the cast lamp irons exceeds that of fabricated lamp irons by far.

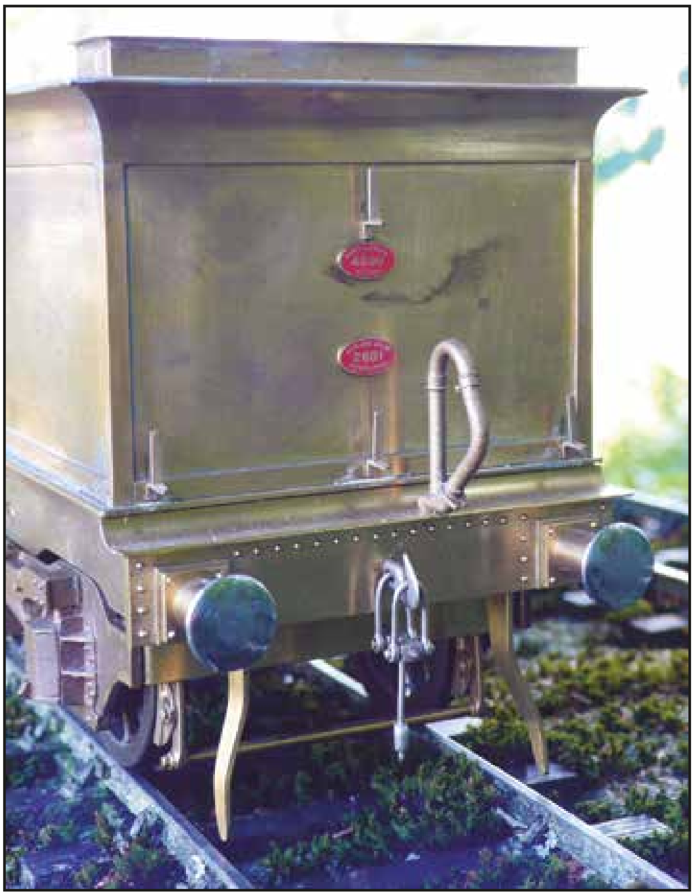
The castings were soldered to the running boards and the spigot running through the boards was used to bolt down the front end of the boards to the frames.

The nice set of lamps which can be seen in the photos was kindly produced from scratch by John Werren, many thanks to him at this point. I still need to make up my mind whether to illuminate the lamps or not.

The brake and sanding gear on the driving axle.

Drilling the smokebox rivet holes on the mill.

On the tender, a slightly different type of casting was used, again with a threaded stub to attach the irons after painting. Where accessible, the lamp irons were secured with a small nut on the inside of the water tank. Everywhere else, a little drop of silicone was used to keep the lamp irons in place and to seal the water tank. Removable lamp irons have the big advantage of not getting in the way when it comes to lining.



The fully fitted-out rear end of the tender.

Panelling of tender and cab

Next I tackled the dreaded panelling on the tender tank. From earlier experience, I knew that soldering brass strips to a larger piece of brass can be a difficult job, because it takes a lot of heat to bring the job up to the required temperature and there is always the risk of total disintegration if not enough care is taken. Discussions with local specialists unveiled different methods, like using glue or even low temperature solders.

Epoxy glues like Araldite were out of question as they start losing strength far below the temperature of boiling water. Whilst this should not pose a problem on the tender, it certainly is a big one on the cab sides. Local member Heinrich Schartner came up with the idea of using a low temperature solder and sourced some soldering paste (Sn42Bi58 to be precise) which has a melting point of only 138°C, far below the melting point of all other soft soldered joints but still high enough to withstand the temperatures the cab is normally exposed to.

The finished boiler casing and smokebox.

Trials showed that the mechanical strength against shearing off was fine, only the peeling strength was not really overwhelming (about same adhesion as if using epoxy glue), but then also this was not really that important.

But before using the low temperature solder, I made a test using my normal soft solder paste (a tin-lead compound) and started to solder the beading along the lower water tank side. As expected, it was difficult to obtain the locally required temperature using my small gas torch. Finally, I switched to the next larger torch and a little later even to my largest one, my 'Big Bertha',to save the situation! I admit I became a little bit nervous and I desperately tried not to think too much about the potential total disintegration of the tender structure...

After this experience (which luckily ended without disaster), I came to the conclusion that the low temperature solder would be my choice for the application of the panels. All remaining panelling was attached using a small gas torch.

The tender bogie with brake hanger castings added.

Smokebox riveting

The smokebox has a line of rivets on the ends, front and back. As the smokebox assembly was produced from a rather thick-walled piece of brass tube, I could not use a rivet embossing tool to simulate rivet heads and had to use real rivets instead. To ensure that the size was right, I chose the smallest rivets I had at hand which were originally intended for '0' Gauge usage but looked exactly right for this purpose (and they could have been replaced with the next bigger rivet size in case they would have looked too small).

Before the rivets could be set, a number of 0.6mm holes had to be drilled into the smokebox. I decided to do this on the mill using a direct dividing head. The smokebox was chucked with a three-jaw chuck and supported with a centre supporting the front end via the smokebox door. The indexing disc happened to have exactly the right number of divisions (60 divisions), so the dividing operation was quite simple. A new 0.6mm drill was chucked with a collet to ensure the drill was running true and machining began.

Loco numbers on rear splashers.

On the rear smokebox end, the rivet holes had just to be drilled through the 2mm thick wall which was not challenging at all, but on the front end, about 40 blind holes with 4mm depth each had to be drilled. To avoid broken drills, I lifted the drill out of the hole every single millimetre. This procedure took quite some time since drilling was done by moving the milling table up and down but it was honoured by accident-free machining, without broken drills.

After drilling, the rivets were soldered into the holes. I applied tiny drops of normal soft soldering paste using a tooth pick where the rivet shafts protruded into the smokebox and the assembly was carefully warmed to the melting point of the solder afterwards. By using this procedure, I succeeded in not having lots of solder penetrating along the shaft and causing unsightly solder fillets around the rivet heads on the outside of the smokebox (if this should happen to you, then just wipe off any excessive solder with a cotton rag as long as the smokebox assembly is still hot).


A steam chest oiler casting.

On the front end, where the blind holes were drilled, I first filled a little bit of solder paste into the holes before inserting the rivets which were cut exactly to length. Then, before heating up the smokebox, I carefully wiped off any solder excess around the rivet heads. The result was again very satisfactory and the riveting was completed afterwards.

To cut the rivets to the exact length I used a very simple tool: a few 0.6mm holes were drilled into a piece of brass of rectangular section with exactly the thickness of the required rivet length. Then rivets were stuck into these holes and kept in position with a piece of wood pressed against the rivet heads. The excess shaft length was then cut off with a side cutter and a few strokes with a flat needle file finished the process. In a few minutes I had the required number of rivets of the required length.

Additional rivets were inserted into the frames below the smokebox saddle. The holes for the rivets were drilled using the cross table of the mill, since my scriber and punch skills were not much used since I took my basic mechanical engineering training. Afterwards, the holes were slightly countersunk on the inside allowing the soft solder to build a small fillet. After soldering, the inside of the frames was filed flush with a Swiss file.

Handrails

With all those details added to the engine it still looked a little bit naked and the main reason for this was the missing handrails along the boiler. I had wisely marked and drilled the holes in the boiler wrapper before rolling it, so I just had to produce the handrail knobs. These were made from free machining stainless steel rod using the same method I described a while ago in the articles about the construction of the GNR Atlantic from a Barrett Engineering kit.

Once machined, all knobs were given a polish with 'Brasso' that I purchased years ago at an ironmonger in Tenterden after a visit to the very nice K&ESR. Next, the one-piece handrail was bent to shape. This is a task that needs some feeling for wire bending, so it is best to practise a little bit with some scrap wire first before ruining lengths of stainless wire. The first step is to bend the wire to a 'U' shape. This was done by bending the wire around a piece of brass rod with a diameter of 30mm in my case. It is easier to make the curve too small than too large since widening is very much easier than narrowing it afterwards.

This part of the work produced the bend above the smokebox door. Next, the bends that redirect the handrails back in line with the boiler barrel towards the cab had to be added and this is where the more difficult part of the handrail forming starts. I worked out a method that works well for me: I drew the shape of the handrail front end on a piece of paper, put the already U-shaped handrail on the paper, pressed down a 10mm diameter brass rod on the handrail where the bend needed to be bent up both ends of the handrail until they stood up exactly vertically. Again, use some scrap wire to practise before ruining your best piece of straight stainless steel wire (been there, done that…).

Tender bogie modification

A brief update on the tender bogies at this point: based on some tests I had the impression that the tender bogies were a little bit too stiff and therefore also too prone to derailments. As the bogies were quite flexible, I assumed that the bogie saddles were too wide, preventing the bogies from flexing in operation. So I milled 1.5mm off the stretchers which were used as bogie saddles and added an 18mm diameter brass disc of the same thickness before reinstalling the bogies. This measure resulted in a much improved running behaviour of the tender and no further bogie derailments could be noticed afterwards.

Engine number and makers plates

The etched engine numbers on the cab sides and all plates were provided by Guilplates of Guildford. I decided to number my engine as No 2901. These numbers were applied before painting using the previously mentioned low temperature solder. Right after painting, I will mask out the numbers and will carefully remove the paint with finest sanding paper until the brass surfaces again, followed by a good polish.

Fall plate

Once the handrails were fitted to the rear cab end, the fall plate was tackled. First, the cab floor was measured from the job and cut from a piece of 0.7mm thick brass, then the fall plate outlines were transferred to a piece of brass sheet and cut out using a fret saw. The fall plate was afterwards slightly curved as per the prototype. This was done by clamping one end of the fall plate against a round bar of suitable diameter, followed by pressing the plate against the former using a piece of flat bar.

While speaking of prototypes, there was an interesting discussion which arose the other day about the question of when diamond plate was first used on British engines. I still have no details about this but judging from pictures of No 763 at York, the usage of unstructured flat sheet metal for both the fall plate and the tender floor seems to have been the standard at the Midland Railways loco works around the turn of the century when this engine was built. The cab floor was obviously still planked with wooden battens that time.

So I went for normal, unstructured brass sheet and glued a piece of scribed ply to the brass cab sheet floor. This sandwich design will keep the floor board from warping if it should get wet and hot.

Finally both cab floor and fall plate were to be joined. My initial attempt to use tiny working hinges for this purpose failed: not because they were not working properly, but they looked a little bit too clumsy and were replaced by dummy hinges which looked far better.

Steam chest oilers

While it is not particularly difficult to produce these items, the proper attachment is always a challenge since the screws used for this purpose are often far too big compared to the castings.

So, I asked my trusted friend Heiri Schartner for a good idea about how to install the tiny steam chest oilers and he duly came back with an impressively simple and good idea. "Just use a small bolt running through the horizontal shaft into the engine frames" he said.

After that idea was born, the production of the oilers was tackled on the spot. Two oilers were supposed to be made from brass. So one cold and foggy Sunday morning in early March 2013, I went to the workshop to produce the oilers. Once I had made a sketch using dimensions taken from a photo taken at the NRM at York, all components were made and assembled using screws and bolts. Then the assembly was silver soldered using the finest silver solder I had at hand and the oiler was given a polish with a felt disk. Producing this first oiler took much more time than expected so I decided – as so often before – to use it as a pattern for the foundry which produced a small number of oiler castings which were fitted to the frames as soon as they had arrived. The final result really looks the part and can be seen in the photos.

With this 19th part of ICNDT, the engine is now mechanically almost completed except for a few minor details and it will be further tested before going to the paint shop.

Therefore this series now comes to an end (finally, some may say) and I hope you enjoyed reading this series as much as I did building the model of this outstanding elegant Victorian engine and writing the articles about it. I especially hope that I was also able to encourage one or other reader to give the model engineering aspect of our hobby a try, too.

Keep up the (boiler) pressure and happy steaming!

Chris Harnett's Swiss compound 2-8-0 metre gauge loco of the RhB from Regner, much modified. The stock is a combination of Brawa and LGB. The loco has hauled in excess of 100 axles and has R/C of gas, reverser, regulator, whistle and drain cocks. Pictured running on Mike Cooper's splendid new line. Photo Peter Armstrong.

A BRITANNIA FROM AUSTRALIA

by Trevor Barber (887)

It was 2004 and I was living in Milton Keynes in the UK; I was working on my fifth ceramic burner gas fired loco, a Martin Evans Green Arrow V2 chassis with a JVR 'B' type boiler, when I received a phone call from my daughter in High Wycombe. She said that she was emigrating to Australia!!! Our son was already a citizen in Australia and we always looked forward to an annual trip to Australia to visit him. Now we were faced with our children being in Australia and my wife and myself left here in the UK.

From this point everything changed and for almost eighteen months we tested every avenue to emigrate to Australia. Nearing retirement meant that we were not an attractive proposition for the Australian immigration authorities. Luckily, our son was able to help and with our daughter being on the way (plus we had already had some funds for our retirement), we found a way and in May 2006 we emigrated. By that time, amongst all of the other things that you need to do to emigrate so late in life, I did manage to complete the V2 but never managed to steam test it. It was completed, painted and boxed along with the other locos ready for the packers. I had been thinking of my next project and decided that the Britannia was my choice and I knew that the model engineering scene in Melbourne (where we were moving to) was light on the ground for parts, so I promptly purchased some Skinley drawings and some fully machined wheels from Walsall Model Industries for the future and included these parts in the container.

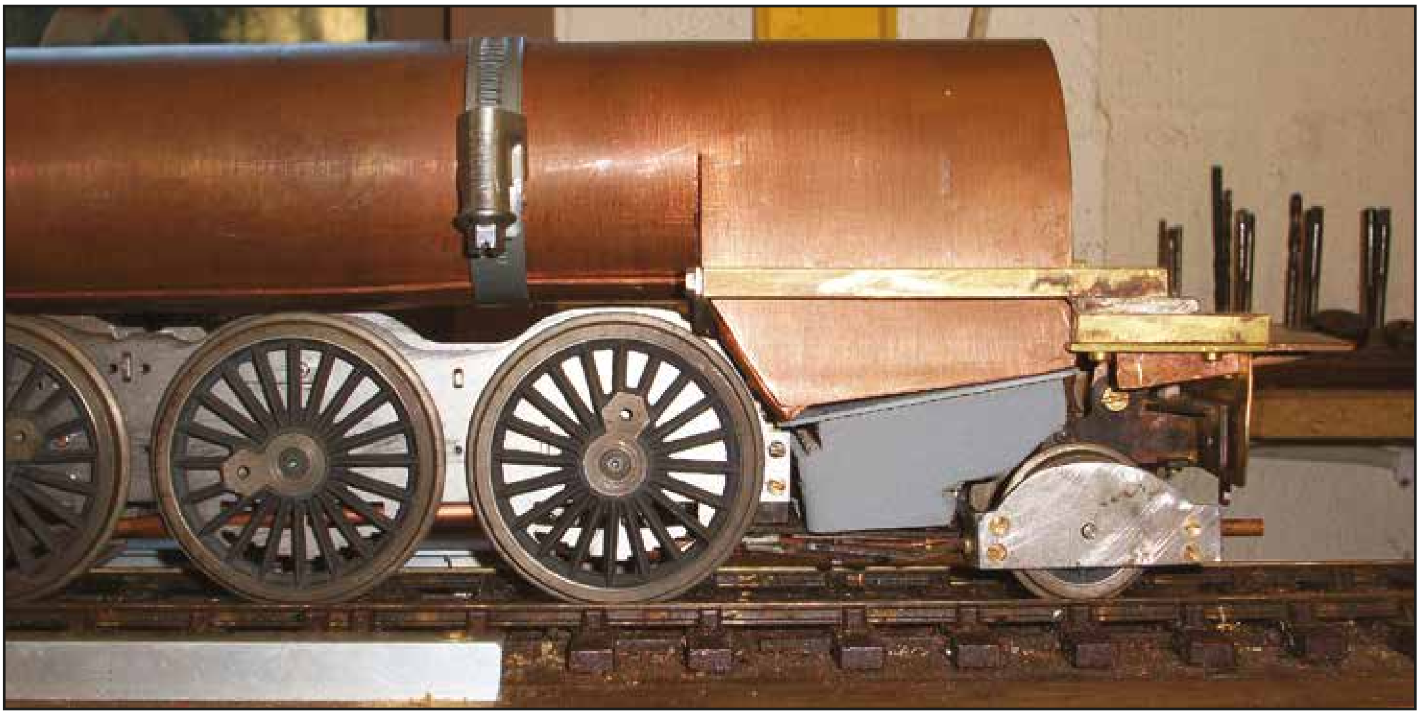
The completed locomotive.

In Australia it was like starting all over again - new house, new contents, new district and a very different culture and, most importantly, a new track in the garden. It was almost two years before the box was opened and the V2 re-appeared. After modifications to suit the new track it actually ran a treat. For the next few months the other locos were re-tested and the workshop was setup. It was time for the Britannia to be looked at. At the same time Model Engineers Laser (MEL) was advertising the parts for my new loco, so I immediately purchased the parts for the loco and tender chassis. The most difficult part of living in Australia is that, apart from some raw materials, most items still needed to be purchased from the UK so there were times when weeks went by before a part arrived from the UK so the next stage could be started.

Being an engineer and coming up from the ranks of a draughtsman, I never lost the design element in my thinking and, as half my working years were associated with a high powered 3D CAD system, I always had a CAD system at home. I did purchase the MEL drawings but chose to use my experience to lay out the loco from the Skinley drawings into CAD and then apply the MEL parts. Because I had some experience of gas firing I had my own thinking on boiler design and I also wanted the possibility of making it adaptable to coal firing with the minimum of effort.

There was also a problem with my new garden track as I was restricted to 7' 6" radius and I had to make quite a lot of changes to the V2 for it to negotiate such tight curves and it was necessary to apply the same thoughts to the Britannia. In the end, the MEL laser cut parts were used with a major modification to the rear frame. Only the last 1/2" of the rear frame was used, as a very different structure was built to enable a ceramic burner to be fitted to the firebox, or the alternative coal fired grate and ashpan be inserted with little effort.

I also found that the MEL cylinder drawings had a square top to the steam chest of the cylinders , so it was back to the CAD system to redesign the steam chest to reflect more of the Britannia piston valve characteristics. The MEL motion was excellent, but I used the Walsall design of crankpins rather than the MEL design. Another issue I found with the V2 which I needed to solve, was cylinder condensate, which meant adding drain cocks to the cylinders. DJB were offering an automatic draincock set which was ideal for G scale locos but was rather overscale, but I purchased a set with the aim of



The much modified rear frames enable a totally accessible coal fired ashpan or a ceramic burner to be fitted. The photo shows a dummy wooden volume to simulate the ashpan.

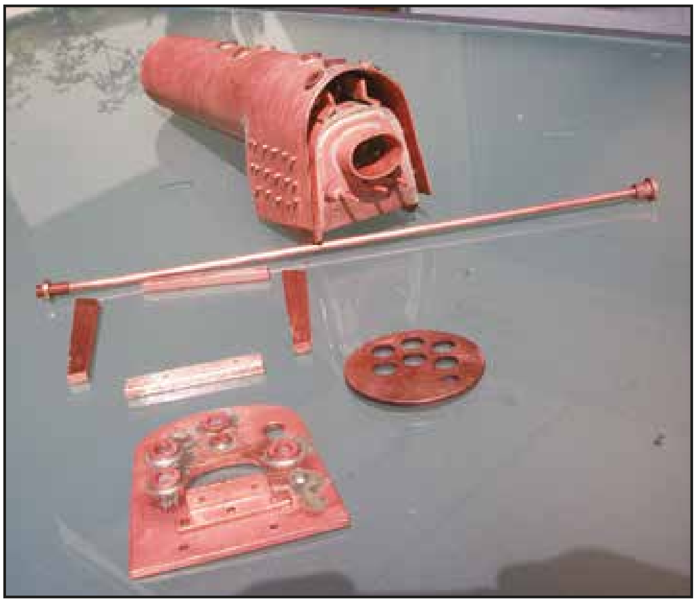
fitting them to my cylinders. I eventually managed to find a way to use the draincocks which meant the front valve pointing inward and backward with the drainpipe looping around to hide it. This gave a quite reasonable appearance but the main point was that they worked successfully. Sticking strictly to the MEL motion dimensions and my own design cylinders (which were built with the same dimensions as the MEL overall dimensions), the chassis was air tested without any trouble .

A push test of the chassis on the 7' 6" radius track proved that the elimination of the rear frame and moving the front bogie 1/16" meant that the loco was able to handle these curves. The loco to tender connection was a problem and a link proved impossible and I had to adopt the 'Hornby' slot and hook approach which also proved very successful.

Next came the boiler. I am strictly a Stephenson boiler advocate and it was found that the Martin Evans V2 boiler overall dimensions were almost identical to the Britannia and would fit with only minor modifications. I knew that my own V2 ran well but was built as a JVR 'B' type hockey stick boiler utilising three 1/2" hockey stick tubes. This would be no good for gas/coal firing which was my plan. As I always make my own boilers and the Martin Evans V2 boiler was a coal-fired design, I set to with my CAD system and came up with a design which was little changed from the original drawings. Building the parts was relatively easy but my biggest problem was the silver soldering! I decided to sub-contract this part out to someone more experienced but being in Australia that was easier said than done. I did eventually find a boiler maker who was also a local boiler inspector at my local club. Sadly, this became a major problem as he would not solder the boiler unless I modified the design and incorporate threaded longitudinal stays which I could not accept. This took many months of discussion and eventually a failure to agree led me to soldering my own - knowing that it would not be accepted to run on the club track. Building a simple copper kettle is one thing, building a large boiler with stays, rivets etc is more akin to a real loco and is another story. I had to build a suitable hearth and acquire a better burner etc, but the boiler was subsequently built. I was expecting a few leaks but this was a disaster. I tried some remedial work and further repairs but was getting nowhere. Sadly, there was no person with experience of building such a boiler that I could turn to locally. So, after more research, I set to and built a completely new boiler, this time being more methodical and with much appreciated advice from friends over the internet and with great care, the second boiler was completed and after a few leaks were fixed the 120 psi test held for four hours.

A new design lubricator was installed as the MEL filler was in the smoke box which meant that, if continuous running, it would have been impossible to add more oil. The regulator and blower are of the Cheddar design using soft seat balls and a gauge glass with blowdown was fitted as well as a low level electronic indicator.

The loco was assembled for a steam run using the ceramic gas firing method and was reasonably successful except that it was felt that the steam was not sufficiently dry. I had chosen not to add a superheater, which would have solved this problem, so I was advised by Gordon Watson to try a heating coil in the smokebox to dry the steam which fed to the cylinders. The difference was astounding. In consequence, after several trial runs it was 'signed off'. The detail



The part-completed boiler.

assembly stage was fairly straightforward with the normal compromises that go with a working steam engine at such a small scale. After painting, it was back onto the track and some performance trials. The loco took just over two years to build. The Accucraft gas tank is in the tender immersed within the water and measures 2" diameter, 3" long and it was found that the gas duration was not ideal and ran for around 20 minutes. Subsequent tests found that once the loco was up to pressure the best method was to turn off the gas, check that the fire was out with a lighter and then refill the gas tank and relight. With a little athletics and adding water to the tender it would run non-stop for 35 minutes.

As I had already built a set of BR Mk1 coaches, one (a full parcels coach) was modified to contain an additional water tank with more water than the tender and, with a water connection between the tender and the coach, it was found that the loco was able to run without any manual intervention. The coal fired grate was pre-built during the boiler stage and could be exchanged in 30 minutes but to date has not been tried. Radio control was also fitted but was found to be more bother than it was worth due to the initial firing and re-lighting.

The loco has a minor problem and occasionally loses its fire when the regulator is closed and the blower is not opened. This is a secondary air problem which differs for each loco design. It requires an automatic blower off/on which is dependent on the blast to obtain the shut-off. This is still a work in progress. Steam raising requires a typical blower.

Overall the loco is a joy to fire and run and has its own characteristics which is exactly what makes one steam engine different from another.

What of the future? Having lived in Australia for over 7 years my love for Australian locos is growing so much so that I have found, sadly, that one of the most prestigious steam locos ever built in Victoria was cut up shortly after withdrawal. My aim is now to capture the history and the glory of these great Pacific locos. Only four were ever built. It has a lot of similarities to the LNER A3s and some of the American Pacifics. The loco was the Victorian Railways 'S' Class built between 1928 - 1930 and the last was scrapped in 1954, days after its withdrawal, and so began the Australian Railway preservation movement. They were designed specifically to run the 'Spirit of Progress' intercity train between Melbourne and Albury (being 5' 3" gauge) as part of the Melbourne-Sydney Express (New South Wales being standard gauge). From 1938 they were all streamlined.

Three model locos are to be built: 1928 original version number S 300, 1937 version number S 301 (the last before streamlining with smoke deflectors and other details) and 1938 streamlined version as it was in its final years, numbered S 303 named 'C. J. LaTrobe'. All will be built in parallel.

All the CAD drawings are complete, as well as all of the laser drawings and the boiler and motion details. The new loco boiler is very similar to the Britannia dimensionally but will have a less complex design and will remain a Stephenson front end and be ceramic gas fired, have automatic draincocks and an automatic blower. The streamlined loco will be 32" (815mm) in length having an enormous 12-wheeled bogie tender. In August 2013 the wheels had arrived and the laser parts were due to be cut in September. It is hoped that they will be running in 2014. The locos will be unique as they are being built to 3/8" to the foot scale with with the exception of the gauge which will be Gauge 1. Only wheels and coupling rods are out of gauge, everything else will be built to scale. The aim is to have these locos running alongside their UK counterparts at the same scale. I will write an article on Spirit of Progress when they are completed. Should any potential Britannia or S Class builders wish to contact me they will be most welcome.

Alan Moreton's Aster 9F on the turntable at Buckfastleigh. The weathering was carried out by Dave Walker. Photo Dave Stick.

WHEEL TURNING TECHNIQUES FOR GAUGE 1

by Harry Wade (2214)

Many years ago, I undertook to turn my first set of driving wheels and my then mentor set me to work without his usual helpful instruction. In very short order, I ruined the first casting and for the first time I experienced that helpless feeling one has after accidentally setting alight a large bank note! Whether in this instance the lack of advice was intentional, to quicken my model engineering education, or accidental I'll never know, but either way a valuable lesson was learned. What I had failed to do, because I didn't know to do it, was look beyond what metal was to be machined away and pay attention to what was NOT to be machined. Years later I found this was no less important in Gauge 1 and our worthy Editor felt a few notes on the technique I developed for wheel turning would be worth passing along.

It's worth keeping in mind there can be as many ways of turning wheel castings as there are wheel turners. Eventually each of us will develop a technique which will produce results which suit us, given our individual skills, experience, equipment, and objective. This is as it should be. My method adds a few steps to the process, some arguably unnecessary, but most of this I put down to my belief that "If I'm going to do this job at all, I might as well squeeze the most and best out of what I have to work with". My hope is to offer some useful ideas or perhaps solve a problem for fellow wheel turners. (Note: My technique isn't applicable to Mark Wood wheels [usual disclaimer] as they require different treatment and Mark's comprehensive machining notes should be followed.)

My objective with any wheel casting, aside from not ruining it, is to have all the visible surfaces of a wheel (flange, tyre, rim face, spokes, bosses, etc,) finish out crisply and in proper relationship to each other, and "all-square", without wobble or eccentricity. I found this is best achieved by first locating the surfaces which will NOT be machined. Sand and shell moulded castings are often not perfectly round or flat so I take time to align the castings as best I can, averaging or splitting the difference of any irregularities. The traditional method in larger scales has been to chuck the casting by its rough tread, machine the back flat, and bore the axle hole all at one go, but this can leave the location, squareness, and concentricity of all the face and the visible features of the wheel somewhat to chance.

To begin machining I chuck the castings in a 4-jaw face forward by the rough flange. Adjusting the chuck jaws I set the inside edge of the wheel rim, where the spokes transition into the rim (Fig 1-A) running as concentrically true as the casting will allow, averaging out ovality. I then set the spoke faces (Fig 1-B) running as nearly axially true (ie, square to the lathe axis) as the castings will allow, again averaging out irregularities. I use a DTI (clock gauge) for this. The rough tread OD and rim face may not necessarily be running true at this point, but that's OK.

Once the un-machined surfaces are established square and concentric, I turn a shallow straight shoulder, or step, on the tread (Fig 1-C) making sure not to cut into what will be the finished tread. At the same setting I skim the crank hub and tyre face just barely to the point of exposing clean metal, in order to check to see if the pattern maker has left enough material to machine both surfaces to the correct relative back-to-front face dimension. If all that looks right, I face the rim and tyre face to about 0.5mm over their final finished surfaces relative to the spokes and each other. I now have several true datum (or 'witness') faces from which to determine tyre width, hub width, counterweight protrusion, and the amount of metal to remove from the back, and that all the features of the wheel are going to end up square and concentric to each other and to the axle.

A wheel with a matching collet.

Changing to the 3-jaw, I reverse the casting and, holding it by its front shoulder, I machine a shoulder or step in the back (Fig 1-D). This assumes there is the usual excessive metal on the back of the casting, to protect the spokes. I make the diameter of this step roughly the diameter of the finished wheel tread. This will help reduce chatter during final finishing stages. I then reverse the casting and, holding it by its back shoulder, I machine the entire front, including axle bore, tread, and flange all at one go. The last operation is to reverse the wheel a final time and remove the spoke webbing and finish off the back flange taper. For this operation I

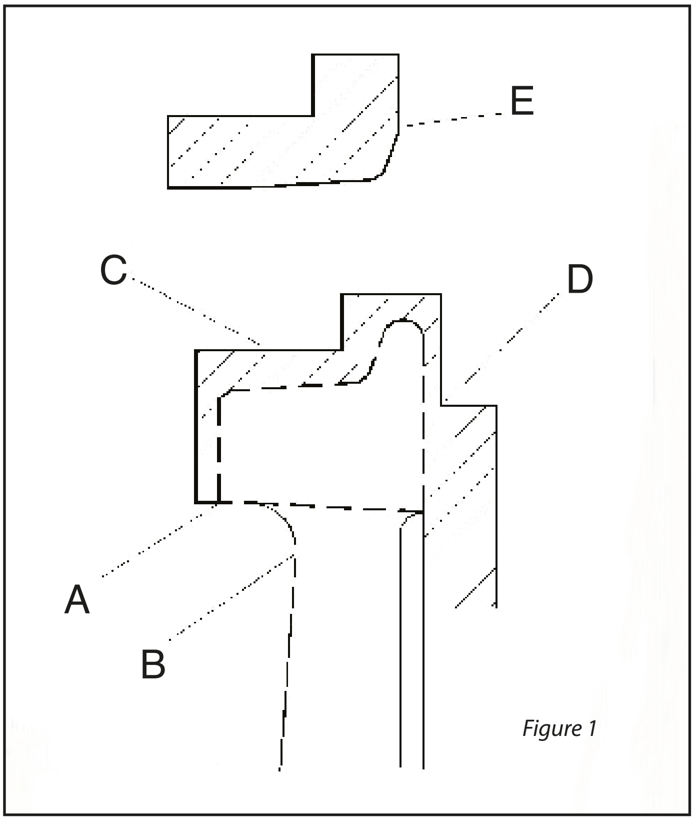


Figure 1

make a temporary split collet from a piece of tube or bar of any material (Fig 1-E). I turn a taper on its ID to match the tread taper (if any) and then cut a straight shoulder for chucking on the OD. In practice, despite the tapers, the collet will hold the wheel without squeezing it out like a wet bar of soap. These steps take time but they result in a wheel which is as true to the eye as the casting allows.

Here are a couple more tips I've picked up along the way which help with wheels in general:

1. From time to time most of us will experience harmonic ‘chatter’, the result of the tool bit wanting to ‘run away’ from the work, which is a symptom of lack of mass and rigidity in both the castings and our smallish and relatively light lathes. There are several remedies for chatter, the first being to reduce the amount of material being taken with each cut. The next option is to slow the cutting speed or the rate of tool feed, or both. Even these may not completely eliminate the chatter and, as a last resort, which I have used successfully many times, using a very light depth of cut you can simply pull the chuck around by hand. The cast iron doesn't seem to care either way.

2. ‘Loctite' with wheels and axles. Most Loctite retaining fluids need a 0.1mm± curing clearance between the wheel bore and the axle, however the 0.1mm clearance is enough to allow a G1 wheel to wobble a bit on its seat. This can easily result in a wheel curing either crooked or eccentric ending up with a visible wobble, eccentricity, or both. My remedy for this is to turn the axle wheel seats to a very close sliding fit to the wheel bores, close enough so that the wheel slides on the axle stub but there is no discernible play. I then turn a ‘trough’ in the axle seat up to 0.2mm deep, leaving a collar or shoulder of 0.5mm or so width at each end of the seat (Fig 2). When the wheels are installed, the trough is filled with Loctite and the wheel slides on but is held concentric and square to the axle by the collars. Potential wobble eliminated.

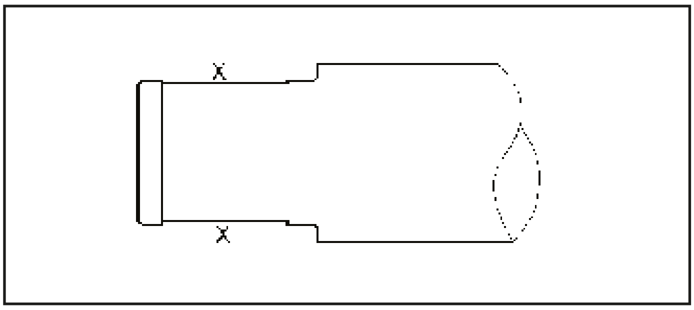


Figure 2

DRIVING WHEEL QUARTERING

by David Eaves (3998)

I don’t know if this subject has ever been covered in earlier editions of the NL&J. Having only built a couple of G1 locomotives so far, one of the conundrums I had was how to quarter the axles and fix driving wheels to them reliably to prevent any future slippage, while still being able to take them on and off as many times as required?

This problem came to a head with my ‘Project’. The G1MRA Project construction book describes fitting the wheels using a tight push fit. That worked fine initially but even during construction I found I had the wheels on and off their axles a few times. Even fixed with locking glue over time and running, plus the effects of oil penetration the wheels eventually started to work loose, causing the axles to spin but leaving the driving wheels with no solid connection. I had to keep caulking the axle ends and re-glueing. I used a centre punch to open the ends of the axle to get a tighter fit. I was surprised just how much thumping these axles and wheels get during running.

A couple of other G1 loco construction drawings I have leave it up to the builder how to tackle this problem. With my second new loco build of a twin cylinder C4 Atlantic ‘Jersey Lily’, where wheel quartering was even more important than on the single cylinder ‘Project’, I did not want to have this problem again. So I tried another method.

I know there are commercially available axle sets with square ends and machined matching wheels with square holes to fit, but that's not for me, I machine my own wheels. The idea was to fit some form of secure locking between an axle and a wheel. The ‘Project’ used axles as in Fig 1.

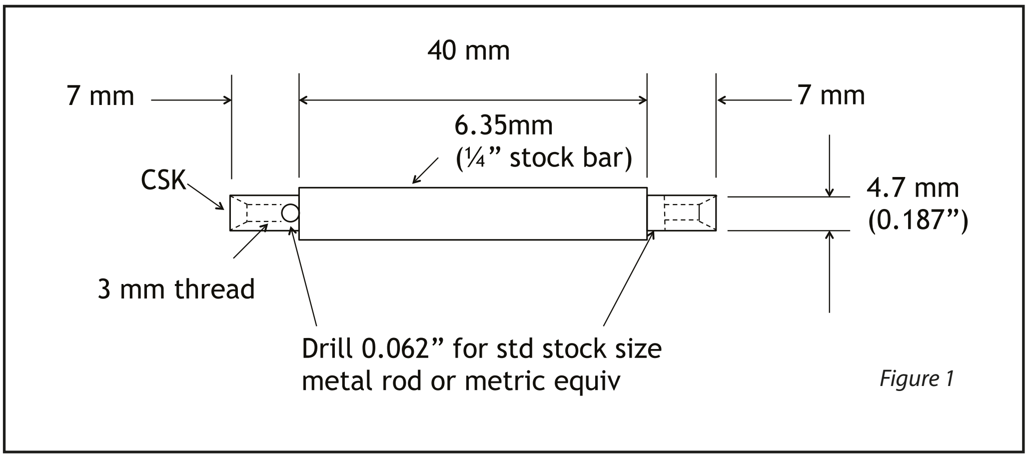


Figure 1

It is not easy to drill an accurate hole through a round bar at 90 degrees and square and on top centre through an axle of 1/4" diameter. I knew this would be very tricky to do freehand every time, especially as I needed more than one axle set. A simple drill jig was needed so that I could consistently achieve drilled holes at 90 degrees to each other at each end of an axle and in the same positions each time.

The axles for the ‘Project’ are 54mm (2.126") total length. The drill jig is exactly the same length of 54mm. Any piece of square metal bar stock would work. I used 1/2 × 1/2" square brass bar. I set the bar up centrally in a 4-jaw chuck and turned the bar to the exact length of 54mm, then drilled a hole right through the square bar to take the axle. Then I marked out the position of the holes on the jig and drilled these as accurately as possible, as they need to be right. Once the jig is completed it can be used as many times as required.

If I was going to drill lots of these axle sets, I could have used mild steel square bar and added hardened drill guides, to give greater wear resistance, but brass will be fine for my needs.

The axles were held in the jig by a fixing screw and both holes drilled at 1/16" diameter each end in one operation. If I have done my sums correctly, marked it out correctly, and made the drill jig correctly, once I have turned down a length of 7mm (0.275") and a diameter of 4.7mm (0.187") at each end of the axle to take the driving wheels, I should see each hole in the axle is at the end of the shoulder I have just turned. No part of the drilled hole should break into any part of the larger 1/4" diameter part of the axle.

Each driving wheel has to be countersunk on the outer face to the same depth for the countersunk head of a socket fixing screw, so I next drilled and tapped a 3mm thread at each end of the axle to take a 3mm countersunk socket screw to hold the driving wheel firmly in place. The end of the axle also has to be countersunk to the correct depth. At this stage there is not much material to play with. With the axle in the lathe chuck, I placed a driving wheel on the end of the turned axle shoulder then started the lathe. The countersunk driving wheel will spin with the end of the axle. As the countersink depth grows the drive wheel will stop spinning once the countersunk cutter has gone deep enough. This will be once it meets the existing countersunk depth already on the driving wheel. It cannot countersink any deeper because the now-stationary driving wheel will not allow it, only removing just the right amount of metal from the end of the axle.

Each driving wheel requires a small slot or recess milled on its back face on the same axis as the wheel centre and the crank offset hole that the crank pin screws into. I figured a slot width of 1/16" and depth of 1/16" minimum to accommodate the locking pin. The overall length of the slot was not critical.

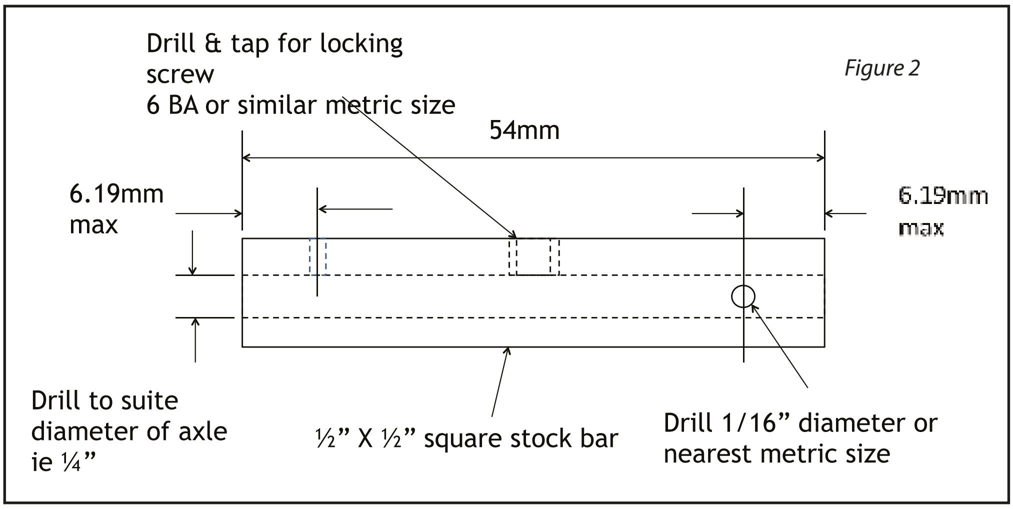
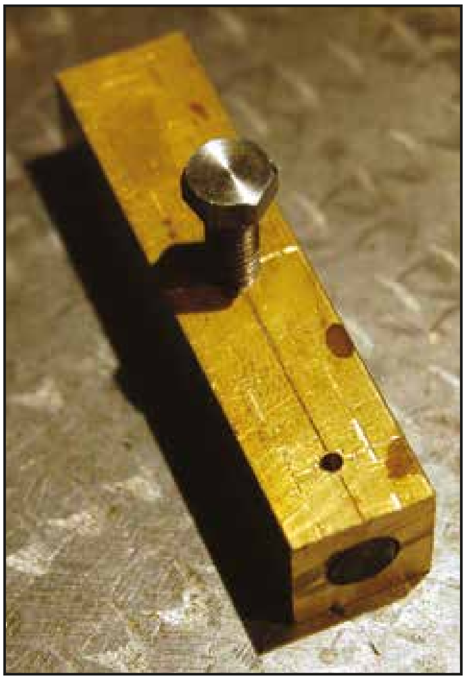


Figure 2



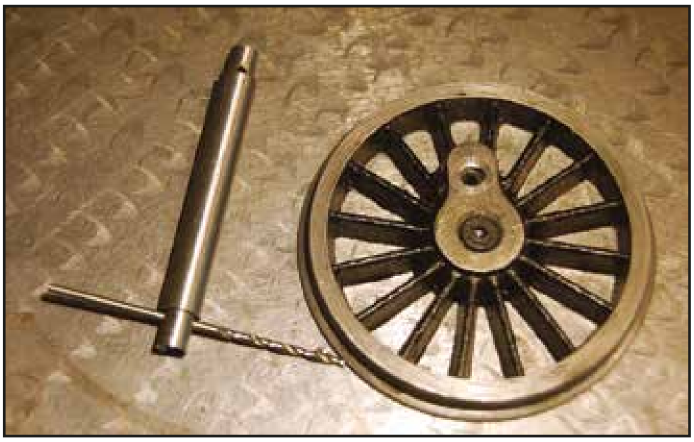
Drilling jig with an axle inside ready to drill holes.

Another jig for milling the slot in the driving wheels was made. The requirements for the milling jig are two drilled holes in line on the same axis plus two threaded holes for the clamp screws. I used some mild steel flat bar about 1 1/2" wide by 1/4" thick. The jig was held in a machine vice and supported by a parallel underneath. The jig accommodates two different crank offset sizes. The driving wheel centre hole locates on a small spigot of the same diameter which is prevented from any possibility of falling through the jig by the parallel underneath. The orientation of the driving wheel is set using a small screw through the crankpin hole on the driving wheel into a threaded hole on the jig. Then all is held secure with a couple of clamps. Each slot can now be milled out for each wheel.

For assembly onto the loco frames the axle with the first driving wheel is located onto the pin and held to the axle with the 3mm countersunk screw. The axle is passed through the frame and the second locking pin fitted at the other end of the axle before the second driving wheel is located onto the axle and locking pin, then held on with the countersunk head screw. For the crank axle on the ‘Project’ the axle has to be in situ in the frames, then locking pins added at each end, then wheels added then locked and screwed on. This arrangement seems to work fine as I have since tested it with my ‘Project’ out on the tracks.

The jig for milling the slot in the driving wheels.

The locking slot being milled in the back of a driving wheel.



An axle showing the two drilled holes and a driving wheel.

A driving wheel with its locking pin fitted.

A CABOOSE FOR MY BERKSHIRE

by Phillip Taylor (2648)

NKP Caboose 1047 waiting for Phillip to steam his Berkshire.

Those within my local Northampton and East Midlands Gauge 1 circles will most probably associate me with my kit-built Aster express locomotives from the 'Big 4' pulling appropriate liveried rakes of BR Mark 1 coaches and so this story really begins with: Why American?

During my career as a chartered engineer, I travelled frequently to the USA through the region served by the former Nickel Plate Road (NKP) including including Peoria and St Louis to the west and eastwards beyond Chicago. My business did not prevent me from observing the varying topography of the industrialised northern States and imagining the hard working Berkshires with their snaking consists each tailed by an ever present little red caboose. Thus it had to be that when Aster's beautiful Berkshire became available I simply had to have my good friend Andrew Pullen supply me with one to squirrel away until retirement provided me with the time to build it.

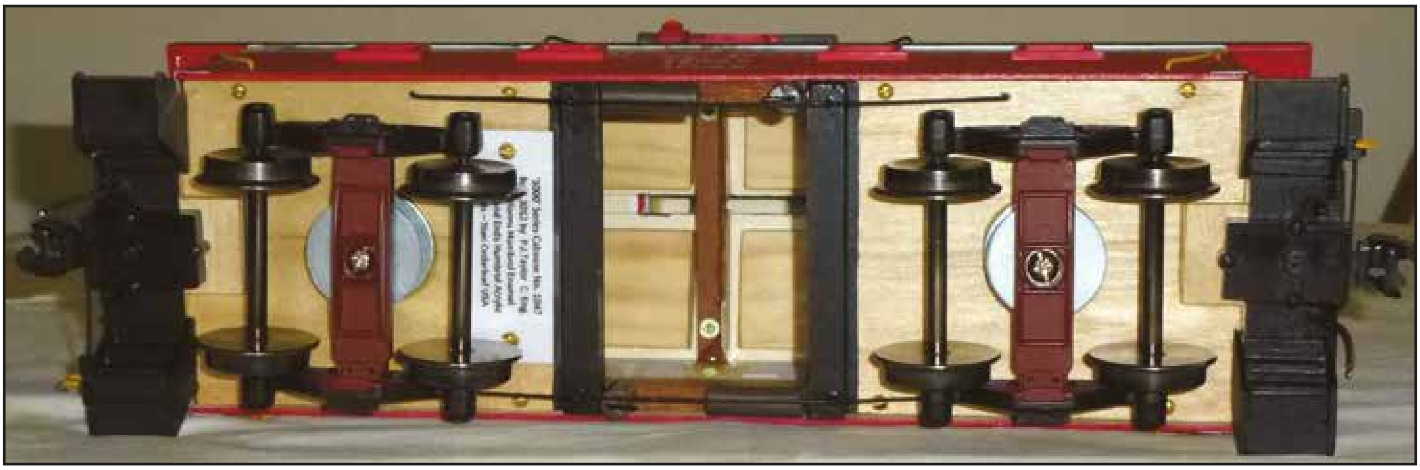
Initially I gave little thought to the 'consist' it might haul since box cars, reefers and a suitable NKP caboose would be readily available, or would they? The cars, yes, but the caboose, well no. As the loco neared completion, I started to ponder the caboose question noting that others had solved the problem by repainting a generic steel bodied caboose into NKP livery. As some in G1MRA will know, I love researching, chasing down the facts and keeping documented records. After studying numerous Nickel Plate Road photographs, particularly of the yard at Bellevue, Ohio, dotted with red cabooses, I just had to recreate one of the original NKP wooden bodied '1000 Series' first built by the Lafayette Car Works in 1881 and which remained in service right through the Berkshire era and beyond.

Modelling

Within my local Gauge 1 groups there are some extremely talented locomotive and rolling stock modelmakers, inspiration indeed to consider scratchbuilding what is a relatively simple shape, demanding not much more than an ability to keep things flat, straight and square! I don't get started with such projects until I know exactly what I am building and how to build it! Thus the NKP Caboose 'Work Book' was born into which went numerous prototype photographs, dimensional data from the internet (who can live without that!) and several schematic downloads from my grey matter of the proposed method of construction.

Armed with all the necessary dimensions, a set of drawings was produced in 1/32 scale during which time it was observed that the angle of the main roof from the horizontal is 7 degrees whilst the cupola roof is flatter at only 5 degrees which has a subtle impact on the appearance.

Next into the 'Work Book' went a series of non-negotiables, as follows:



The underside showing the mahogany framing, internal access, cupola securing screws and bogies mounted against sprung washers.

- It had to run true and as smoothly as track would permit.
- Couplings and coupling height could not cause problems.
- The interior had to be accessible for bogie and weight adjustments if necessary.
- Colour scheme and decals had to be of high accuracy (More of that later!).
- Dimensional accuracy was not to be compromised.

Method of construction

Having purchased well detailed Accucraft reefers and box cars, an early decision was to use the same knuckle couplers and very appropriate Bettendorf bogies which at a stroke would eliminate coupling problems and should permit smooth running if suitable bogie mounts were developed. A three-ply birch plywood body would be mounted on a hardwood underframe. Soft soldered brass work would complete the hand railing, stanchions and flue stack. Slater's plastic materials would be used for window and end door detail. I scratched my head regarding handrail knobs until Paul Forsyth surprised me at a local GTG by producing just what I required from his magic box of bits and pieces.

Challenges and solutions

Correct height of the base of the bodysides above rail height had to be considered, as did the height of the end platforms, whilst

One end of caboose 1047.

Brass collar and coil spring height control.

accommodating the method of bogie mounting. It was the platform height above railhead that was to prove to be the correct datum starting point. This set the height of the hardwood underframe to which bogies could be mounted and adjusted, setting the platform floors at the correct height. The caboose body could then be engineered to straddle the floor such that the bottom edge of the body sides sat at the correct height above the rail.

The 'Workbook' was revisited to amend constructional details. (Make it up as I go along!)

Bogie mounts, in the form of internally fine threaded tubes, were secured into the caboose floor into which securing screws would be set to a depth governed by brass collars placed over the threads thus dictating the floor height. Suspension would be provided by a soft coil spring over each collar working against large diameter washers. The screws were set in position using Loctite 222. This arrangement produced a very smooth riding underframe. Both bogies have the same mounting arrangements. The bogie retaining screws are tightened up to the collars but with just sufficient clearance to permit them to swing freely. The springs, obviously, are just a little longer than the collars and compress to the length of the collars when the screws are tightened which is sufficient to absorb shake and wobble when running. I was a little surprised as to how smoothly the arrangement worked and had designed in the large diameter washers top and bottom with a view to inserting some form of rubber mounting shock absorbing material between them to ensure a smooth performance on the track.

Should wear on the collars take place over time resulting in wobble there would seem to be two possible solutions; either

The excellent decals are by Stan Cedarleaf in the USA.

and soldered brass was routine enough, but I wanted to ensure that the whole could be dismantled if the need ever arose. I decided to make the roof detachable rather than have the whole body assembly lift off the underframe. To facilitate this, the cupola once detached via the central underside opening reveals roof securing screws.

Painting and decorating

There was something familiar about Caboose red. John Squire seemed to be near the mark when he suggested British Railways locomotive buffer beam red was a close approximation. Some experimenting with a Humbrol tinlet of enamel convinced me that it was about as close as I was going to get. The upper body white band should be slightly off white and I convinced myself that several top coats of varnish might just take the edge off the white.

But first, the paint system to use. Although in possession of a capable airbrush, I often resort to aerosols having achieved good results with the acrylic system, where thin rapid drying coats can be applied in quick succession. I have also had excellent results with acrylic car primers which Humbrol acrylics seem to take to readily. That being so, I decided to paint the whole body in acrylic. The results were quite satisfactory.

Finally came the matter which I was dreading - sourcing correct decals. The perfect solution came from Terry Foley who put me in touch with Stan Cedarleaf of Dewey, Arizona, USA. Upon receipt of images from me, Stan produced four sets of astonishingly accurate waterslide decals, very reasonably priced, in less than one week. Well done Stan! I am the first to admit that these superb decals really do add some authenticity to the model. To protect the decals I finished the whole body with Humbrol enamel satin varnish which I considered would be durable and waterproof. I must say I was satisfied with the result, with only the

break the Loctite 222 joint securing the screws and tighten slightly before resetting them or insert some circular shock absorber material between the washers.

Close examination of the image of the bogie will show the spring base seated against a small wearing washer with a little grease applied to permit smooth rotation whilst at the same time exerting just a little pressure to take out any shake.

Progress!

Once a running underframe was created progress with the body was swift. Measure twice, cut once, scribe straight and constantly refer to the prototype photographs and the 'workbook' dimensions. "If I couldn't do that I shouldn't be in the business of modelmaking" I repeatedly told myself. The width of the vertical planking was simply calculated by observing the detail on the NKP museum's 1047 over the scale model dimensions. Scribing of the vertical bodysides was opened out by running through with a defunct ball point pen. The remaining work with plastic detailing

Phillip's Aster Berkshire was finally finished in 2013 and is seen here running on John Squire's track.

fitting of windows remaining.

I cannot fail to mention the assistance I had from Chris Beamer of the Mad River and Nickel Plate Museum, Bellevue, Ohio, who kindly provided some digital images of the small lettering on preserved caboose 1047 in order for Stan to correctly create these decals for me. These, incidentally, relate to the overhaul and repaint dates in NKP's Fort Wayne shops – 24th March 1955!

So yet another Gauge 1 project is completed. I must say that whilst these cabooses were some of the largest in the USA, it seems such a small but yet conspicuous red marker trailing at the end of the train and positively diminutive when paired up against the Berkshire's huge tender! Just like it was in the 1950s!

SCRATCHBUILDING

by Rodney Maennling (224)

I like to model right from scratch using raw materials. This article might be of interest to our new members, or those needing rolling stock, and keeners wanting to simply hand build. Over the many years reading the Newsletter I have picked up many tips and ideas. I hope this article may add to this resource.

I do not particularly enjoy assembling kits, and I know I'm influenced by my father, a gifted model maker and radar researcher who set me onto the path of working from scratch. There is also a unique sense of personal satisfaction in completing a modelling project that evolves mostly from ideas in one's mind. The other reason for me is limited economic resources that can be seconded to this hobby, due I reckon, to other recreation and sports interests in my life.

Having acquired two Aster live steam locomotives, a Schools and a Shay, in the late seventies, I set about building rolling stock specifically for each locomotive. In doing this, I quickly realised I had tapped into various skills I had acquired during the younger years decades ago, and I have gained much pleasure in putting these dormant skills to use.

I had limited machine shop metal working experience that would be beneficial, so as I enjoy working with wood, I put this medium to use. I work mostly with hand tools especially my beloved Hobbies fret saw, assisted with a bench-top drill press and small band saw, all accommodated in my thirteen by thirteen foot workshop.

There's nothing complicated in what I'm saying here but an inherent desire to produce something made by hand. I was trained on the drawing board as a mechanical draughtsman, so during winter months I unpacked a dusty box of drawing tools, and using a small set square and dividers, I organised a small draughting pad on my lap. I dug out wagon drawings from a collection of pre-war Model Railway Constructor magazines, and drew them to 1/32 scale on 10mm squared graph paper. For that matter, I assume the same can be done using modern digital equipment, with more accuracy I bet.

Using tracing paper on top of my graph drawing, I pencilled out my working plans from the drawing underneath, using dimensions and lines of suitable wood I thought would work. What came to mind was 3mm plywood, door sheathing, or balsa bits from model shops. I also consulted carefully the G1MRA specifications data sheets for rolling stock for correct dimensions.

For my first attempt I decided on a five plank four wheeled wagon, sketching in the side view, plan, and end view. The datum line I use is the 1/4" Baltic ply floor, and from that arrange wheels and buffers, following the G1MRA specs. This is the basic approach I use for all rolling stock designs.

At the time when I started building I couldn't obtain G1 hardware pieces so I improvised in any way I could. Here is an example: roofing nails with their large heads became buffers which, suitably filed and trimmed, did the trick. Wheels and axle guards were purchased from the limited G1MRA parts members had made, and were kind enough to sell to other members. The challenge in finding alternative materials for a job, though much less difficult, is often a challenge. I reminded myself that this being a hobby means it is part of the enjoyment quotient!

Here are other examples of my scrounging. The photographs show examples of my improvisation. Take a close look and you'll see: drawing pins snipped off as lamp shades; dolls house pea lamps

Rodney's scratchbuilt Devon Belle observation car.

for lighting; model aircraft plywood for roof sections; emery paper as roofing material; drinking straws as an accordion chimney; and baling wire for handrails. I stripped copper wire for decorative grab rails; jewellery findings used as chains and hooks; tin cans and

Inside the Devon Belle observation car.

Rodney made a travel box for his Devon Belle observation car that doubles as a display case.

Part of the detail inside the CNR caboose seen on the front cover during construction.

newsprint aluminium sheeting as cladding for a water tower; and to mention again, 1/8" mahogany door sheathing for walls and sides, 1/4" Baltic plywood for flooring/solebars. Some items you cannot replicate, and nowadays we're blessed with an enormous array of parts, transfers, and accessories via our brilliant G1MRA NL&J.

As my experience grew, I tackled more complicated projects. However, I drew the line at hand building coaches with their many windows. There is the odd exception. My 'Devon Belle' coach happened to have windows for which I was able to use one piece of acrylic on each side. Project made do-able! I used hobby plastic sheeting (styrene) fixed over the wood to replicate 3 inch vertical wooden planks. In this particular coach I used liquid rubber to make up moulds for the plaster of Paris furniture. While visiting the actual coach at Paignton I photographed the wall map, then pencil sketched it to scale size for the model.

Modern adhesives are astonishing. A fast curing wood glue (Tite Bond) is perfect for major joinery on a model, but cyanoacrylate gel (as per Krazy Glue) from model shops cures tight in 10 secs and is perfect for delicate attachments. I found that epoxy (its consistency is a gel) can make a nice rivet head if a tiny blob on the end of a pin is placed strategically one by one, if nerves can endure this personal abuse…!

Last but not least. British fasteners for modellers (especially BA brass and steel screws, nuts, and brads etc.) are the best in the world. Mine came from Clerkenwell Screws who carry a very wide range. Check their website. But note: I think G1MRA members are the best source of advice, bar none !

I've probably bored you once and for all, so I suggest you take each piece of information, think on it, and place it in your mental contemplation box. I wish you every success with a project of your own design and making.

Some of Rodney's goods wagons made almost 25 years ago.

TABLE DECORATION ON THE CAYTHORPE LINE

by John Squire (1852)

The level crossing and Caythorpe North signalbox.

Some time ago I described how, with forward planning, I found it possible to build and operate a G1 garden railway while at the same time remaining blissfully married. Having accomplished this onerous task, I was inclined to sit back and bask in the sunny days of my retirement. However, not all my fellow playmates were impressed with my efforts and my railway became one of those included in the description 'Table Top Railway' where tailchasing was the rule and very little of interest to the real railway aficionado occurred.

To me, this seemed a little unfair as the layout consists of twin 240 foot circuits, one for coming and one for going, 30 sets of points, sidings and passing loops, storage for numerous rakes of rolling stock and a steaming bay with a turntable able to accommodate all but the most expensive Asters (who needless to say are catered for on their own exclusive sidings). There are even fireproof places where coal fired locos can be steamed. Never-the-less, it is true to say that all of this is set at a height which dismisses the need for back bending and damp knees. Crawling under the track (a nasty habit at best) is positively discouraged. The purpose built pedestrian over-bridge gives clear access to the inside of the track, which denies the track crosser the opportunity of inadvertently kicking that splendid blue Duchess with her rake of bespoke coaches into the lawn as she speeds past, although it must be admitted that this can detract from that frisson of anticipation on a busy running day.

But the plain truth is, and one must always be truthful, that this entire railway track was indeed laid on a rather large almost circular picnic table, so what to do? I decided that away from the steaming bays and hubbub of the marshalling yard (seven sidings to those who despise exaggeration) a nice country station was needed with a loop, a bay platform and a goods depot, so plans were made, holes dug, cement mixed, posts planted, ply cut until Lo! and Behold; a splendid extension to the table top area arrived and sat there awaiting action. And so I rode forth and acquired some second hand buildings intending to decorate the whole thing a bit and achieve a sense of a railwayness in order to allay the doubts of my critics and so it would have been but for Messrs MS and IN.

In my village there is a retired scientific engineer, one Mike Smith, much addicted to detailed modelling, who has to my everlasting shame been suborned from his previous existence in the accurate world of 0 Gauge into the slapdash world of G1 high speed live steam. However, unable to shrug of his old ingrained and outmoded ideas of prototypical modelling to a high order of accuracy, he cast a scornful look at my modest bits and pieces and, in a moment of foolish kindness, offered to make me a goods station. Then no less a person than Ian Nugent, a wanderer from the more remote parts of East Anglia who allows his splendid scratch built steamers to whizz round my track from time to time, recognised that the two bits of unplaned 2×4 timber planned by J J Squire would not be adequate as platforms for Mike's models and offered to build some platforms for the layout. Recognising in an instant that I

Newly overhauled 47xx on a pick-up goods.

Caythorpe Station footbridge from the up platform.

Platform 1.

was on to a good thing, I gratefully accepted their services and I concentrated on making a pathway and laying gravel to improve the wet weather access so necessary for those cloud filled skies of high summer and laying in supplies of cold beer for those rarer days of overheated train chasing. No railway shed should be without its own dedicated refrigerator.

Now all this would have progressed quietly and steadily but for a terrible warning from the Spoerer of Norfolk. "THE SWISS ARE COMING" he cried. "On Friday 13th 2013 of September you will be invaded by a hoard of Swiss G1 inspectors for a close examination and trial of the track quality and general standards of your layout. England Expects, etc, etc". Naturally terrified at the prospect of exposure as a slap dash layabout, I threw myself on the mercy of my two stalwart friends who rose magnificently to the challenge and threw themselves into the creation of the 'new' Caythorpe station. Ian's splendid platforms which sit level on my baseboard, a feat of fitting and skill with a plane seldom surpassed, as my levels were of a notional nature, coupled with Mike's extraordinary dedication to detail and huge output have created not just a goods shed but a country station complex on my track which will to my mind stand up to any comparison for its excellence. I did put up the shelves in the shed where all the bits and pieces are stored away safe from the dangers of pigeons with digestive problems and huge falling apples shaken out of their trees by passing low flying RAF planes (Ed please note). *(Not guilty! At least not there -Ed).*

In addition we were also blessed by a gift of some signal gantries by Don Evans who acquired them from the estate of Robert Head some years ago. Mike refurbished them and added them to the signals which he had already made up and instantly the railway atmosphere was added to immeasurably (There are two emergency buckets of water kept to hand, one to put out any large conflagrations and one for the first driver who knocks one over).

In any event by Friday 13th, which shall from now on be an auspicious date in my calendar, all was ready and set out for the first time in its entirety. Accompanied by music from Don Gurney's beautiful street organ, our Swiss inspectors all had a jolly good time as did we locals, 40 of us in all. Trains were run, wine was taken and at last my garden table tops were used for food while the trains ran on the RAILWAY.

As a postscript, I must say that for me this event displayed the very the essence of the special quality of G1 live steam in the garden. Those present covered the whole spectrum of our area of the hobby, all the popular forms of firing were used side by side; even battery powered radio control shunting took place. Fine engineers played alongside mere beginners and even the dreaded cheque book modellers who some find so hard to accept. Friends and neighbours visited and helped with the catering and the local cricket club chipped in with a splendid marquee just in case it dared to rain.

Finally, as can be seen from the pictures, in future anyone wishing to come and play trains will now have the pleasure of running through Mike's wonderful station and past his signals (with care) and all-in-all will have a more railway-like experience. To that end, I do invite individual members or groups who would like to come, to phone me to arrange a date. Parking is not too stressful and, while I publish regular running days, I am happy to have extra running days for groups who travel from foreign parts. Some have already made successful visits from such remote far flung places as Yorkshire despite the language barrier and the perils of crossing the border into England.

Empty garden railways are no fun.

SIMPLE SLIP ECCENTRIC VALVE SETTING

by Roger Thornber (1124)

I have been assisting Martin Hulse in producing Edition 5 of the Project book by producing various 3D drawings and pictures. There was considerable discussion when we got to the subject of valve setting. Looking into this, I felt that there was some misunderstanding that warranted a small article on the matter. I hope I will not be accused of teaching grandfather to suck eggs.

I will start off with a simple look at the sort of valve motion that I have used for most of my locos and I will only consider slip eccentric, although the methodology can be applied to other gears such as Walschaerts. The diagram below shows the dimensions that I use. Notice that the valve overlaps the ports by 1/16". This is known as lap. Dead centre is when the piston is at its maximum travel at either end of its stroke. If the cylinder centreline is through the driving axle centre, then dead centre occurs with the axle rotated 0 degrees or 180 degrees. Lead is the amount that the valve opens before dead centre and lag is the amount the valve opens after dead centre. In Diagram 1 there is neither lag nor lead.

Diagram 1 is an easy example because every dimension is a multiple of 1/16" and results in angles of multiples of 30°. I will assume that there are no non linearities, which is nearly true, except for short connecting/eccentric rods. The motion imparted by an eccentric is simple harmonic so the valve movement can be represented by a sine wave. An eccentric with a throw of 1/8" will give a total travel of 1/4". If the valve is set just to open at dead centre it can be seen from Diagram 2 that the valve is fully open at 60 degrees of rotation and closes at 120 degrees. The opposite end starts to open at 180 degrees and closes at 30 degrees. Simple geometry shows that, in order to achieve this, the eccentric needs to be 120 degrees from the centre line of the crank axle. Whether above or below the centre line determines the direction of motion. Diagram 3 shows this (for forward gear).

Diagram 4 shows the valve positions during a stroke starting at Fig 1 with the piston at left dead centre. In this position the valve is moving to the right and just starting to open admitting steam to the left of the cylinder. When the crankshaft has rotated 60 degrees the piston has moved through 38% of its travel and the steam port is now fully open (Fig 2).

The valve has now started to move in the opposite direction until it closes the port at 76% of the piston travel or 120 degrees rotation (Fig 3). The crankshaft continues to turn until at 180 degrees the piston is at right dead centre. During this time the valve has moved further to the left until it is just ready to open the right hand port (Fig 4). Fig 5 has been included because it shows that the right hand steam port, which up to now has been exhausting, closes to exhaust. This occurs 9% before the right hand dead centre. For the remainder of the stroke compression occurs in the right hand end of the cylinder.

In NL&J 100 John van Riemsdijk introduced a simple diagram (Diagram 5) that illustrates this in a different manner. I personally prefer to use the sine wave but his alternative has some interesting aspects, and can be used later with other models. Line A-B represents the valve opening. Line C-D shows the exhaust valve closing at 8% travel. Note that the angle between the horizontal and line A-B represents the amount the eccentric must be advanced from the vertical (in this case 30 degrees).

This diagram has been drawn for the dimensions above. It shows again that the valve, with this very simple layout, closes at 120 degrees (75% cut off). So far we have assumed a fixed eccentric i.e. one way of rotation only. Looking at Diagram 3 the eccentric is 120 degrees in advance of the crank. If the eccentric is turned anticlockwise through 120 degrees the crank will work in reverse. By using a stop collar fixed to the axle, with a pin on the eccentric and letting the eccentric rotate about the axle through 120 degrees, the loco will reverse. In practical terms, the angle on the stop collar has to be increased to allow for the pin dimensions. A 1/16" pin at 5/16" radius gives an additional 6 degrees each side, a total stop collar dimension of 132 degrees. This is shown in Diagram 6.

Summarising so far, a stop collar has been derived that, with valve dimensions as Diagram 1, moves the valve from just opening at dead centre and closing at 75% cut off in both forward and reverse. Now to look at Project. The valve and ports are very similar to Figure 1 except for the overall length of the valve which is 0.522". Thus there is an additional lap of 0.011". Diagram 7 shows the resulting JVR diagram.

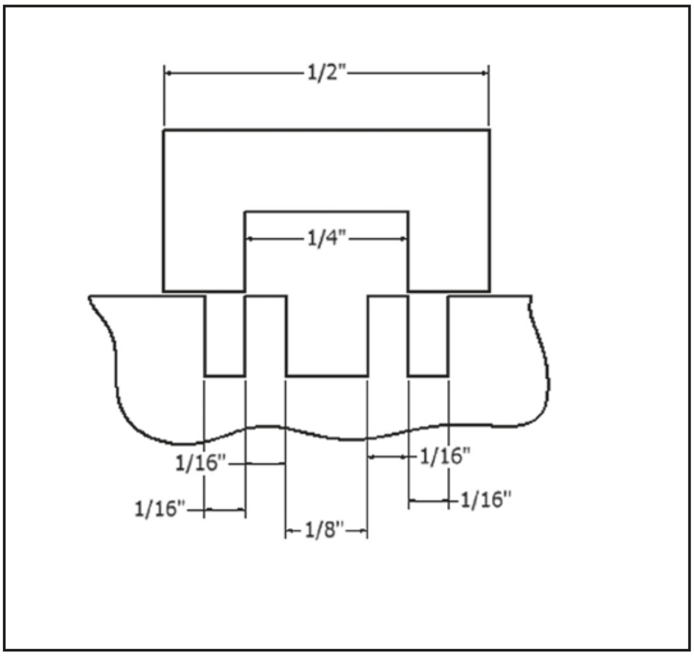


Diagram 1.

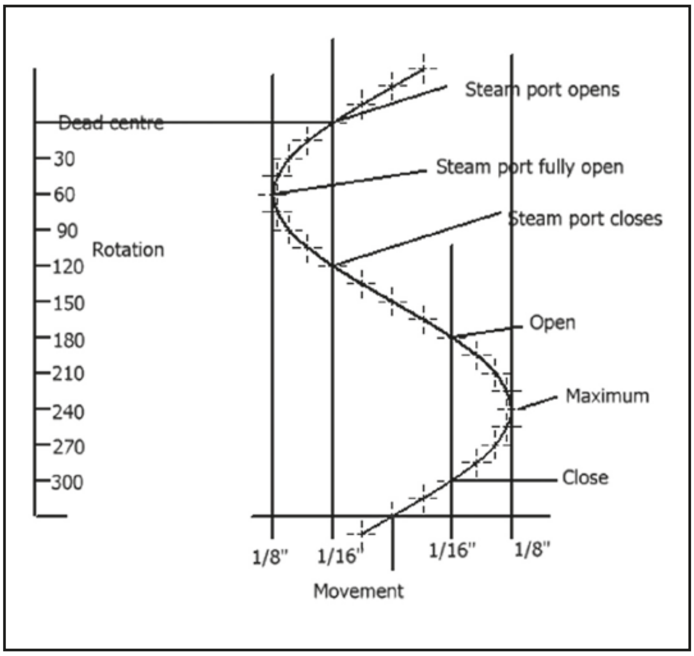


Diagram 2.

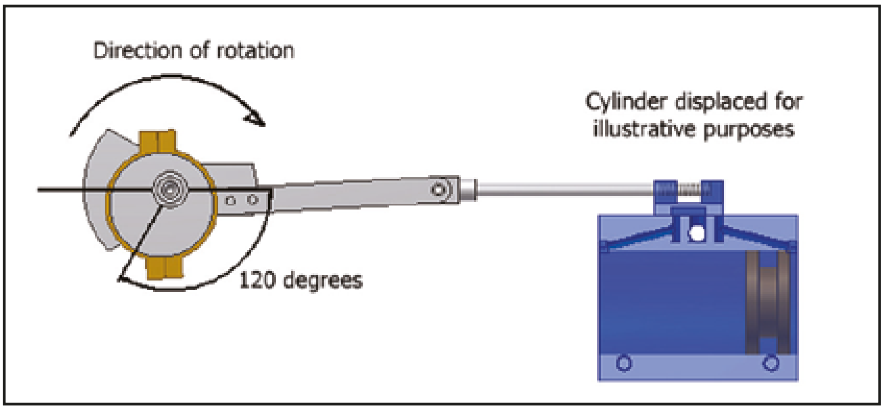


Diagram 3.

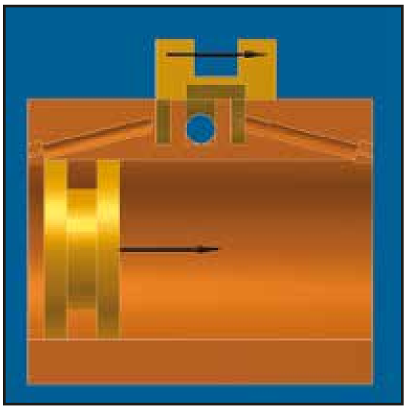


Diagram 4. Figure 1.

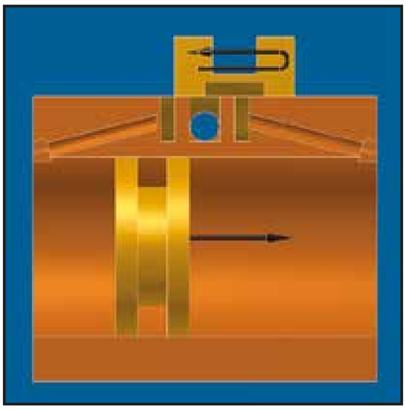


Diagram 4. Figure 2.

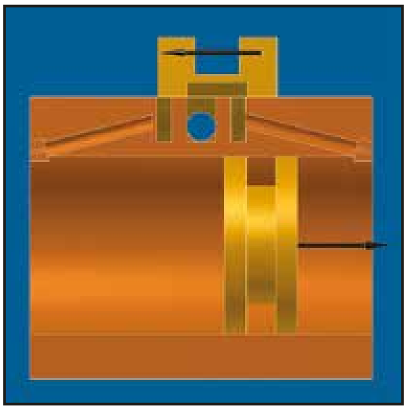


Diagram 4. Figure 3.

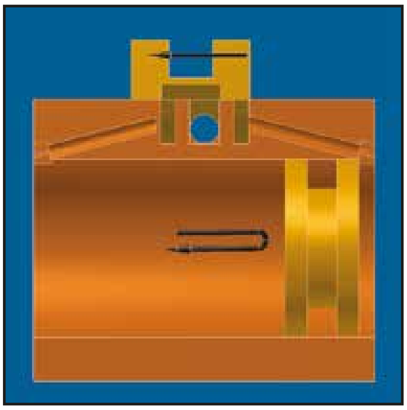


Diagram 4. Figure 4.

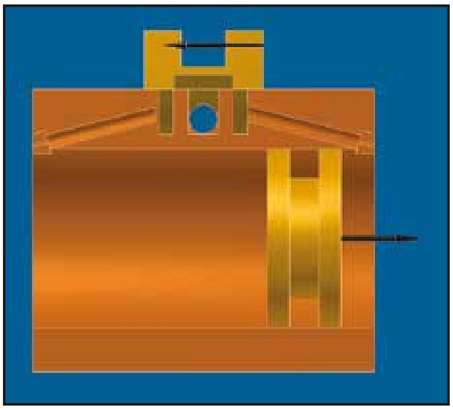


Diagram 4. Figure 5.

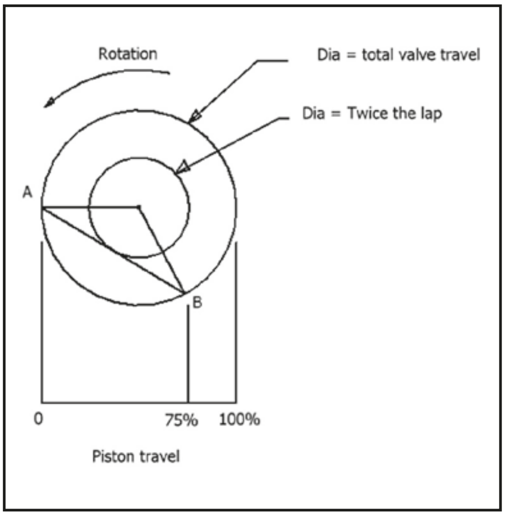


Diagram 5. Figure 1.

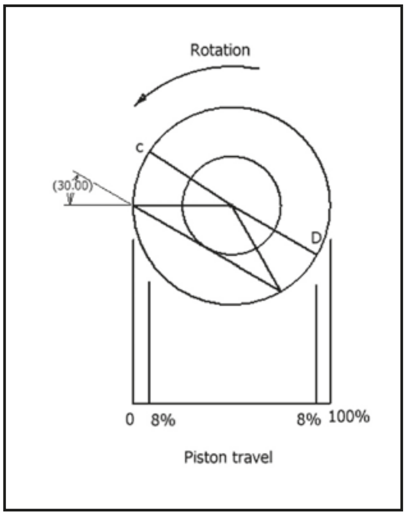


Diagram 5. Figure 2.

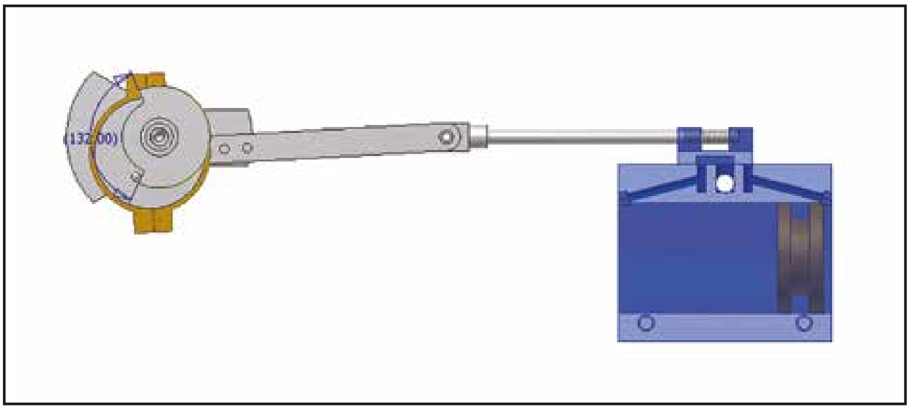


Diagram 6.

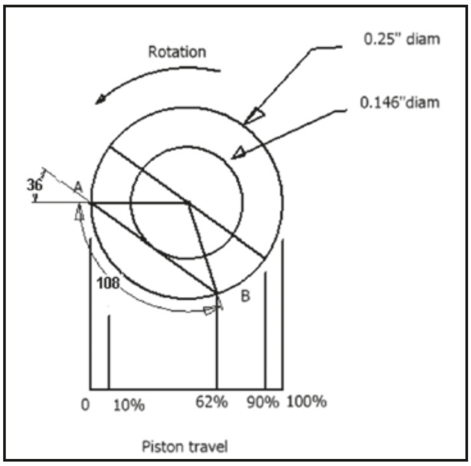


Diagram 7.

Using the same principle as applies to the simple example, to get equal performance in both forward and reverse gear, the stop collar angle should be 108+6+6 = 120 degrees. The drawings for Project give an angle of 135 degrees. This is a difference of 15 degrees. If the valve is set to just open at dead centre in forward gear then in reverse the valve will not open until 15 degrees after dead centre (15 degrees lag).

The alternative is to set the valve so that it opens at 7.5 degrees after dead centre in both forward and reverse gear and the performance will be the same in both events.

So this simple analysis has shown that there are three alternative settings for Project:

1) Set the valve in forward gear so that the valve is just opening and accept the lag in reverse gear.

2) Replace the stop collar with one with an angle about 120 degrees and set the valve to open at dead centre. This will give equal performance in either direction.

3) Retain the existing stop collar, but set it centrally about the crank axle. This will give a 7.5 degree lag but equal performance in both directions. (note that because the eccentric rod, in its mean position, is slightly (about 1.5 degrees) below the motion line the stop collar needs rotating 1.5 degrees)

Detailed 3D drawings of Project, which include all the non linearities, confirm that the analysis gives close agreement with the above.

My belief (until proved otherwise) is that it is desirable to have some lag on a single cylinder engine and that Option 3 is the one to go for.

TROUBLES WITH GAS

by Alan Norridge (3556)

Some of you may remember from a previous article that I have a number of Accucraft gas-fired locomotives, some American main line and others British narrow gauge. The smaller main line locomotives and the narrow gauge ones all have single poker burners and these work satisfactorily once lit, the others have twin poker burners and these give problems. Now each of these locomotives has one burner in its combustion chamber on one side and the other burner shares its combustion chamber with the steam/superheater pipe and this is where the problem occurs. To make it easier to refer to the two burners I shall call the burner with the steam/superheater pipe in the combustion chamber the LHB and the other the RHB as that is where they sit in the boiler.

The problem is that the LHB will not stay alight. Once lit, it will run for several minutes, become unstable and go out. It has been so bad on my Accucraft GS4 Daylight that I have not run it for several years but, having solved a problem with its axle pump, I thought it was about time to fix the burner problem. I should point out that I solved the axle pump problem with a tremendous amount of help and encouragement from Ryan Bednarik in the USA - thanks Ryan for all your help. Back to the burners, I have spoken to a number of people more expert in gas firing than myself about this problem and had a wide range of answers but all revolving around secondary air, so this was the problem to investigate. My understanding of the air required by the burners is as follows: primary air is the air supplied to the burner via the four holes in the back of the burner and secondary air is the air around the outside of the burner poker. If this is not correct I am sure someone will put me right.

I first checked both burner pokers pokers, and they are identical, and ran them on the bench in free air and they both performed the same. I then ran the burner giving the trouble, the LHB, in the RHB position and it ran perfectly. I did not repeat the reverse experiment as it is a real pig fitting the LHB in its combustion chamber. Next I decided to see if there was too much secondary air in the LHB. Now looking at the back of the boiler there are two 2mm diameter holes one in each burner flange to provide secondary air, so I drilled and tapped the LHS hole M2.5 and closed it off with a screw, so no secondary air. I then lit the LHB, it went out almost immediately exhibiting the same instability as before but more so, so I needed more secondary air. I had another look at the back of the boiler to see if I could see any differences between the two burners and every thing is identical, so what to do? Looking yet again at the back of the boiler, I noticed both burners have a 3×5mm slot at the bottom of the burner flanges which go through to the combustion chambers. This slot is for the steam/superheater pipe but there is only one pipe and this is with the LHB so the slot on the RHB is supplying more secondary air so I concluded I needed a similar slot on the LHB. I drilled and tapped an extra 3mm hole in the LHB flange. It was tapped in case I had to block it. Before filing a slot from this hole I decided to test the LHB and it lit and ran perfectly. I set the locomotive up on the rollers and ran for at least 2 hrs with both burners working perfectly, the first time this has ever happened.

So if you have twin burner Accucraft locomotives and have a poor performance from the burner with the steam/superheater pipe in the combustion chamber have a look at its secondary air supply.

The two poker burners can be seen at the bottom of the boiler in this view of the GS4 Daylight disassembled.

FROM OLD SCHOOL TO RC

by Wouter van Tiel (4114)

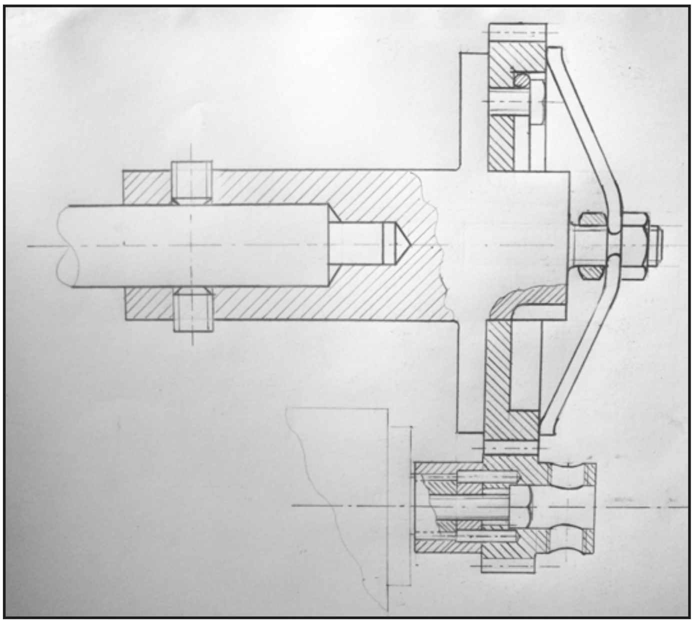
Running a live steam locomotive always needs experience and the driver's attention. However, not all beginners have been lucky enough to see prototypical steam running. That means for me more than 60 years ago, waiting and waiting in cold and warm weather until the shunting engine came into movement, either pushing or pulling freight wagons.

A few years ago I started myself as driver of my first Aster 140C. I built her in a couple of months and my first experience was on the rolling road. The steam and the noise from the chimney was wonderful. It is still one of my most reliable engines.

I learnt to drive this loco on the Krogen track and she also ran very successfully on Chris Ludlow's marvellous old track in Maffliers. Now I continue my learning at the Krogen track in Aalborg. In the last few years I have built more than 10 live steam Aster locomotives which give me great pleasure to run, especially on the track in Aalborg, where David and Lotte can see how I spend my time in the garden.

My first locomotive, the French 140C, as well as my AD60 Garratt are easy to drive. A locomotive such as the Aster 232U1 needs more experience, because (as David described in NL&J 237) she needs much time to achieve stable constant velocity. She is a heavy engine with great power. With her big driving wheels, after starting you need to be quick to reduce the regulator setting as the engine starts compounding. If you are not quick enough, you will wish that your layout had large radius curves, or the engine will derail and stop alongside the rails.

The best train for the U1 is heavy coaches, so you have to open the throttle so that the lower pressure cylinders warm up quickly enough so that no condensation occurs in the cylinders. It can take a long time to set the engine running just right. Five to six revolutions per second of the driving wheels give the most beautiful performance. To achieve this quickly without cycling around the track, radio control is the best solution.



The large gear friction device and how it is attached to the regulator spindle. Tightening the nuts increases the friction.

But at the Krogen track, David and Axel built an extension to the original level track. They built a slope downwards at 1% and a corresponding climb of over 2%. Even with the throttle fully shut the train, running 2% downhill and assisted by gravity, held her velocity. After passing the lowest point of the track, the train stops unless the regulator is opened again. It is clear that running on a track going up and down you need full RC on your engine. So the U1 became my first RC engine.

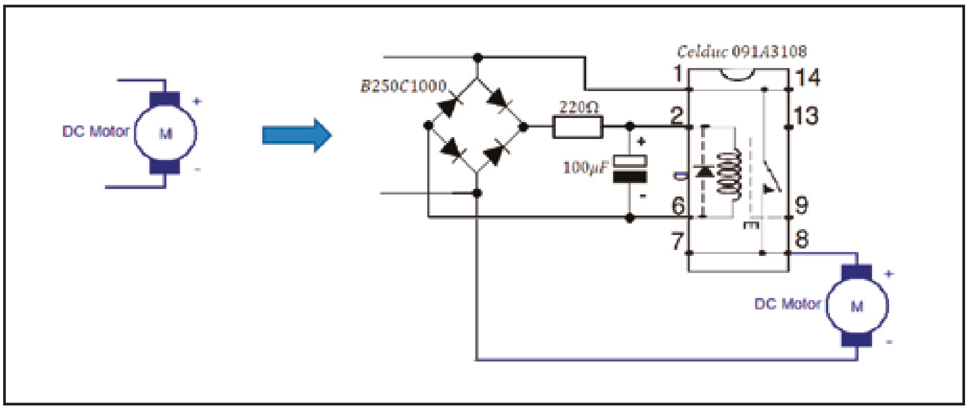
The servo and electronics mounted to operate the regulator.

I used a HT-225 MG servo and a simple Jamara 2.4Ghz four channel transmitter. In this combination the servomotor shaft makes a movement of about 80 degrees. Under most circumstances that is sufficient to get enough steam into the cylinders and cause slipping during the 2% climb. The pulling power of the loco is 8-9N. With the weight of the coaches and the rolling resistance you can determine the weight of the total train with which the loco can climb a hill. A very important argument in favour of RC is if a heavy train climbing a gradient had a coupling break, the loco could reach astonishing velocity very quickly.

The servo and the regulator handle are directly connected with a paperclip. So, if the RC fails you can disconnect the servo very easily and continue with manual control. Therefore the original regulator handle is retained in the cab. The joystick on the transmitter corresponds with the position of the regulator handle. So you can see on the transmitter the position of the regulator. The servo follows the transmitter stick 1:1 from shut to open.

Now I come to my blue Duchess. A beautiful four cylinder simple expansion locomotive developed in the 1930s, but painted as running in the 1950s.

After I built the loco, the first runs took place on my rolling road. It seemed immediately that I have to open the regulator spindle over one revolution to get the engine running at a suitable speed. The reason for that is the thread machined on the valve spindle. Most Aster locomotives have a normal metric M5 thread. In the Duchess, they chose an M5 fine thread, giving finer control of the locomotive. However, that meant that a direct coupling by a simple paperclip was no longer possible. I had to find another solution. On my rolling road, the boiler delivered enough steam if the blower was open so that you could hear the burner a little. On the Krogen layout, hauling a train, the steam production was not good enough. The boiler pressure fell below 1.5 bar. However, the engine kept running but not enough for hauling the train at speed. It seemed that the airflow through the firebox was very poor. After discussion with David, I removed a part of the firebox insulation and made the wicks smaller in the burner tube, which increased the rate of steam production. Then she could haul a six coach train at speed without the help of the blower.



The Integrated Circuit for the servo control.

Of course, I wanted to install remote control on this engine. But how do you realise more than a quarter of one revolution of the regulator spindle with a servo only acting over 80 degrees? I needed a 4:1 ratio gear system. The biggest wheel on the servo and the smallest one on the spindle. The internal forces would be greater, but with a simple servo I could turn the regulator spindle. By the transmitter joy-stick position I know also the position of the regulator.

However, in practice it did not function very well. I needed a greater regulator opening when climbing a 2% gradient. That meant over one revolution of the regulator spindle. I decided to design another device so that the servo could turn the regulator spindle several revolutions.

I developed the following proposal. Modify the servo so that you can run the motor in both directions. Build a gear system and couple the servo to the regulator. However, with this system you cannot see from the transmitter joystick what is the regulator position. The train is travelling round the track, so you cannot look into the cab. When you want to stop the train somewhere on the track, you cannot see the regulator position and risk damaging the servo if you continue to command the servo to move when the regulator is already shut. If the servomotor cannot turn further, overheating damage occurs. Your train stops but might not start again.

I found the solution in a friction coupling between servo and the regulator spindle.

For me, most easy to design is the friction coupling. I used the original gear wheels 1:4 for low internal forces on the teeth and torque on the motor axles. With the big spring you can

Wouter explaining his modifications to David Clement's wife Lotte at the Hotel Krogen.

adjust the friction force by turning the nuts. The small spring inside has two functions: first the gearwheel could turn even if the regulator spindle is already in the closed position. When you command the servo to open the valve the small spring blocked into the part connects with the regulator spindle and turns it to the open position. The problem with friction is that the friction force in either direction is more or less the same. When the regulator was closed and the opening torque needed a little more friction force, the motor turned but the valve did not open. Therefore, I built in a small spring to release the regulator from the closed position. The big advantage is that now you are able to adjust the lowest force for closing the regulator and the minimum loading of the servomotor to open and close the valve.

The electronic solution I discussed with my servo retailer (Quartel Pynacker Holland). My colleague found an article named 'Hacking the servo', in which it was explained how to transform the servomotor to turn in both directions without restrictions. John followed the description and we tested it with the transmitter. The problem now was that the transformed servomotor no longer had a stop position. The so-called 'dead band' cannot be found - although the motor turns very slowly, you cannot drive your locomotive when the loco is constantly accelerating or slowing down if you can't set a steady valve position.

After analysing the signal sent from the transmitter and the receiver to the motor my colleague John designed an Integrated Circuit to realise a 10 degree 'dead band' on the transmitter. The joystick on the transmitter has a mid-position set by internal springs. This is also the servo stop position. So you can push the joystick in the direction you wish and the motor will turn until you release the stick and the motor stops immediately. The valve stays in the position you want. The big advantage is you don't have to look at the transmitter, just keep your eyes on the train.

This is the most realistic situation you can imagine to drive your train. Looking at the train, downhill, uphill, crossings, points and running into curves. You give as much steam as you want, providing the boiler can produce it, of course. Open or close the regulator, the servo will follow you without problems. Even when the regulator is fully closed, you won't kill the servo!

The RC setup performs very well but we are working to make it more simple in future.

http://www.youtube.com/watch?v=lHRRLd-9TWw&feature=youtu.be
The 140C climbs the 2% incline of the Krogen layout manually driven.

http://www.youtube.com/watch?v=B70LUzE3BZ8
The blue Duchess running without Blower assist, climbs the 2.5% incline controlled by the new RC device.

I WAS FED UP WITH IT - AN AUTOMATIC 'VENTILATING-VALVE'

by Ulli Holtmann (843

In the NL&J 182, I wrote an article entitled 'I was fed up with it'. The subject was the first ceramic burner for a Gauge One steam locomotive; the ceramic burner for the Aster SNCF 232U1. Fourteen years later, I had to realise that I was fed up with my even older BR 01. All the engines without an axle pump and so without a bypass valve tend to suck the water from the water tank when they are cooling down. The friends I talked to about it told me to open the blower, leave the regulator open or take the safety valve out. Even those with engines with an axle pump forget sometimes to open the bypass valve - perhaps under the pressure of an exhibition. So there was a request for an automatic valve. The design is so simple and it is so easy to make, that this will be more a picture story than a technical description.

The valve is meant to give the boiler an access to air, instead of restricting it to the water in the tank. I decided to replace the plug on the top of the hand pump with the valve shown (Fig 1). The valve shown in the pictures works very well for about 10 litres of consumed water. The problem is to find a stainless steel spring that is soft enough to just keep the ball up and not to hinder the air valve from being sucked open. So I thought of a toggle valve, where just one ball jumps up and down.

Then I remembered the hand pump I made from car tyre valves (NL&J 190), achieving more than 32 bar at 290°C. I went to my tyre dealer, got some old valve bodies for tubeless wheels and some old valves. Then my lathe came into action, with a new sharp tool I removed the rubber and gave the body the necessary shape. The problem was that I had to get rid of the spring inside the valve. There are two possibilities, either to file off that little bob on top of the valve rod and use brute force and ignorance to

The first vent valve installed on the tender hand pump

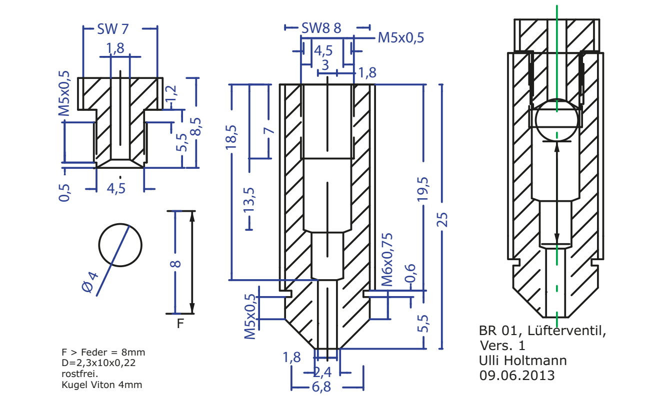


Figure 1.

tear the valve apart or to twist the shaft against its body so that the spot-weld breaks. Then I filed off the welding marks on the shaft so that this can move freely. I squeezed the outer end of the shaft flat so that it can't slip through the valve body and achieved a lift of the valve of 0.3-0.5mm. This makes sure that the valve will shut properly.

From my car dealer I also got some very posh valve bodies used with the 'show-off wheels' of the SUVs. The material is very good to machine. On some of them are lock nuts which you can use to adjust the ventilating-valve if necessary.

Construction

Machine the SUV valve body down to 7.5mm, the diameter of the thread for the cap and the body of the rubber valve down to 6.5 mm diameter. Then cut it off to a length between 25 and 30mm. Screw in a valve number 3573-1 (the common one) and measure the internal depth to the tip of the valve-rod. To that dimension add 1.5mm and cut off the body. The valve rod should protrude 1.5mm to form the stopper for the valve ball of the pump. Then cut the thread M6x0.75mm 4mm long (Fig 2). With these dimensions, the ventilating-valve will be able to replace the plug of the outlet side of the hand pump. If you could get hold of a SUV valve-body with a lock nut, you may cut the thread longer and use the nut to adjust the stroke of the pump ball.

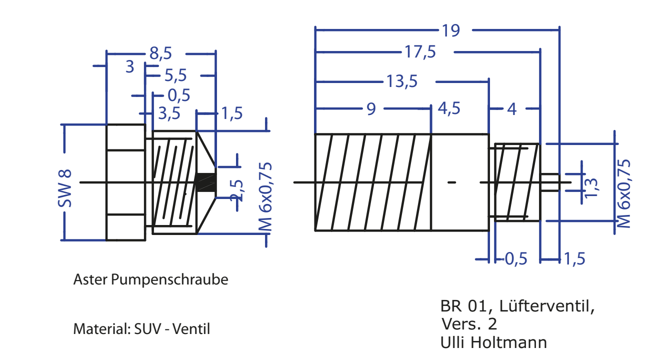
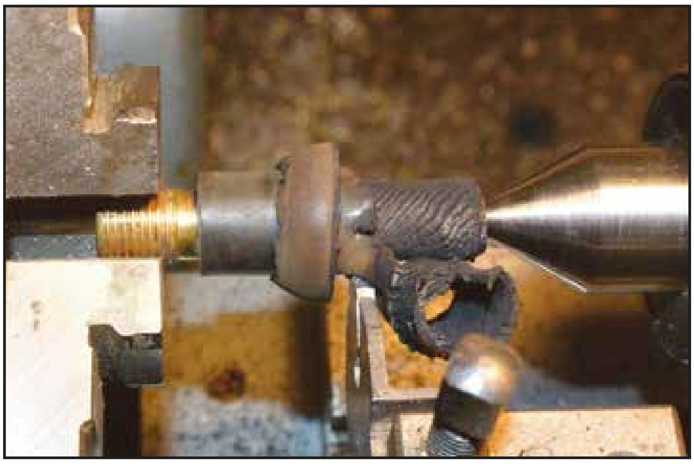
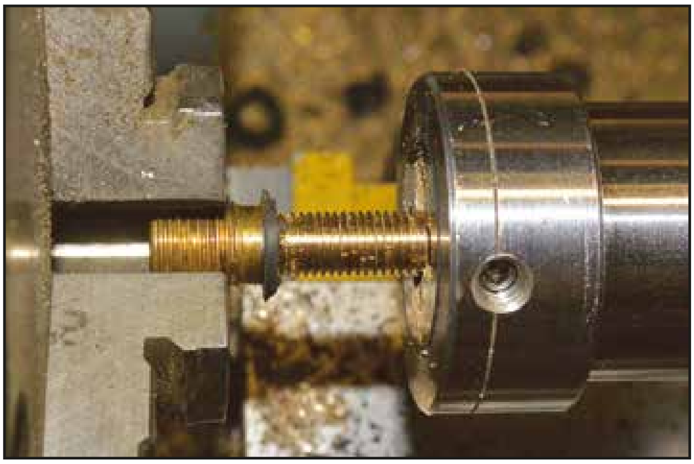
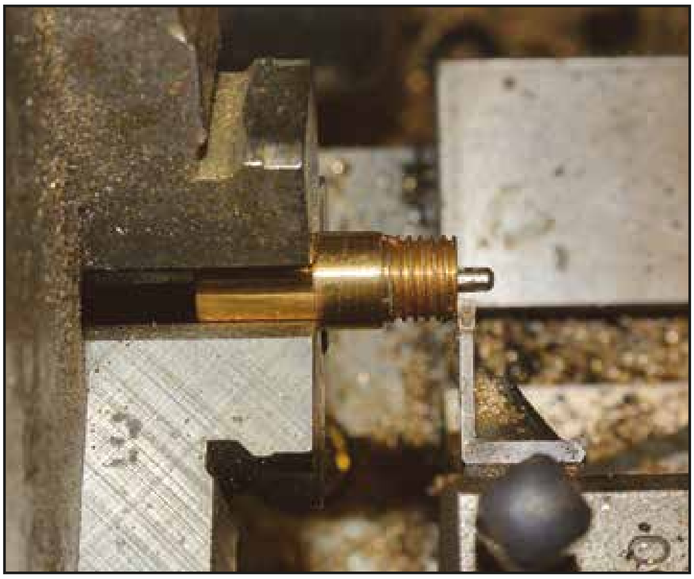


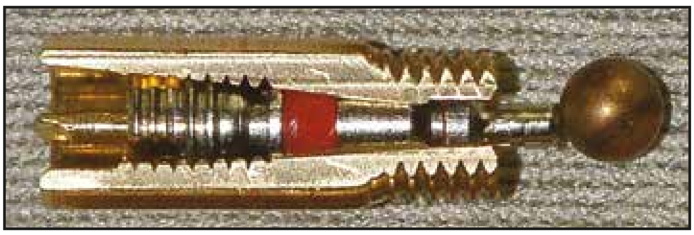
Figure 2.

The modified valve is on the left, with a valve and dust cap fitted to SUVs in the middle.

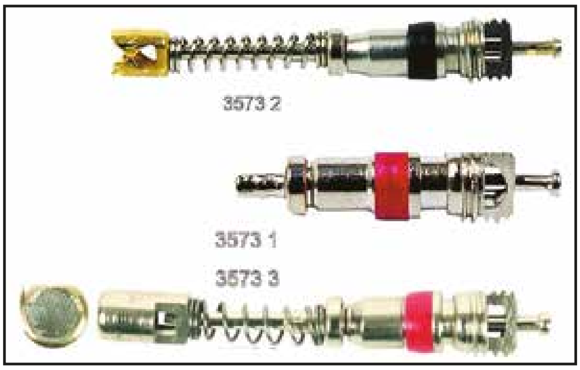

Removing the rubber from a car valve.


Cutting the new thread on the car tyre valve.


The SUV valve being machined to leave 1.5mm of the valve stem protruding.

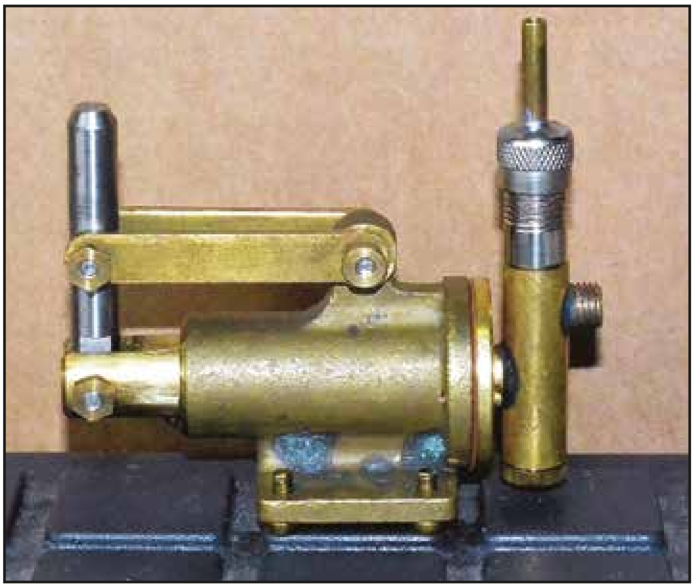

A sectioned valve showing the valve stem protruding 1.5mm against the ball.

The various vent valves developed from tyre valves.



The various valve innards referred to in the text.

As the valve number 3573-1 is difficult to use, my friend David found another one on the Internet, number 3573-3. Here, you cut off the rod as long as possible (make sure the spring doesn't disappear under the bench, it's quite usable for many other purposes.) Then cut down and grind the rod to a length of 16mm between the top of the body and the tip of the rod, see 3573-1. Number 3573-2 (marked black) cannot be used because the dimensions are different. You squeeze the outer end of the rod flat to restrict the lift or leave it as it is. In that case, it is possible that you lose a tiny drop of water while the valve is shutting. The round thing holding the spring is a little air filter which one can use on the tube extension to make sure that the valve reaches the surface of the water in the full tank. The final photo shows the valve with an extension so the air intake is at the top of the tender water space. The standard valve caps, whether metal or plastic, get a hole of 1 to 2mm diameter on top fixed with 'Nut Lock'. They give the chance to get the air intake up to 2mm higher. If this is not enough a thin tube of respective length soldered or glued in can reach the surface of the water.



The Aster BR01 tender pump fitted with the SUV tyre valve vent valve.

Peter McLaughlin preparing his coal-fired 'Duchess of Hamilton' for its first run having just taken delivery. The loco was made by Pemberton Models. Photo Garth Bridgwood.

THE CURVILINEAR SIGNAL BOX

by Dick Comber (2500)

You may have seen the Curvilinear signal box advertised in the NL&J. A kit was bought and assembled and you can see from the photographs it makes up into a very nice little building. It comes with comprehensive instructions (16 pages), most of which were followed. The kit is largely brass and my natural instinct would be to soft solder brass objects together, but the advice is to glue many of the parts, and much epoxy was used. One of the reasons for recommending glue must be that the parts are etched so precisely that the smallest amount of heat, even localised heat from a resistance soldering unit, is enough to expand parts so much that they no longer fit together.

When it came to the staircase the instructions were ignored. It is intended that the vertical posts should be laminated from the supplied brass strips but square brass tube from Eileen's Emporium was used instead. The tops of the tubes had rivets glued into them to round them off.

Getting the levers lined up is not so easy. They were set in a bed of 24 hour epoxy on a strip of brass and lined up using a temporary balsa wood frame. This was done outside the box, and the unit was later glued into the box.

The finished box was mounted on a sheet of brass which was not part of the kit and the very neat sign was made by Bill Whiting.

The etched kit uses tab and slot construction.

The interior with added details.

The completed signalbox.

Dave Brutnell has now completed the Australian NSWR Class 32 he displayed at the Expo. Photo Peter Trinder.

SIGNAL REFURBISHMENT

by John Perkin (4054)

My project at Brighton Technical College in 1981/2 was microprocessor controlled four light colour signals, long before British Rail got in the act. I know little about signals and signalling, but have been always been inspired by the LNWR array at Rugby required for the opening the Great Central Railway which my great uncle, the GCR Mining Engineer, must have passed many times.

Seen at Garth Bridgwood's GTG recently were four Home and two Distant Gauge 1 signals in an old plastic box. They came from the late Ronnie Cross but were apparently never installed on his layout. Never one to resist a challenge, and as Garth only wanted ten pounds, I wrote out a cheque before someone else realised their potential. It has cost far more to buy the brass strip and wire and to replace the missing signal lamps with loco lamps from GRS.

Although they are probably the LMS version, they could pass for early LNER before lattice posts were introduced, but would not be suitable for GWR or SR layouts.

The first task was to completely dismantle those parts still not in bits, clean off all the rust, and remove and dispose of any old signalling wire.

Six new bases were made from brass strip 100 × 25 × 1.5mm with holes drilled for the signal post and for two screw or bolt fixings to the base board. The signal post hole was pre-tinned with solder. The lower section of each signal post was cleaned with the bench grinder and pre-tinned with solder. Signal posts were then soldered into the brass strip base using my metal set square to ensure that they were vertical in both directions parallel to and perpendicular to the track.

All six ladders were cleaned of rust and pre-tinned with solder for soldering to the upper, and centre ladder supports and to the brass strip base. New ladder supports were made from nickel silver strip as required to replace missing or damaged units. The brass base strip, ladders, and ladder supports were then pre-tinned with solder and the ladders held in place with small crocodile clips whilst all solder joints were completed. This results in a very rigid structure of brass base strip, ladders, and the upper and centre pole ladder supports.

The brass base strip was then sprayed with grey primer and permitted to dry overnight. The next day the base and lower section of the poles were then sprayed with satin black. When dry the upper sections of the pole were painted by hand in satin black and matt white.

The signal arms only required the red, yellow, black and white areas either hand painting and touching up.

All six lamps were missing, but GRS have white metal 'G Scale' loco lamps which are the correct size. Only four were in stock and I have ordered the other two for collection later. I intend to experiment with white LED illumination and as the signal poles are hollow brass it should not be too difficult to feed two small thin wires up the pole to the lamps. A pair of AA batteries mounted under the base board with a small ON/OFF switch should provide sufficient power for the white LED.

As explained above, any remaining signal wires were removed and disposed off, and new signal cables of small diameter brass wire were cut to length, fitted in place and the ends bent over to retain in position but permit movement. Operation by other remote cable or local solenoid should be straight forward.

As I do not have a layout, the signals are available for purchase by G1MRA members with offers over ten pounds each plus delivery at cost, otherwise they will be sold on Ebay later.

The back of the distant signal showing a G Scale loco lamp.

MAKING A CYLINDER FOR ROGER THORNBER'S GWR 2251 LOCOMOTIVE

by Malcolm High (3098)

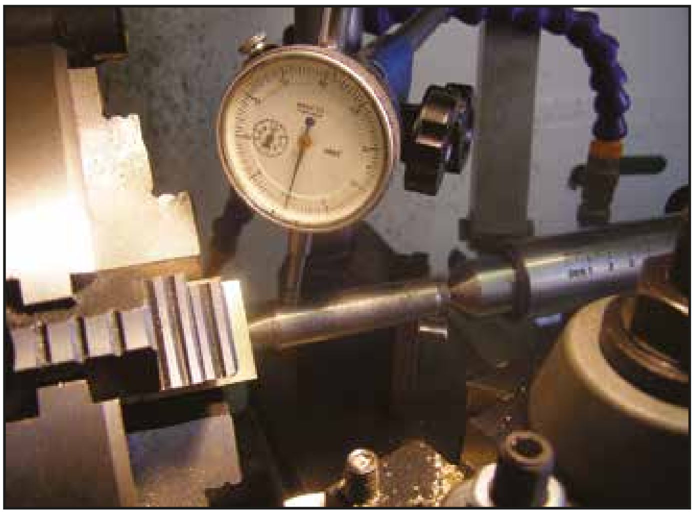
Currently, in Engineering in Miniature magazine, Roger Thornber has a series on building a GWR 2251 Collett goods engine. This is a simple single cylinder locomotive that he built a number of years ago. It could be looked upon as a Project in GWR clothing. What attracted me to it was its simplicity and how easy it would be to produce a number of laser cut parts that would reduce the build time considerably. One of these parts is the cylinder block and what follows is a description of how I built it.

The basic principle was to take a block of brass and make a jig into which this could be placed, lining up all the holes. Assembly of the jig was straightforward. Instead of silver soldering it together, the ends of the tabs were just peened over. The fear was that a cusp of solder might stop the block from fully entering the jig.

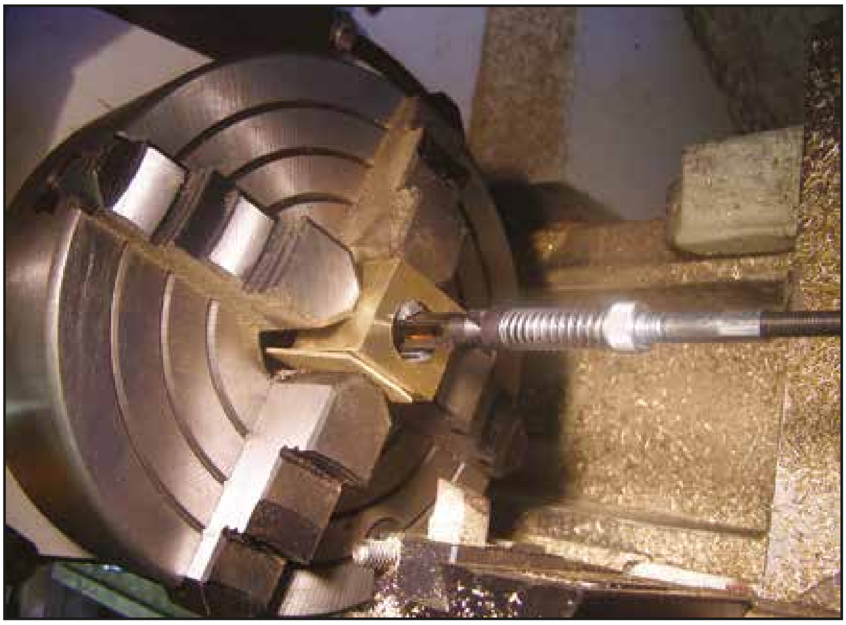
The cylinder was made from a piece of 1 1/2" square brass bar 1 1/2" long. The first operation was to face off the cut ends in the 4-jaw chuck, not to length but just to get the ends square. Then the other opposing sides were faced off. After measuring, these were placed back in the lathe and faced off until the desired width was achieved.

The length of the block was next. The final surface finish on the top of the block has to be good so the valve makes a good contact with it. Having decided which surface this was, it was protected by a piece of thin ply when the block was returned to the lathe. Once the cylinder was to length it fitted easily into the jig.

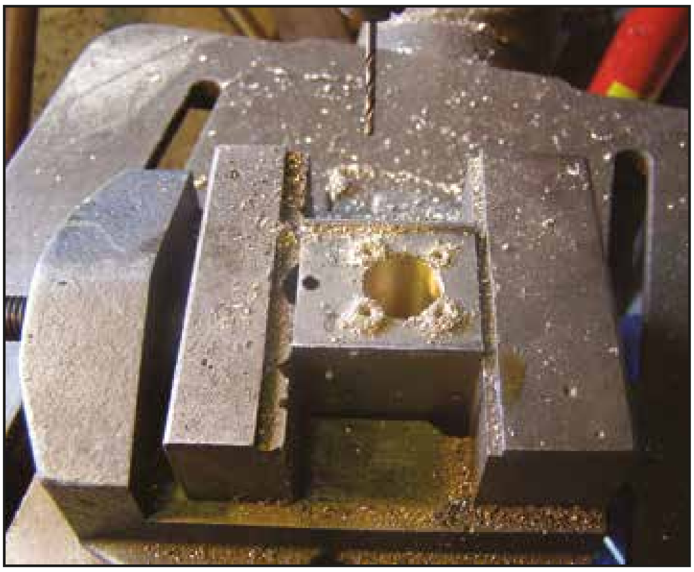
One end of the jig has a small hole in it; this is the centre of the cylinder bore. With the brass block completely in the jig, I drilled through this hole and into the block, approximately 1/8" deep. The block was removed and placed in the 4-jaw with the recently drilled hole facing the tail stock. It is very important that the cylinder is bored in the correct position. The hole was first lined up by eye using a centre. Then another centre was



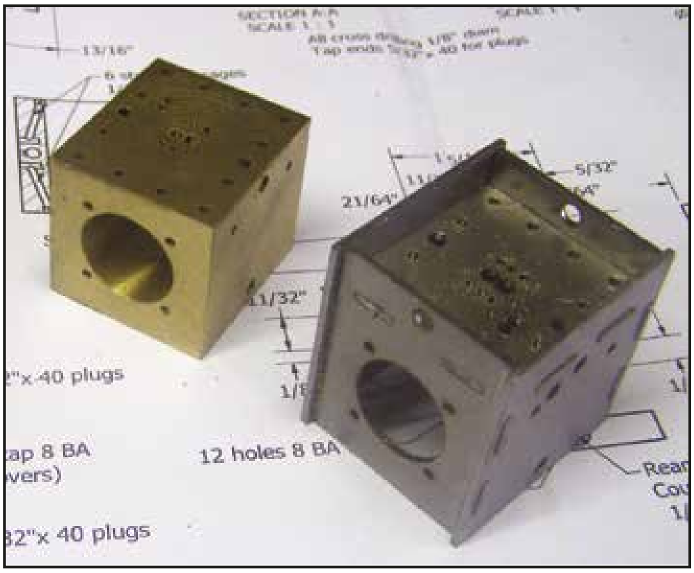
Lining up the cylinder bore prior to honing the bore.



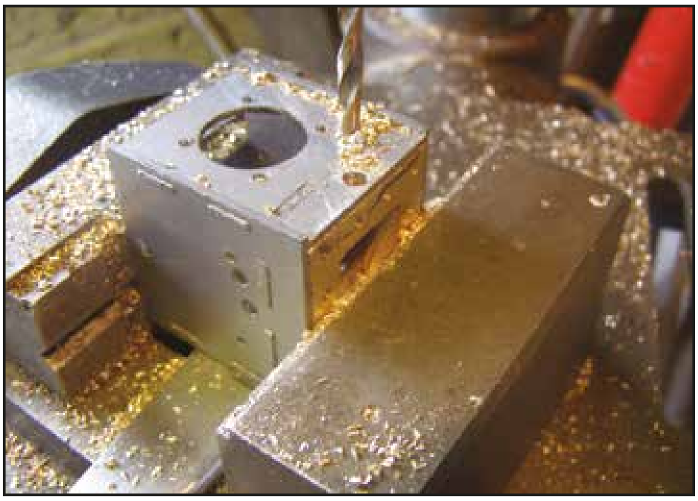
Honing the cylinder bore.



Drilling the end cap holes.



The almost completed cylinder and jig.


Drilling the valve rod holes in the steam chest.

Completed steam chest.

placed between this centre and the hole as shown in the image. Using a dial gauge the block was centred.

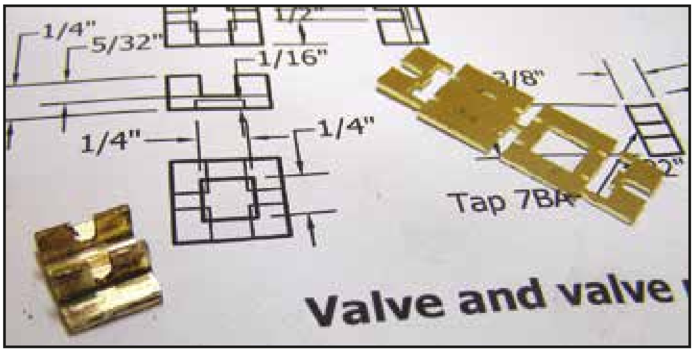
The cylinder bore was started with a centre drill and then a drill through the block with a drill slightly smaller than the finished size of 3/4". Using a boring bar, another few thou was taken until the hole was 15 thou undersize. Finally the boring operation was completed with a reamer. The surface finish was improved by using a honing tool.

The brass block was returned to the jig, making sure the holes lined up. All the 72 thou holes that are to be tapped 8BA were then drilled. It is very easy to drill into the cylinder and thus ruin all your work. The depth stop on the drilling machine was set each time. It only takes a few seconds to set up and can save you a lot of work.

The 1/8" cross drillings and the six 1/16" ports were then drilled. The two small holes that admit steam to the cylinder were drilled next. There are two holes not on the jig. These are the exhaust and the steam inlet which are on the open face. These were carefully marked out on the relevant face and drilled.

With the cylinder still in the jig, the steam passages from the ports to the cylinder were drilled. A flat face was put on where the holes were required and then two 1/16" diameter steam ways were drilled in each end.

The steam chest is another piece of brass. A similar principle was used here - first facing it off to size and then using the jig to



The slide valve and the kit it came from.

drill the holes. The centre was removed by chain drilling and filing. The cylinder end caps are a simple turning job but the jig was used to drill the four retaining bolt holes.

That completed the cylinder, but it is not a lot of use having the ports right if the valve is not correct. Therefore I devised a kit for this as shown in the image. Basically it is four parts that slot together. These are then silver soldered using low melting point 0.7mm diameter wire. Finally the face is finished off with a piece of fine cloth on a flat surface.

The advantage of the jig is not only the time saving, but it also guarantees all the holes are in the correct position which goes a long way to ensuring the engine runs first time.


A GCR Robinson loco trio, Jersey Lily, O4 and D11. Photo David Pinniger.

CUTTING EDGE TECHNOLOGY

by Keith Bucklitch Technical Secretary (2424)

An accident with one of the commercially available electric steam raising fans or 'blowers' caused me to take a close look at the device. It appears to me that there is an inherent danger in the fan as supplied. Basically, there is no guarding of the fan blades, and the gap (measuring some 21mm wide) is large enough for a finger to be inserted accidentally coming into contact with the moving fan blades. Closer examination of the fan showed that the blades are curved, but rotate in a direction that would appear to draw air centrally rather then expel the air out of the fan. It was this direction of rotation that had made the fan doubly dangerous, in that the sharp edge of the blade was driven into the inserted finger causing greater damage, whereas if the fan had been rotating in the contra direction, it would have bounced off the finger with little injury.

Although the instructions with the fan mention the use of gloves (and goggles) when using the fan, how many of us actually take such precautions when steaming our locos? So, a better solution is to fit some protective shielding around the fan blades. It is interesting to note that the commercial fans available for 5" gauge and similar scales either have a shield surrounding the blade housing with a small side orifice for the hot gases to escape, or have blunt fan blades that are less likely to cause injury.

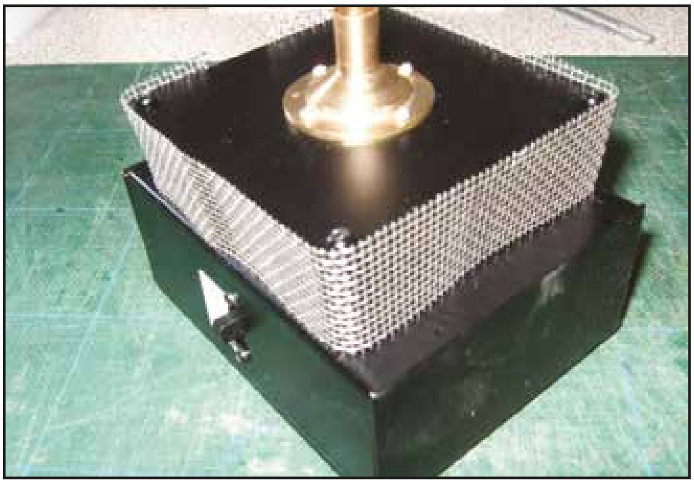
The easiest way to fit some protection around the fan is to wrap some wire mesh around it. I happened to have some 2mm stainless steel mesh in my material store, which is easy to cut with snips and bend to fit around the fan. See the attached picture showing the mesh in place.

As mentioned, the rotation was such as to make the sharp edge leading. I therefore took the opportunity to reverse the direction of rotation by resoldering the wires from the battery switch. Before I did so, I had 'measured' the suction effect by seeing at what height the fan would lift a piece of paper from the workbench. After reversing the rotation, this suction effect was again measured with little apparent difference.

Finally, the switch on the side of the battery box was arranged so that the off position was with the switch upwards. Now, it is normal practice in the UK that switches are operated downwards for on and up for off. I therefore removed the switch block and reversed it. That required a new label to indicate off/on.

There is a further benefit to this change, in that it is easy to switch the fan off and remove it from the loco chimney with one hand at the same time.

The result is that we have a suction fan unit that is safer to use and easier to operate. If a child was to insert a finger into the previously unprotected rotating blades, it is easy to envisage the finger being amputated. I wonder what the manufacturer's liability would be in such an instance.



Mesh was fitted around the fan to keep fingers safe.

The switch block was reversed and relabelled.

EMMA

by Martin Veasey (2970)

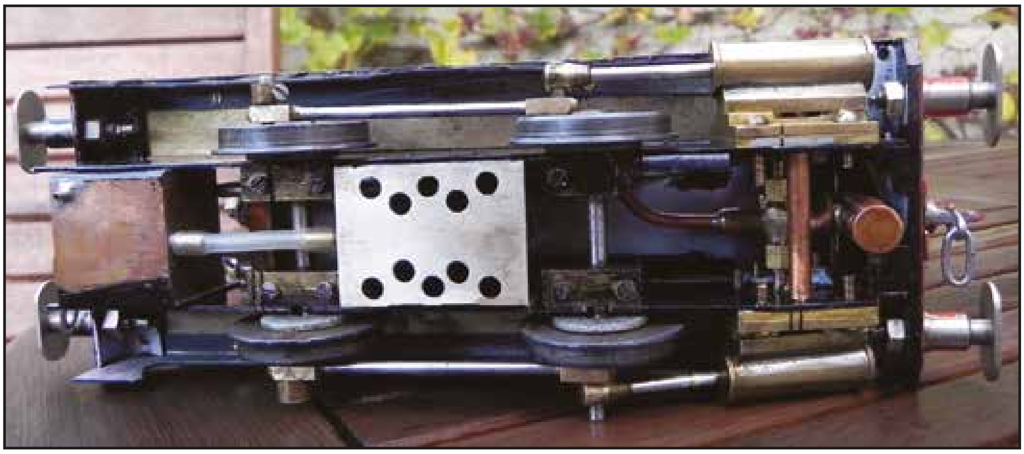
How many locos have you actually got then Dad? my daughter asked me as I was loading up the car to go to a GTG. I realised at that point that neither her or her brother had any idea of how many locos I owned apart from the long green one and the scruffy black one and that when I finally get my own box for that journey to the great GTG in the hereafter, the lot will probably be sold off or left to friends that I know will appreciate them. So I decided to make one that she might actually like to keep and, as I enjoy making my locos from scratch, it would be a unique item rather than a production model that just has a nameplate stuck to it. I had plans to put a small oval of track in the garden so settled on an 0-4-0 that would go round small radii easily with a meths fired pot boiler and simple oscillating cylinders.

I used some spare G1MRA coach wheels and a standard mild steel frame with a set of cylinders bartered from Alex at Camden Books who often has various useful bits and pieces to hand. The cab roof is set at a jaunty angle, just like my daughter's hair and was cut from brass sheet. The front windows are shaped like eyes and were cut again from brass sheet, the rivet holes drilled in the window frames whilst back to back, then soldered in place before the rivet holes were drilled right through and some 1/32" rivets put in and resoldered. There are two water feed pipes to the boiler, the

Emma poses for her portrait.

The simple cab. The try cocks for the boiler water level are to the left of the regulator.

top one being a try cock so when both taps are open, water is syringed into the lower pipe until it comes out of the top pipe and this indicates that the boiler is nearly full. The taps are then screwed shut ready for firing. There is a small displacement lubricator behind the front buffer beam and a meths tank under the cab floor. Meths is syringed into the pipe on the top of the tank until it comes out of the overflow beneath and the tank is filled to the correct level. I have hung the meths tank on two bolts from the running boards and connected it to the burners with silicone tube to reduce heat transfer after the first tank I made overheated and the flame blown around by a windy day ignited the jet of meths fumes coming from the breather hole! It looked like a mobile candle running round the track and sadly melted part of the front of a wagon coupled behind (sorry Colin...). There is no reverser fitted, just a regulator handle painted bright red and the most expensive part, a pressure gauge. I have set the safety valve to release at 30 psi and pumped the boiler up to well over 100 psi to test with no problems. Now that I have a small track in the garden and I am able to fine tune the loco, I have fitted a full width tank to give longer running time and suspended it loosely again on two captive bolts to minimise frame contact and heat transfer with great improvement.



A look at the lady's underpinnings reveals the simple construction.

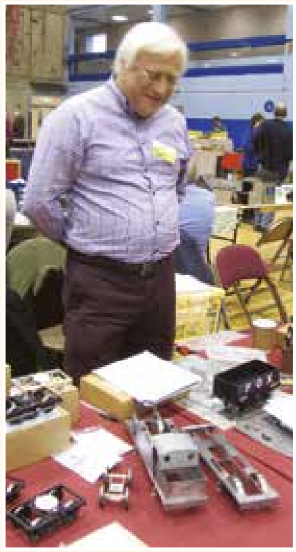
It is not a very heavy loco so can only manage about seven wagons without slipping, but unfortunately the little pot boiler is unable to keep up with the two double-acting cylinders and soon runs out of puff. With only three lightweight empty wagons it runs continuously with the regulator just cracked open. I expected to be laughed out of the garden when I showed it to the Group but most members were quite captivated and comments like "Cor, she can pull my wagons any day!" and "Can I buy her some meths?" have been heard! My daughter loves her and has now taken her to put on show in their new house promising to bring her back for a run now and then. Not a problem, as I now have some locos that run ok on my track. I must make one for my son now...

Emma with a maximum load.

WOKING AGM PICTORIAL *A selection of photos by the Editor*

The forthcoming Accucraft A1. Clearly the Editor's favourite......

A general view of the hall.



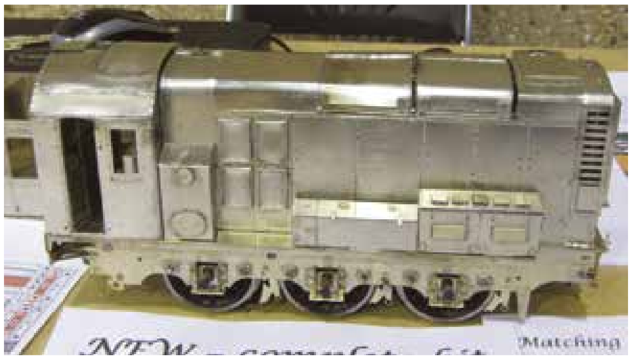
Peter Korzilius had his new SR M7 parts on display.

The Bring & Buy saw plenty of frenzied action - nobody would stand still for the photo!

Adrian Johnstone and Elizabeth Scott were awarded the President's Cup for their work digitizing the NL&J archive.

Lots of goodies on the JTT stand.

Fred Phipps displayed some of his 1/32nd scale diesels and coaches.



After several requests via the G1MRA Yahoo Forum, Orion Models have quickly produced an etched kit for the ubiquitous 'Gronk'.

Displayed on the Metalsmith's stand, this LNER P1 was one of a pair made by Carl Wadkin-Snaith to Finescale standard and is fitted with a working booster engine under the cab.

Almost a full house for the AGM.

Barrry Applegate chatting to Neil Butcher about one of Neil's wagons.

Tony Whibley displayed his meths-fired, RC ARMIG in 1923 livery.

The Committee are delighted that Andrew Barnes has decided to withdraw his resignation and continues as Membership Secretary

OVERSEAS GROUP CONTACTS

In addition to the fixtures shown at the back of this Newsletter, each of the Groups below organises its own events. Please contact the members listed below for details of a Group's programme. (Advise changes to peter.trinder@coldwaltham.net)

Australian Group Contact Tony Cotton - Email is: locowork@bigbutton.com.au

Danish Group: Meets first Sunday of month. 13.00 - 17.00 at the Hotel Krogen layout, Dining at 2 times through the year. (March and November) Contact David Clement Email: don@clement.dk

Dutch Connection A number of GTGs at members' homes in the Netherlands are organised each summer; contact Fred van der Lubbe +31 (0) 255515236 Email: fred.van.der.lubbe@planet.nl

France, Longjumeau Group meets every month plus other special events or exhibitions. Contact Hugh Hutchinson: 01 69 34 41 38 - E-mail: evc45.longjumeau@yahoo.fr

German Connection Contact Joachim Wolf +49 (0)761/290539 Email: g1mra.d@googlemail.com

Great Lakes Group, Canada have a comprehensive and varied calendar of two-day get-togethers, Friday pm/evening steam up every other week for the Ottawa area at David's, and Dick Abbott has regular steam ups for Toronto area members. Contact David Morgan-Kirby (613) 836 6455 inside Canada

New England Group (US) Monthly get togethers April-November at various members' houses. Contact Don Jackson 207-633-4335 email: scotia77@roadrunner.net

New Zealand Group Members have frequent, informal GTGs, both live steam and electric (track & battery). Aiming for 'open to all' GTGs bi-monthly. Contact Geoff Hallam Email:tirausue@clear.net.nz

Pacific Coast Live Steamers (Central Division) meets once a month March to November, on a rotating basis, at various members' homes. Track times are usually 11am until dark or hunger overtakes. Single dish lunches are provided. Some beverages are available, but it's best to bring what pleases one's individual palate. Guest steam powered locomotives are always welcome. Contact Tony Dixon 1-925-454-1245 Email: tony@mfcomputing.com

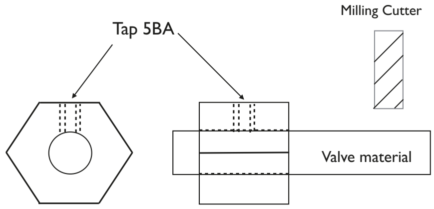
Seattle Live Steamers meet fourth Saturday of each month at members' homes, 10am-5pm (coast time). A pot luck goodies table is provided, each member brings something. We also meet on the second Saturday of each month at Boeing Field, Georgetown. Contact Peter Comley, 253-862-6748, sales@svrronline.com

Swiss Group Contact Christian Schmutz +41 61 711 82 04 Website at :- www.g1mra.ch - Email: csc@csc-schmutz.ch

TIP FOR MILLING DEE VALVES WITHOUT A DIVIDING HEAD

by G Gregson (2514)

Get a piece of hexagon bar around 3/4" across flats 1" long. Drill and ream 1/4" diameter. Drill and tap 5BA from the top into the bore. Place 1/4" rod into the hole, protruding around 5/8" and secure with a 5BA grub screw. Hold the hexagon bar rigidly to machine each of the three flats and the 1/4" slot. Cut the valve to length. The valve can be held in the hexagon bar to be faced off to length.



Taken at an Oulton GTG in 2011. Photo Dave Alexander.

GROUP NEWS

Compiled by Geoff Uren

Having just arrived back from the AGM, I have been galvanised into action by Bill, who has reminded me that my Group News for this NL&J is running late. I can't blame leaves on the line or the wrong kind of snow, but I had wanted to wait until the AGM was over so that I knew that I still had a job! It was good to meet many of you at Woking, and I'm grateful for the reports that you've provided.

Roger Hayward of the Guildford Model Engineering Society has advised me that they are planning to build a Gauge One fixed track over the winter. They plan to have a Gauge One Group, and initial plans are for joint meetings with their 16mm members on Monday evenings, as a joint 'Garden Railway' Group. I hope to be able to give more information in a future NL&J, but for anyone interested, please email me at g1mragroups@gmail.com and I will let you have contact details for Guildford MES.

NORTH LONDON GROUP

David West has updated me on the Group's activities, and tells me that they have taken full advantage of the good weather this year holding running sessions every Wednesday. They have also spent time improving facilities on site, including building a shelter which they have fitted out and christened 'The Bothy' (from the Gaelic Bothag meaning a hut, and also a drinking den in the Scottish Islands!!). Unlike Scotland, however, members can now retire after their run to drink tea (instead of uisge beatha!) and discuss future projects.

These have included a major effort by some members to remove brambles and weeds, rotavate the ground, sow grass seed and plant new bushes, resulting in a much improved appearance. Thanks to a generous donation of a quantity of paving slabs, the Group has also been able to pave the steaming areas and they are now extending a pathway around the track. Other modifications have been made to the raised track, being between 2ft and 4ft above ground level, approximately 300ft long, and consisting of three continuous circuits with passing loops. This enables six trains to be run simultaneously. The NLSME site also has raised and ground level 3 1/2", 5" and 7 1/4" tracks approximately half a mile in length, and a 60 × 30ft boating lake (some of their G1 members are also into boats).

The Group has hosted visitors this year from neighbouring groups, including Peterborough, East Anglia, Kent and Essex. David tells me that G1MRA Groups and individual members are always welcome any Wednesday at the NLSME site near St Albans, Herts, but check before setting off in case they are meeting elsewhere. For contact details see UK Group Contacts in the Newsletter or GIMRA website. I am already planning a visit next year!

VINTAGE TINPLATE GROUP

Dave Orchard has sent me a copy of the Vintage Group's October Newsletter. As those who went to Gauge One Expo will know, the Vintage Group put on a spectacular display, and Dave tells me that the Group was looking forward to the planned 2014 Expo. At the time of writing, your Committee is reviewing various options for 2014, but evidently many members and Groups still support our efforts to raise the profile of G1MRA by staging events like Expo.

The Vintage Group will be attending the two-day Large Scale Model Railway Show at the Fosse on 15/16 March 2014, with an even larger tinplate layout measuring 30 × 15ft. Already the Group have plans for May and June 2014, and I will report details of these events nearer the time.

EAST ANGLIA GROUP

I'm grateful to Robert Anderson for an update on East Anglian activities. In addition to the usual end of the month runs at on Anglia Roads at Hepworth, the East Anglia Group has introduced a monthly weekday run on the second Wednesday of each month. The inaugural run in September was well attended. If these runs continue to prove popular they will be continued into the New Year.

The visiting Swiss Group took advantage of the under-cover facilities at Hepworth in September, when bad weather made it impossible to use the intended outdoor venue. At the end of September East Anglia members accepted an invitation from the North London Group to run at their splendid track in Colney Heath.

It has been another busy year for Anglia Roads, and the layout made its annual visit to the Lions Club model railway charity show in Felixstowe in October. In November, members of the Ipswich MES made a visit to Hepworth to run on Anglia Roads - this was reciprocal visit following a second visit by East Anglia Group members to the IMES track in Ipswich. Robert tells me that both visits went very well.

Amongst the trees at Colney Heath during an EAG visit. Photo Robert Anderson.

The runners at Felixstowe. £200 was awarded to the EAG who nominated the East Anglian Air Ambulance Service as their beneficiary from the show. Photo Dale Thornburn.

Contact the EAG Events Coordinator, Robert Anderson, for the latest details of East Anglia Events. (rw.anderson@waitrose.com 01763 208764).

NORTH WEST GROUP

The track extension at Southport MEC continues to progress well, and the outer loop of the extension has now been completed. The first trains ran around the whole circuit on 25 September, the first being Andy Mitchell's Black 5 hauling some BR stock, accompanied by Alan Sanderson's ARMIG.

On 2 October, the Group held a meeting at Euxton, and in addition to plans for future events, the possibility of procuring a trailer for storing and transporting Withnell Junction was discussed. Maurice Rushby talked about his current project, a GCR Class A5 4-6-2T, including details of some ingenious engineering for the crank axle and valve gear. He is also adding provision for radio control of regulator, blower, whistle, drain cocks and reverser. That should keep Maurice fully occupied this winter, and the project would make a very interesting article for a future NL&J!

GROUP FUNDING

In November, I sent out to Group contacts a document concerning G1MRA funding for Group Attendance at Exhibitions. Your Committee decided that some rules were required to cover what had, previously, been dealt with on an ad-hoc basis. The document explains the level of G1MRA support available, and the kind of shows and exhibitions that we would support from central funds. Please have a look at it, and if you have any comments or suggestions, let me know. While I'm wearing my 'G1MRA Groups Liaison' hat, I'd just like to remind you that Groups liaison is a two-way process, and I would welcome your contacting me about any matters that you think your G1MRA Committee should be addressing. We are here to help Local Groups get the maximum enjoyment out of our hobby, so don't save it all up for the AGM!

Now, back to work on all those projects that I bought at Woking................

David Pinniger's Robinson Jersey Lily was built by John Lawrence from a Keith Cousins kit and beautifully painted by Liz Marsden. The elegant chimney was made by Dave Brutnell. Photo David Pinniger.

MIKE WROE (686)

I am very sad to report the death of a dear friend and fellow G1MRA Yorkshire Group member on August 24th. I met Mike about 15 years ago at a GTG and we struck up a friendship. As we lived within a mile of one another we started to travel to meetings and exhibitions together. Mike also had a wonderful G1 layout in his garden where we spent many hours drinking coffee and trying to work out why my engines wouldn't steam. As local interest grew in Gauge 1, Mike, myself and a few others formed the Yorkshire Group and had meetings several times a year on Mike's track where his wife, Avis, made everyone welcome.

Mike and I spent many hours travelling together or in my workshop or his garden and he talked a great deal of his family of which he was enormously proud and devoted. Avis and he were very close as were his son, daughter and granddaughter.

My thoughts are with Avis and the family who, having supported and nursed Mike through a long illness now miss him greatly.

I miss and will continue to miss his eternal optimism that my loco would eventually run without one of us pushing it.

I will miss our friends running trains on his track, particularly on New Year's Day. I will miss him visiting my workshop for advice, help or just a chat.

It's going to be lonely in the car without him!

Tony Halford

GEOFF AUSTIN (177)

On joining G1MRA in 1978, I found to my surprise that I drove past a member's home every day. That member was Geoff. At the time he was Head of Physics at Walsall Queen Mary's Grammar School.

As a boy he attended that same school and his father was head of Public Transport for Walsall, which still ran electric trolley buses. When he came home for holidays from Liverpool University, Geoff's holiday job was as conductor on the Wolverhampton route, a source of many amusing anecdotes.

We had both built Gauge One locos to LBSC's designs, his being an 0-6-2 called Mona and mine an 0-6-0 called Chingford Express. Neither worked! We got our heads together and soon had them running. Following an exhibition at Kettering where we knew that G1 would be running and reading about GTGs in newsletters, we invited members within a forty mile radius to a meeting. About twenty turned up. That was the point in 1979 from which the first of the formal regional groups, the Midland Group, developed.

Geoff was one of the original committee formed to design an exhibition track for use in the Midlands, Stanley Junction. Geoff took on responsibility for the provision of a stud contact system complete with electronic speed controls, powered points and section switching. This track made its first public appearance in 1985. In 2001, the track was revamped and re-appeared as Stanley Midland, this time without any electrics as it was felt that the trend was towards battery powered electric locomotives. Geoff put Mona into retirement and concentrated on this form of propulsion and its control systems. He became the leader of a sub-group who played a valuable part in our exhibition schedules.

His contribution and dedication to our exhibition track has given pleasure to countless people, both members inside the track and the sea of faces on the other side of the barriers, for almost 30 years. We, his friends in the Midland Group, miss his clear thinking, attention to detail, and above all his ready, if dry, sense of humour.

John Barrett

THE GRAND SWISS TOUR
11 - 21 September 2013

East Anglia Tour UK Co-ordinator, Peter Spoerer (679) reports:

It all started two years ago at Adam Houghton's house. It was a warm late summer evening on the patio. We had been playing trains all day long and we were enjoying a meal together with the Swiss 2011 tour party. Wonderful! I turned to Charles Simon, and asked him why the Swiss touring party never came to East Anglia. "We have not been invited", he replied, followed by, "Have you got some nice tracks in East Anglia?"

Charles Simon and Peter Spoerer at Denham. Photo Markus Hofmeier.

A lively discussion ensued. I told him about the many great Gauge 1 tracks we now have in the East, and also the excellent preserved working railways we have. Charles told me to make him a proposal, and to show him where the tracks were, and what they were like. Many things need to be considered, such as minimum track radius, as well as parking for the tour bus and cars, etc. I went back to Norfolk determined to put East Anglia back on the Gauge 1 map again by attracting the Swiss tour to our part of the country.

WHICH TRACKS TO CHOOSE?

We have several decent sized double track Gauge 1 layouts in East Anglia, and many other smaller ones as well. Charles had told me that they would want to run on only four tracks during their tour, as they would like to visit other railway attractions on alternate days. Which four tracks to choose was the question. I contacted all the large track owners, and told them about the idea of the Swiss Grand Tour, and would they like to participate. In the next few months I visited the offered tracks to check them all out.

SURVEYS AND ENTHUSIASM

Accurate surveys were made of each track to determine the various track geometries and scale drawings were made to send to Charles for his consideration. Regrettably, one of the tracks was eliminated after this stage, due to the fact that a part of it had a radius under 3 metres. Above all things however, I needed enthusiasm, and I needed commitment. All this for a project in two years time that might not even come off! After this stage I was down to five tracks to choose from. Well four tracks really.

FOUR TRACKS

The first track was John Squire's at Caythorpe, the second was my own White Horse Railway at Norwich, the third was Merv Myhill's very large track at Hindolveston, and the fourth was Peter Chrisp's brand new track at Denham.

CONTINGENCIES (The Fifth Track)

You should always have a contingency plan just in case something goes wrong. If one day was stormy and wet, we would use the East Anglian Portable Track, then housed in an industrial unit at Thetford, kindly arranged by Richard Hill. So we now had four running tracks, and a fifth emergency indoors track ready as a foul weather contingency.

THE PROPOSAL

With eighteen months to go I sent a written proposal with drawings and photos to Charles Simon detailing the suggested running tracks, along with details of other railway attractions for the Swiss party to visit while in East Anglia. He came back with a very positive reply. They would first visit The National Railway Museum in York, and then come down through East Anglia, then go on down to Kent for Adam Houghton's and Paul Abrams' GTGs, to finish off their tour. Great! It was on!

Charles Simon's 'Swiss specification' WYKO Br65 exits the tunnel at Iden. Photo Charles Simon.

TRACK TEAMS

I wanted to involve as many East Anglian Group members as possible with the Swiss tour. Every EAG member was given the opportunity to participate, and to experience the wonderful international G1MRA 'esprit de corps' engendered by such tours. EAG track teams were set up to assist the track owners in preparing their tracks for their day. Anyone who has a track will fully understand the enormous amount of work involved in preparing a track, and any help, however small, is much appreciated. Each track owner had well over a year to get their tracks in tiptop condition and, with the help of their teams, to make all the improvements that they wanted for their special day. These teams achieved some amazing feats of building and G1 engineering during this time. There is nothing quite like have a dead-line to work to!

SPECIAL ARRANGEMENTS

The following year and a half just flew by. Charles and I exchanged many emails. I regularly visited the three other East Anglian host tracks to offer support, and to check all was proceeding well. Special arrangements such as food and drink had to be coordinated, and pure alcohol had to be delivered to the tracks for the Swiss locos as well! I had lots of help on my track for which I am very grateful, especially with buildings and stations. Then all of a sudden it was September 2013, and the Swiss were on their way.

COME AGAIN

For our part, we felt that the Swiss Grand Tour of East Anglia was an unqualified success. It was with great sadness, and lots of hugs, that we bad farewell to the Swiss party. Many new friendships were made, and many old ones reinforced. It was a steep learning curve for us all, and a lot of hard work, but it was very worthwhile. On behalf of the four hosts, I would like to say to the Swiss party, thank you for coming to visit us in East Anglia, and you are always welcome to come again.

Markus Hofmeier's French 141R and Gerald Lackner's coal-fired 232.U1 crest the summit at Hindolveston. Photo Heiri Schartner.

Markus Hofmeier's French double header at Caythorpe. Photo Heiri Schartner.

Charles Simon (1663) on behalf of the Swiss Group writes:

Our fantastic trip to England is now history. It had involved almost two years' preparations by Markus Neeser, Lukas Schweizer and myself, as well as by our English friends, in particular, by Peter Spoerer. And it was worth it: in spite of some irritations such as mixed weather and one delay due to a traffic snarl-up, everything finally turned out fine thanks to the flexibility of the organisers and participants. A delay on the German autobahn meant that we missed the Rotterdam-Hull night ferry on Wednesday 11

Gerald Lackner monitors his radio-controlled Br52 at Caythorpe. Photo Peter Spoerer.

Rolf Engler's Wainwright C class with his scratchbuilt Birdcage rake at Denham. Photo Heiri Schartner.

September, with the result that, on Thursday, we enjoyed a visit to a windmill and a tour of the port area in Rotterdam, Europe's largest seaport. One consequence of this was that we were forced to skip our scheduled visit to the Great Central Railway.

The most important thing for me was that our Swiss group enjoyed a thoroughly successful trip to England, where we were able to steam our locomotives on six superb tracks, which are beautifully integrated into their owners' gardens. We received a warm welcome from our hosts and enjoyed delicious food. We had a lot of good, technical discussions with old and new friends from among G1MRA's ranks. And for our bus driver, Werner Steinlin, it was the third time (following our trips in 2009 and 2011) that he chauffeured us safely and carefully round England. Every day, he skilfully stowed our valuable array of bags and cases into the hold of our coach: a big thank you to Werner, too!

Our first meeting (at John Squire's on Friday) was accompanied by perfect weather. Recently expanded by a small through station, his track nestles beautifully into an immaculately kept garden. We enjoyed our runs on this visually beautiful and well-maintained track as well as the contact with our host and other G1MRA members. The lunch prepared by John's wife Robbie and her assistants was also excellent.

Silvio Gohl appears happy to have completed the long lap at Staplehurst with his Br50. Photo Charles Simon.

The visit to the National Railway Museum in York was another highlight that many of us were looking forward to. Some made straight for a separate department, where they could study pre-ordered locomotive drawings: a wealth of detailed information for loco builders. We are keen to see the models that the 'watchmakers' in our Swiss Group will be rolling out over the next few years.

Stephan Wirz prepares his A3 at Hepworth watched by Markus Hofmeier and Ron Pointer. Silvio Gohl records the scene. Photo Heiri Schartner.

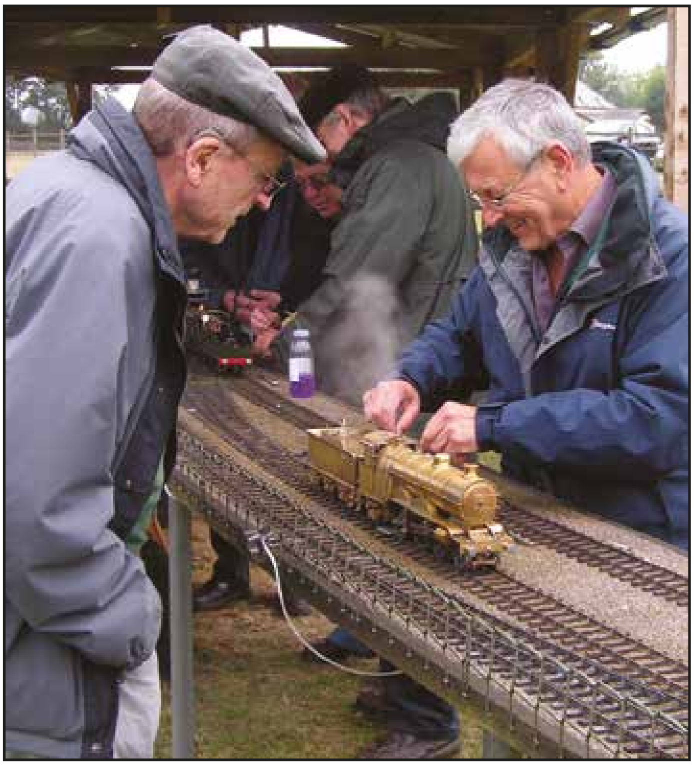
Unfortunately, heavy rain and strong winds meant that on Sunday our GTG at Peter Spoerer's had to be transferred to the East Anglia Group's hall in Hepworth where we were cordially welcomed by some live-steaming members of the EAG. We spent a truly enjoyable day running in this large hall, which is large enough that the 'Anglia Roads' track can be set up in all its glory including the decorative side components. There is also enough space for tables and chairs for spectators, which enabled us to appreciate and enjoy in comfort the lunch brought from home by Sue Spoerer. In the evening, we moved to Peter and Sue's home, where we could at least admire Peter's track in the drizzle. A visit to his well-equipped workshop was very inspiring. The evening came to a close among friends with a wonderful dinner prepared by Sue and her helpers including Bram Hengeveld and some great entertainment. In particular, the magic tricks by the Chairman of the EAG, Ron, got us all laughing. Marcus Neeser still can't work out how one of Ron's tricks works with an elastic band!

Werner Jeggli's Dampfsprinter at Iden. Photo Heiri Schartner.

Way out in rural North Norfolk, Mervyn and Sarah Myhill were expecting us on Monday in their large, romantic garden. The large Gauge 1 track is difficult to take in initially as its curves and tunnels blend in well with the garden. Together with several other G1MRA colleagues, we enjoyed running our trains in this beautiful and appropriate setting. In between, Sarah served us a tasty lunch together with wonderful cakes in the afternoon.

We spent the next day on the Bure Valley Railway and a boat trip on the famous Norfolk Broads, a wetland area that is largely a nature conservation area. On the return journey by steam train, some of us were allowed a special treat, taking turns riding in the cab of the miniature railway locomotive. Special thanks are due to our accommodating organisers Andrew Barnes and his wife Susan!

Peter Chrisp was the last track on our East Anglian list: once again, a warm welcome from Peter and his wife Sarah and other G1MRA colleagues. We enjoyed this Gauge 1 track, which was also very well integrated into a large and varied garden and its woodlands, and the extensive lunch. Then it was time to take our leave of East Anglia and our old and new friends including Peter Spoerer, who had contributed so much to the success of our trip. Everything went perfectly thanks to Peter's choice of tracks, his meticulous preparations and the active help of the East Anglia Group!



Werner Jeggli watches Roger Olding adjusting his Tom Barratt-built (from a Barrett kit) coal-fired LBSC Atlantic. Photo Charles Simon.

After spending the night in Ashford, we received a warm welcome in Staplehurst from Adam Houghton, Ralph Bagnall-Wild and other G1MRA friends. We immediately took possession of this fabulous Gauge 1 track (always a must for us), which is the largest track we are familiar with. With our locomotives and carriages, we all tested many times the forty-metre long, 1:100 incline constructed to perfection by John Butler: a highlight! Unfortunately, we had to dispense with night running due to the rain. This gave us an opportunity to enjoy a great dinner at the George Hotel in nearby Cranbrook.

And then it was Friday again: in splendid weather, we steamed to our hearts' content on Paul Abrams' jewel of a track, which is beautifully integrated into the garden, where we held numerous enriching discussions with a host of G1MRA friends. Brenda's buffet lunch was again a delight. And then it was finally time to say goodbye if we were to get to the Calais ferry on time.

To all of our hosts, their wives and supporters: our sincere thanks for the generosity with which you received us, allowed us to run on your tracks and looked after us so well. We appreciated it very much and do not consider the welcome so generously extended to us to be a matter of course. As Peter Spoerer said himself – we are aware that a visit like ours needs a lot of preparation: to host and feed 16 live steamers and four ladies all day long is a big thing! I think that the greatest benefits for all of us are the regular cultivation of our cross-border friendship and the exchange of experience within G1MRA. We would all like to express our thanks and I can say that the trip was a great success, another pearl in our G1MRA activities. We are all still talking about the days we spent with you: we keep this trip as a precious memory. And yes - we are looking forward to our next trip to England!

We will certainly be seeing you again sometime in the future, be it here in Switzerland or back in England. You can be sure that we will keep the flag of worldwide G1MRA comradeship flying!

Thanks to Bram Hengeveld, a video of the Swiss visit to Caythorpe by can be viewed at
http://www.youtube.com/watch?v=NsJpkifBNp0

AN ELECTRIC SUMMER

by Alan Leslie (4073)

A few like-minded G1MRA members in the South Eastern area took advantage of the amazing summer weather last year to get together to run their battery electric trains on a regular basis. As the meetings progressed there was the pleasant realisation that, using the capabilities of radio controlled locos, not only could trains be run in the usual manner round a line for the standard 30 minute rostered slot but also some other railway operating practices could be attempted.

The first attempt was at double heading. The preparation of coupling up the locos and then backing them down to the train was most evocative of times past on the real railway. Then, after coupling up to the train and setting off, keeping just the right tension in the coupling between the locos was challenging and most rewarding. More evocation of the past occurred when the train was carefully slowed to a halt prior to the locos being uncoupled and run off together to the shed area and a fresh loco put at the head of the train. The need for double heading was real since we attempted to run almost scale length trains - we had a 41-wagon van train on one occasion and then a 32-wagon coal train on another. Other trains run included long parcels and passenger trains.

The appearance of double headed Black 5s on a long train on my own single track line with its natural vegetation background really reminded me of the Highland lines in the days of steam.

Other things we tried were running to a sequence with morning goods and passenger trains followed by main passenger trains and

A local passenger train. A Bachmann Prairie and Finescale Brass coaches.

express goods trains.

The idea was interesting and has great operating potential for the future, but we need to work things out a bit more, since our initial plan proved too complex to carry out successfully. There is also the opportunity to use radio controlled battery locos to shunt and marshal the trains we intend to run. In fact, a quiet corner of the line would be suitable for this.

Great fun was had last summer and we would like to invite other owners of battery electric locos to come and join us in this informal Battery Electric Group. As well as my own line, several other owners have offered their tracks to host future meetings of the group. We intend to arrange meetings at fairly short notice depending on the weather forecast and would contact interested members by phone or email to give the date and location of the next meeting.

Anyone interested in joining us should contact Alan Leslie - details on the membership list.

A short goods train headed by an electric B4.

An Accucraft Flying Scotsman with vintage teak coaches.

Shades of the Highland main line. Double headed Accucraft Black Fives on a 41 wagon train.

UK GROUP CONTACTS

In addition to the fixtures shown at the back of this Newsletter, each of the Groups below organises its own events.
Please contact the members listed below for details of a Group's programme. (Advise changes to peter.trinder@coldwaltham.net)

Bristol Group of some 25 members with about 10 tracks in the area meets on most Saturdays and a few weekdays from Easter to late September. New members welcome. Contact: Martin Veasey 01225 460367

Cambridge and Suffolk Group. An informal group of local members arranging running sessions and events throughout the Summer months. New G1MRA members are particularly welcome and boiler testing is available. Contact Derek Pollard 01473 311384 or Geoff Calver 01638 721238.

Chiltern Group is a friendly group with the majority of members situated between Luton, St Albans, Chesham and Aylesbury. Between us we have a number of tracks and have regular running days during the month. If you are interested in joining then call Bob or Garth for meeting dates and times. Contacts: Bob Gamble 01582 696820 or Garth Bridgwood 01442 214441

Cornwall Group meets generally on the fourth Sunday of each month (phone beforehand) and also on every Thursday from mid afternoon. Meetings generally held at Nick Wilder's, phone to check. Visitors welcome. Contact: Nick Wilder 01726 61502

Cotswold Group We are a small group, please contact Robin Mosedale, 01285 862440, or e-mail robin@mosedale.mail1.co.uk

Essex Group meet every Thursday between 10am to 1pm at various local tracks. Contact Cliff Barker 01702 217422

East Anglia Group meets on the last Sunday of every month, indoors at Thetford during the winter months and at various East Anglian venues throughout the rest of the year. Contacts Ron Pointer (Chairman) 01553 827163 or Robert Anderson (Events Coordinator) rw.anderson@waitrose.com.

East Dorset and New Forest Group Meetings held on the second Saturday of the month all year.
Contact Peter Armstrong 01425 276587 Email: peter.armstrong@talktalk.net

East Midlands Group For running days, see http://G1eastmids.club.officelive.com Contacts: John Squire 01400 272270 (squmus@btinternet.com) or Terry Geeson 01400 281893 (g1mra.dobsonbridge@gmail.com)

Hants & Sussex Group meets on Sundays and/or Tuesdays. Contacts: Bill Whiting 02392 483030 or Martin Hughes 02380 402103

Isle of Man Group meets every two months. Visitors to the Island should phone ahead and bring a loco. Contact John Messenger 01624 628081

Lincoln and Humber Regular programme of events, runnings and visits. Contact Phil Sheard 01472 291934 (day) 01472 697334 (eve)

Midland Group Contact Phil Shrimpton 07935 248 355 or Jeff Towe 02476 733845

North East Group Contact David Nutt 01423 733074 email dnutt@btinternet.com

Northampton District Group Contact Paul Barford 01604 457558

North London SME Gauge One Group - Meets and holds running days every Wednesday between 11am and 4pm throughout the year at Colney Heath near Hatfield; guests welcome. Contact Malcolm Read 020 8723 1606 or David West 020 8504 5192 (dapamwest@btinternet.com)

North Norfolk Steamers - We have regular running days organised throughout the year on local outdoor tracks. Contact Mervyn Myhill 01263 860559.

North Staffs Group at Keele. Large G 1 twin track 20 Volt DCC garden layout - which can be switched for analogue running if desired. Steam line and steam bay facilities. Anyone wishing to join us for running sessions please contact Pat Honey: Tel 01782 516887 or e mail: pgh87@talktalk.net

North West Group Fixed meetings are second Saturday from April to October but we have a lot of ad hoc meetings. Meetings arranged for visiting GIMRA members – Contact John Lindars 01942 816902 See our details on the G1MRA web site.

North West Kent Group meets regularly on the second Friday of the month from September to May.
Contact Graham Shutter 01959 565238, email shutter@btinternet.com

Peterborough Group Regular meetings. Contact Peter Bird 07747 824113

Salisbury and Stonehenge Group meets second Monday of each month at the Andover & District MES site.
Contact Richard Boast 01980 652921

Scottish Group - Contact Stephen Millward 01542 841237

Scottish Group (Central) Contact Les White 01236 761872

South East Group - The group meets every first and third Wednesday of the month and details of additional meetings are circulated weekly by group e-mail - Contact Julian Barton for details. julian.barton@hotmail.co.uk, 01622 763069

Surrey Group meets on the fourth Tuesday of most months, and on a few other dates. Contact Alan England 01932 400282

Vintage Tinplate Group operate about 12 working vintage train layouts per year at Model Railway Shows, Toy Fairs, Exhibitions, Fetes, in the Greater Midlands area. A newsletter is issued quarterly between Journals. Dave Orchard 01527 874988 or Craig King 01622 851807

West Oxfordshire Group Contact Ken Wilkes 01993 842016 or e-mail kenwilkes861@btinternet.com

Yorkshire Group normally meets every second and last Sunday in the month from April to September at various venues in the region. For a full running list please contact Peter Vincent 01246 590259 or email: secretary@gauge1north.org.uk

LETTERS

Dear Bill,

I am writing to you in reaction to the official part of the last AGM, in particular to the many comments regarding the Fosse event of last summer. I was greatly surprised by the tone and by what were sometimes harsh comments. In my eyes, the loss incurred must be seen as an investment in G1MRA's future. What else should we do with £60,000 in the bank other than promote the idea of Gauge 1 and G1MRA? Personally, I am convinced that new members are essential if we are to keep this noble association alive. Who will care about all our great models in future if there are no younger members? Events open to the general model railway public are one of several ways of attracting new disciples. The focus, of course, should then be on the needs and possibilities of a more general public. This is undoubtedly more important than, for instance, giving everybody an opportunity to run.

I think that many things within G1MRA have improved considerably since the new Chairman and his Committee took over: a digital membership list, a much improved website and new events to mention but a few. I have been a member for nearly 30 years now, and all our Committees have done a splendid job and invested a lot of time and energy in their duties. All our members should be grateful for that. Nevertheless, it is important to bring in new ideas from time to time. Here, I am expressing my own personal viewpoint, but I am sure that the vast majority of longstanding members over here are of the same opinion.

Allow me to make a few general comments about our trip to the AGM. For many years now, a group of Swiss members has attended the AGM on a regular basis. As long as the meeting is not too far away from Heathrow, we can do this in one day, which is quite important. We have to get up at 3:30 am (British time) in the morning and we arrive home again late at night. Because there are no cheap flights at the times in question, it is also a financial effort. We do it because it is a good opportunity to keep in contact with old friends and an opportunity to meet the many traders. Our bags are empty when we arrive, but they never are when we leave. We buy a lot of items, not just for ourselves, but also for a number of colleagues who are unable to attend.

Let me say thank you to the Chairman as well as to the other members of the Committee for all your continuing efforts on behalf of G1MRA. I am very grateful that G1MRA makes it possible to be part of the worldwide fraternity of Gauge 1 enthusiasts.

John Werren (450)

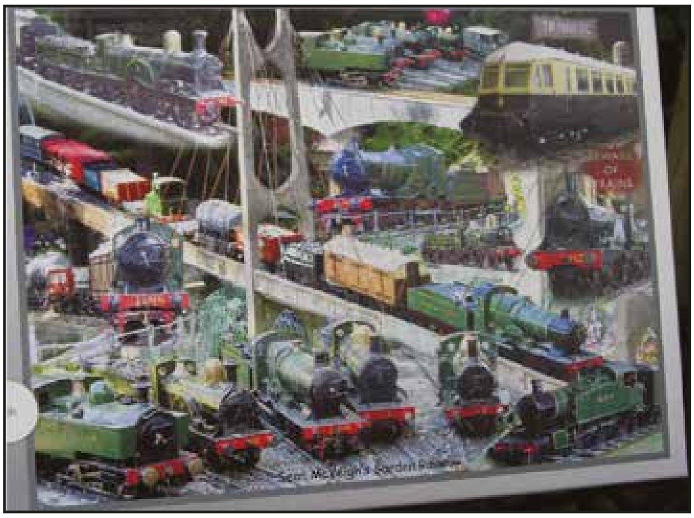
Dear Bill,

Here is a good idea for that elusive Birthday or Christmas present; a jigsaw puzzle of your own favourite engines. My eldest son Richard has a great talent when it comes to photography. He also knows my zest for jigsaw puzzles.

It was never-the-less a wonderful surprise to be given a 1000 piece jigsaw of 'Sean's Garden Railway' on achieving my 21st Birthday for the 4th time!

The puzzle is a montage of most of my engines taken in my garden. My wife told me he arranged the shots as I was having my afternoon siesta after hosting one of our local GTGs.

As a matter of interest, all except the tank loco in the bottom left hand corner and a few open wagons are scratchbuilt. About £150 would be the maximum cost of any one engine, hence the good selection I have managed to build.



The reproduction of the photos on the puzzle is first class and the quality is excellent, printed on strong card, sharply cut and of course fully interlocking.

The company that made the puzzle can be found at www.myphotopuzzle.co.uk

Sean McVeigh (962)

Dear Bill,

My wife Yvonne and I enjoyed a wonderful day in September at John Squire's track on the occasion of the Swiss Group visit. Perfect weather and very interesting visitors, even more fantastic was the food laid on, so thanks to John's wife Roberta and lady helpers.

Don Gurney (3289)

Dear Bill,

Whilst in some respects I may be considered as a new member after about 10 years I have been concerned for sometime about the location and time of the AGM. Why is it in November (the start of the winter) with dark mornings and evenings for any who travel any distance? If it is the accounts, then any accountant will quickly show how financial years can be changed. Why is it Woking?

I thought the event at the Fosse was by far and away the best event I have been to for Gauge 1 and that includes AGMs and thanks to all who organised it. Why can the Fosse not be the AGM? Light mornings and evenings for travel and a larger venue than Woking or Stoneleigh and close to the M40, M5 and M6. I think that the turnout at the Fosse was much larger than any AGM at Woking or Stoneleigh based on my observations.

During the AGM at Stoneleigh a vote was passed to hold the next AGM at Stoneleigh. This was studious ignored by the Committee and the AGM returned to Woking, since then votes held at Woking have been to keep it at Woking. I consider it a fair observation that such votes are geographically biased and therefore undemocratic. In light of the work done by Adrian Johnstone with 'The Mountains of G1MRA' the centre of gravity of the membership is somewhere close to Daventry, much closer to Stoneleigh than Woking. It may be fair to assume that the Stoneleigh vote is a better representation of the wider membership than those votes held at Woking.

I think it is time that the Committee cancelled the vote for the next AGM at Woking and showed some leadership in taking the AGM to the regions. It is nice to see thanks for the work of the regions in the NL&J, but I think it is time to recognise the work of the Groups in the regions and move the AGM around the country - that way everybody will get a show.

I think Question 13 on the survey is undemocratic; it proposes two choices for the South East. Why can the South East not be part of 'other' in 13.3, the question shows bias. The AGM needs to come North of the M25 and along either the M1, M5, M6, or M40. Avoid going along the M4 - Swindon and Reading are hopeless for access from the North by car or train.

Alan Williams (2512) Warrington

You have raised a number of valid points. Woking has been used for many years, but we are seriously in danger of outgrowing it, purely in terms of numbers attending versus the capacity of the building (recent building changes have not improved the venue for our type of event). Attempts have been made in the past to move the AGM elsewhere. You mentioned Stoneleigh: Your Committee have certainly not 'studiously ignored' the show of hands at Stoneleigh. We did consider using Stoneleigh again, but the operators would not take our booking due to their uncertainty about the future of the site. Public transport to the Stoneleigh site is non-existent - as soon as we announced the location, complaints were received because of that. Your Committee would dearly like to hear of suitable venues for our events, to include details of size, costs, refreshments, car parking, but nobody seems bothered enough to check out the details before complaining that we are not listening! Of interest, one response in the recent Members' Survey stated that the member would have attended the 2013 Expo if it had been more central...... Ed.

Dear Bill,

Thank you to both you and the Committee for all your efforts on our behalf and particularly for the way in which you approached and handled our recent AGM.

Having considered with some dismay the financial aspect of our recent Expo in Peter Wood's pre-AGM report I was concerned that the meeting had the scope to 'unravel' on various fronts, but nonetheless felt compelled to attend in order to fully understand the situation and see what the Committee's thoughts were on the way forward.

My fears were happily unfounded. Upon entering the AGM I was impressed with the clarity of the additional information presented to those attending and, once the meeting was under way, was further impressed with the way in which Chris and all of the Committee members who spoke dealt with all of the the issues. The whole meeting was handled in a straightforward, honest and candid way and I thought that the points made by speakers from the floor were well responded to and the issues raised properly acknowledged. I was also very pleased that Andrew Barnes has decided to stay as our Membership Secretary, great news - thank you Andrew.

Going back to the main topic, I attended and enjoyed our Expo very much. I thought that opening it up to non-members was exactly the right move and I very much hope that we do something very similar again in the future. I was, however, pleased that the decision was made at the AGM to take stock for 2014 before organising the next event and hope that this gives us breathing space to try to avoid some of the problems experienced with the Expo this summer. I have to say that not having the next event at The Fosse would be a positive step in the right direction in my opinion. I really do not think it offers good value for money and we (G1MRA) deserve, and can, do better.

All that said, the finances of G1MRA are otherwise in remarkably good health and I completely agree with Dick Moger's sentiment expressed at the AGM regarding the Expo, it was a brilliant and very enjoyable event of which the Committee can be rightly proud. I'd also be pleased to get involved in helping to organise the next event myself, having a little experience in this area.

Very best wishes to you all, Simon Castens (2165)

Dear Bill,

Firstly, many thanks for an excellent publication.

As a fairly new member living in North Wales I have to agree with Terry Geeson that there is very little G1 action in my area.

I am a member of the North West Group but the monthly meetings are a 2.5 - 3 hour journey so not really on in the evening. I did get to Peter Woods recent GTG but that was a 4 hour round trip.

I am in the process of building my track with 61 posts concreted and most of the cross timbers installed before Autumn slowed things down! The track will be around 150ft double track oval with branch and storage sidings in a shed and hopefully something will be running by this time next year.

I am situated in North Wales between Mold and Denbigh so if any members in the area would like to meet up or just phone for a chat my details are in the membership list.

Looking forward to hearing from like minded folk!

Regards, Chris Forster (4687)

A Jersey Lily running with an appropriate train. Photo David Pinniger.

MEMBERS' ITEMS FOR SALE

Members are strongly recommended to take their own precautions before parting with money in response to any advertisement.

Tower Models Mark 1 Coaches, Maroon. Five painted and lined, unweathered. Run only once briefly, so as new. £610 each or £2950 all five. Prefer buyer collects.
Martin Hearne - (4023) - 56 The Park, Bookham, Surrey, KT23 3LS. 07866 975866

Quantity of model engineering tools and equipment. List available. Buyers to collect or postage extra.
John Swift - (4634) - 01704 574430 (Southport)

Aster C12 locomotive, restored by Aster Japan. Operates, but may need to wicks adjusted. Will ship world wide, asking for £750.00/ USD$1200.00 plus shipping. Locomotive will ship from USA.
Adam Barr - (4721) - 440 785 8194, Email: co90221@yahoo.com

Tenmille Track:-
Track 1 yard lengths....................£5
Right-hand Points....................£40
Left-hand Points....................£40
Signal Box Brass Kit
G1 Signal Box complete (unopened) kit....................£75
Myford ML7 Lathe in nice condition with a selection of accessories....................£650
Andrew Rawe - (1159) - 01543 254 653
Email: andrewrawe@tiscali.co.uk

Complete contents of retiring G1 modeller's workshop for sale. Cowells ME lathe, milling machine and jigsaw with numerous attachments. Bench drill, small compressor. Numerous handtools, measuring equipment, metal, nuts and bolts plus partbuilt G1 loco. Model engineering books. Far too much to list here and the accumulation of 30years of modelling. All must go as one lot at £1500, no offers. Phone call to view.
Fred Barkby - (329) Leicestershire, 0116 2415161

Aster Streamline Commodore Vanderbuilt, electric, new boxed ....£3900 o.n.o
Ivatt 2-6-2t BR Tank No 41229 (built by Ian Cherry from kit), electric ....£1000 o.n.o
David M Dickens - (1026) - 01323 734731

Accucraft Live Steam BR (Ex L&SWR) B4 in BR Black No. 30089, early emblem, radio controlled 45/32 gauge 10 months old, run 4 times £695
Ashley Redman - (1450) - 01865 875422

Class 14 (teddy bear) Diesel Loco. Etched parts in 10mm scale, Gauge 1 with four packets of etchings unopened and one packet of detail white metal castings, also unopened. Also a set of six wheel castings. An electric motor with a worm gear fitted each end. A set of Build instructions....................Price wanted £195
Nigel Gettings - (1320) - nigel@gettings.eclipse.co.uk

ROD 2-8-0 heavy freight locomotive. Keith Cousins Kit, built by John Lawrence, extras by Dave Walker and painted by Liz Marsden. £2,500
Patrick O'Donnell - (3786) - 01279 412943
email pulbah44@yahoo.co.uk

Aster King George V - Factory assembled, unsteamed, in original box etc., c/w all operating and technical instructions, as new condition. £3500
Lewis Goodacre - (2014) - 024 7667 7847 or 07790 665104

South Bend 32" V Bed Lathe Mounted on a wooden cabinet, 16 Collets, 2 - 3 Jaw Chucks, 1-4 Jaw Chuck, Fixed Steady ..£600
150mm Bench Grinder £10
Chesterman 14" Height Gauge boxed £20
Hartlow Depth Gauge 0-6" £20
For further details please ring:-
Frank Norton - (1186) - 01843 842120

Aster BR 5MT (black) with special BR1B tender with rounded top by Cromford Designs. £3,950
With matching set of 7 Tower Models brass BR Mk1 coaches in BR Maroon....................£3,950
All v. little used.
Geoff Mogg - (1062) - 01727 839111
e-mail: geoffmogg@hotmail.com

The Estate of Tony Hall Patch
GWR Saint David built by THP £2000
GWR 2-6-2 tank loco by Bonds No 1304....................£750
GWR 0-6-0 tender loco Dean goods....................£1200
M&SWJR Beyer Peacock 2-4-0 part built £50
GWR class 3600 2-4-2 tank electric with r/c control van.....£250
LNER Gresley 'teak' baggage van (F Newman?) £200
GWR Hydra low loader 4 wheel wagon from Barrett kit £40
LNER EM1 Bo-Bo electric, "working pantographs", with GW bogie-battery van, ex Victor Harrison, illustrated in G1MRA NL&J No.1! very historic £1000
All items can be inspected in Elstree, sold as found, sensible offers considered.
Contact Graham Colover for more information 0208 953 8143 or 07768063289

ARMIG Engine Unit £150
G Gregson - (2514) - 01664 561414

New PROJECT Boiler, multi-flue, silver soldered copper, G1MRA boiler certificate, photos available £150
Garth Bridgwood - (980) - Tel: 01442 214441 Hemel Hempstead.

ETAT (O.C.E.M) 242A1 This engine is unique. It has been built according to the original drawings of the ETA by model maker Peter Schnabel in 1986. It is running on DC....................€13500 Negotiable
Bavarian Gt 2× 4/4 14254, (BR96) test run only. DC electric, Bavarian green ....€1200, Negotiable
SBB Be 6/8 13254, Swiss Crocodile, original box, Swiss Green DC catalogue number 5763....................€999, Negotiable
Contact Ulli Holtmann - (843) - horst.u.holtmann@t-online.de

B4 Steam Loco, almost new, with boiler certificate, instructions, steam oil and remote control....................£450
Terry Metcalf - (3766) - 01753 869646
Email: annmetcalf@btopenworld.com

Aster Evening Star £3800
Project 4F - finished BR Black £550
David Stubbs - (4005) - Email: destubbs@btinternet.com

WANTED

Wanted: - Tender for Bassett Lowke G1 Mogul required. Any condition.
Stuart Rose - (4477) - Northampton 01604 709267 or sicrose@btinternet.com

BING Tinplate "Titley Court" or "Sydney" in any condition.
David Daw – (2261) – 07581-034968

Aster/Fulgurex SNCF 232 U 1 electric version.
Malcolm Pallant - (128) - 01395 515049 or mwpallant@btinternet.com

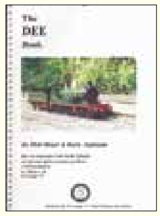
Members' SMALL ADS for Gauge One items are free. Send them to Peter Trinder, 19 Silverdale, Coldwaltham, West Sussex, RH20 1LJ or by e-mail: peter.trinder@coldwaltham.net by the press closing date. Please quote your membership number with your advertisement copy.

THE GIMRA BOOKSHOP

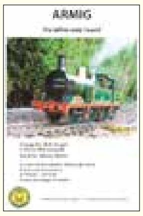
The Project Book edition 5 created in 2013

This brand new version will offer improved help to beginners who want to build from scratch. This latest edition of a book which has stood the test of time over 40 years contains almost 100 pages, more colour, up-to-date drawings and more accurate descriptions based on feedback from users of the original editions. Those who want to build this meths fired single cylinder LMS 4F 0-6-0 tender engine will be able to buy parts from G1MRA suppliers if they wish.

The Dee Book



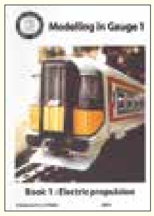
Members who already possess loco constructional skills could move on to the Dee 3. The Book contains 96 pages including full drawings and constructional descriptions to build a live steam SR D class twin inside cylinder tender engine. The loco design can be modified to run on meths or gas. A separate book of metric drawings can be purchased, but you will still need the Dee book. Sets of lining transfers are also available.



The ARM1G Book

Those who may not have the time to work from scratch may wish to choose the new gas fired ARM1G, for which most parts are available off the shelf. It is a scale live steam working model that involves very little engineering and is simple to run. The book contains 70 pages including full drawings and constructional descriptions to build a twin cylinder 0-4-4 Twin Tank.

Modelling in Gauge 1 - Book 1 - Electric Propulsion

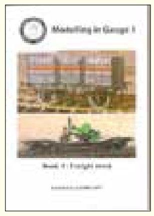


The first book in the Modelling in Gauge 1 series. Draws on articles from 200 Newsletters to bring you the collective experience and wisdom of G1MRA members on the subject of Electric Propulsion. Motors, gears, locos, bogies, controllers, batteries, tracks and much more. 110 pages.

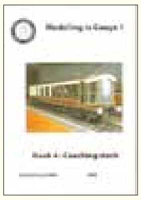
Modelling in Gauge 1 - Book 2 - JVR's Contribution to Gauge 1

This book draws on the many articles and letters written by John van Riemsdijk that have been published in more than 200 Newsletters over a 50 year period. John's work is legendary. Many of us will know of his input into the design of the many Aster locomotives running throughout the world. His work is published in this volume of over one hundred pages including photographs and drawings.

Modelling in Gauge 1 - Book 3 - Freight Stock



A compilation of articles on Freight Stock that have appeared in Newsletters since 1963. 96 pages.



Modelling in Gauge 1 - Book 4 - Coaching Stock

A compilation of articles on Coaching Stock that have appeared in Newsletters since 1963. 102 pages.

Also still available A Pictorial History of G1MRA

A 96 page celebration of G1MRA's Diamond Jubilee. An amazing selection of photographs from G1MRA archives, showing garden get-togethers and exhibitions between 1947 and 2007.

Price per item including p & p	Members in UK	Members outside UK
Dee	£7.50	£8.50
Project or ARM1G	£12.50	£13.50
Electric or JVR or Freight or Coach or Pictorial History	£8.00	£9.00
Dee Metric Drawings	£4.00	£4.50
Dee lining transfers	£35.00	£36.00
Special offer Electric book plus Pictorial History	£10.00	£11.00

Cheques made payable to G1MRA – Please quote membership number.

Overseas payments preferred by PayPal or in £UK drawn on UK bank.
Add £1 for Pay Pal handling charge or £8 for non-£UK cheques handling charge.

NOTE - Overseas members may be able to order their books through their membership co-ordinator.

All books and lining transfers are available from:

Mike Bland, G1MRA Books, 153 Nork Way, Banstead, Surrey SM7 1HR

For further information email: g1mrabooks@virginmedia.com

TRADE NEWS

Compiled by Bob Carter

Amongst others, this issue sees some interesting new release announcements from Accucraft, including their US outline passenger cars. At £180 each these look to be remarkable value, and one can only speculate what may happen when they begin to look at producing coaches to run behind British locos! The LNER A1s are a personal favourite, and I was delighted to see that Accucraft will also be releasing a live steam model of 'Tornado' next year. I gather Peter Spoerer will also be offering a radio controlled version in due course.

I am now beginning my final year in compiling this bit of the Newsletter, and will be starting to look for someone to take over. Being in the know, in advance, as to what your suppliers are planning makes this a fascinating job, and I have also met a whole bunch of interesting people. So why not give it a try? Please don't hesitate to contact me if you would like some background to the job!

Firstly a few brief news items:

ARISTOCRAFT CLOSURE

At the beginning of October the Polk family, who own Aristocraft, announced that they were no longer able to keep the company afloat and that it would be closing from 31st December 2013. In business since 1935, they have been major suppliers of G scale models. Their near Gauge 1 offerings included the celebrated Class 66 model, a staple form of motive power on many Gauge 1 tracks around the country. They cite the recent economic climate both at home and abroad as the root cause of their failure, having invested many millions of dollars in product development in recent years, but without the sales to match this.
There are rumours around that the Bachmann behemoth may take over production of some of their lines, but I have seen nothing to substantiate this to date

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RAILWAY MODELLERS' RESIDENTIAL COURSES

Bill Read, our editor, drew my attention to the range of short residential courses being run by Missenden Railway Modellers at Missenden Abbey in Buckinghamshire. Most of their courses are angled towards the smaller gauges, but a couple this year will be of interest to G1MRA members:

6-8th June - Tim Shackleton presents "Airbrushing for Modellers" a hands-on course for beginners embracing all modelling disciplines.

24-26th October - The Autumn Weekend usually offers courses on electronics, programming workshops, and train control. The special subject for 2014 is CAD technology and 3D printing.

For further details go to:
www.missendenrailwaymodellers.org.uk

LOCOMOTIVE DIAGRAMS

Well known railway film maker, Rob Dickinson, has asked me to inform members that his website now contains (free) offers of locomotive diagrams for Indonesia and the former East African Railways.

http://www.internationalsteam.co.uk/java/ford/PNKAdiagrams.htm

http://www.internationalsteam.co.uk/eastafrica/eastafrica.htm

Rob Dickinson 33, Baynham Road,
Mitcheldean, Glos GL17 0JR, UK
E-mail: internationalsteam@gmail.com

ULTIMATE WICK MATERIAL

Peter Armstrong, who has now retired from the business, has reminded me that the Ultimate Wick is still available from Ms Anne Rolfe (his daughter) at 35 Ridgefield Gardens, Highcliffe, Christchurch, Dorset BH23 4QG. Tel: 01425 276587 email: annerolfe@talktalk.net.
The price is still £8 including P&P. You will also be able to purchase it from Northern Fine Scale at GIMRA shows.

MTH CATALOGUE

Steve Jackson of DP Associates has advised me that despite his initial pessimism, MTH has in fact put out a 2013 catalogue. This can be accessed online via this link:

http://www.mthtrains.com/sites/default/files/catalog_files/2013_rk1_v_1/index.html

By the time you read this I am afraid that a 15% price increase will have been applied.

Contact D.P. Associates, Melrose, Route Orange, St. Brelade, Jersey JE3 8GQ if you would like to place an order.
Tel: +44 (0)1534 485113, E-mail:steve@railking.co.uk

A1 PACIFIC 'TORNADO'

The latest Gauge 1 locomotive offering from Accucraft and The A1 Steam Locomotive Trust, builder and operator of Britain's newest main line steam locomotive, will be a live steam model of 'Tornado'.
'Tornado' is a major milestone in the British railway preservation movement, becoming the first main line steam locomotive to be built since 1960. Since making her first moves in 2008, Tornado has hauled the Royal Train three times, appeared on the BBC's Top Gear programme and run tens of thousands of miles on rail tours and preserved lines.
I am a great fan of the LNER A1s and long time supporter of the A1 Locomotive Trust, and consider that Accucraft's model does a great job in capturing the elegance and power of the Peppercorn class A1 design. The model is gas-fired with slide valves and has all the features G1MRA members might expect from an Accucraft locomotive. The two cylinders are fitted with drain cocks. The chassis is constructed from stainless steel. The boiler is copper, the cab and tender are constructed from etched brass. The UK RRP will be £2895.00 with a release date of 2014 and the model will be available in apple green, blue and BR green. The image shown here is of the engineering sample. The retail profit from sales of this model will go to The A1 Steam Locomotive Trust to complete the financing of Tornado herself.
Speaking on behalf of the Trust, chairman Mark Allatt enthused "Just as Tornado has brought pleasure to countless numbers of steam enthusiasts, we hope that this wonderful miniature Tornado will do the same".
The model will be available from the A1 Steam Trust, Accucraft UK and

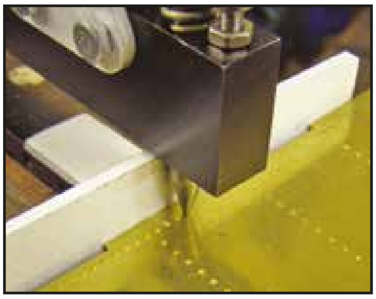
selected retailers. Versions available (subject to batch):
S32-12A – lined apple green, rimless chimney 'BRITISH RAILWAYS'
S32-12B – lined BR blue, rimmed chimney, early emblem
S32-12C – lined BR green, rimmed chimney, early emblem
S32-12D – lined BR green, rimmed chimney, late crest

STREAMLINED PASSENGER CARS

Accucraft Trains have announced the production of streamlined passenger cars in 1:32 scale. The cars are of metal and plastic construction, with a metal body and die cast trucks. A range of road names is to be offered including: SP Lark, UP Gray, NYC, SP Daylight, UP Yellow, N & W, Pennsylvania and CP. They will also be available unlettered. The UK price is expected to be an amazing £1080 for a rake of 6, or £180 per car.

See the Accucraft advertisement in this issue for more details and contact information

RIVETING PUNCH



Engineering in Miniature magazine has just started a series by Roger Thornber describing his latest loco, a GWR Collett Goods. This is a very simple engine, essentially a Project in GWR clothing. Many of the parts will be available laser cut from Model Engineers Laser making it suitable for the beginner. The plate work comes as tabbed and slotted kits whilst there will also be a jig to help build the cylinder.
Naturally being a GWR loco there are lots of rivets. To enable these to be portrayed accurately and inexpensively a riveting punch, which has also been described in Engineering in Miniature, is now available. The picture shows the tool being used to simulate a row of rivets in the tender side panel of Rogers's new design.

See the Model Engineers Laser advertisement in this journal for further details.

TWO STAGE MOTOR/GEARBOX

MSC Models have produced a Gauge 1 version of their MT Helical range two stage 30:1 motor/gearbox unit (pictured) with 1/4inch axle. Despite its apparent small size, this is a very powerful unit which takes full advantage of its very efficient helical gearing. They anticipate that this unit will be of interest to those Gauge 1 modellers building or re-motoring small or medium sized locomotives. Price - £97.00 plus p&p.

They are also producing a Gauge 1 version of the double reduction range of JH motor gear box units with 1/4inch axles. These will be available early in 2014 with ratios of 14:1, 25:1 and 33:1.
Price – around £80.00 plus p&p.

The established range of Gauge 1 Crailcrest motor/gearbox units continues to be available with ratios of 14:1, 25:1 and 33:1.
Price - £84.00 plus p&p.

Please see the MSC Models advertisement in this issue for further details

HAND HELD TRANSMITTER

Peter Spoerer Model Engineers have introduced a new range of neat hand held transmitters for use with electric radio controlled locomotives: the Indi-Handset range. Compatible with the complete range of Spectrum DSM2 equipment, these allow you to replace a big Spectrum Dx5e transmitter with one of these little handsets. You do not have to change any of the old R/C equipment in the loco. They are compatible with both the AR600 and Orange receivers, and all makes of speed controllers and sound cards. They can be bound to more than one locomotive. Indi-Handset Receivers are also available. Pre-production handsets were exhibited with great success at the Woking show last November. Production models will have smaller and neater push buttons, and also slightly different labels. By the time this magazine reaches you these handsets will be on the market and ready for sale. Please see Peter's advert for photos of the handsets.

There are three Indi-Handset models, which give direct control (Tx-20), selectable inertia for acceleration/deceleration (Tx-21) or allowing multiple (up to 12) locomotives to be controlled simultaneously (Tx-22). The function buttons will allow soundcards and servos (for, say, uncoupling) to be operated. Multi loco operation will require the use of a Selecta compatible receiver. Prices range from £47.00 - £64.00

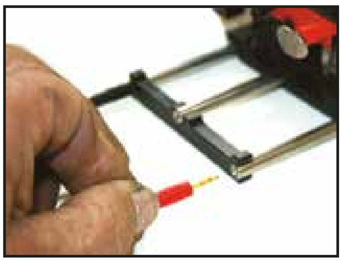
SELECTA RECEIVER

The Selecta 102-22 receiver will work with all models of the Indi-Handsets and all Spektrum transmitters. Measuring only 18x14mm it has seven channels and is DSM2 compatible. Priced at £12.00.

Peter also informs me that he will soon start working on a live steam version of the Indi-Handset for garden railways, and a postage stamp sized receiver and speed controller for the very smallest of locos.

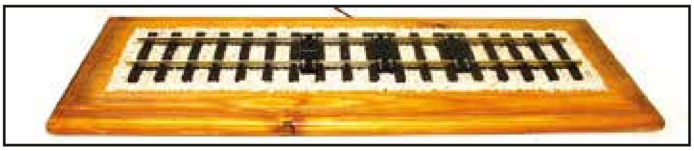
MAZERO ROLLING ROAD ACCESSORIES

As intimated in Trade News in NL&J 239, Peter Spoerer Model Engineers are now introducing a range of accessories for their Mazero rolling roads, featuring single-, twin- or triple-axle (for tenders) supports. Prices range from £2.50 - £10.00.
Also available are power connection plugs, which allows 12vDC to be connected to a Mazero Rolling Road so that a 2-rail pick up locomotive can be run. It consists of a power wire with two colour coded, gold plated, 3mm plugs on the end. The user must drill 3mm holes in the ends of the rods so that the plugs can fit into place. Also available in 3-rail. Priced at £4.00



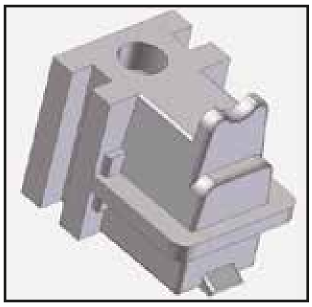
MAZERO TRACK ROLLER BLOCKS

These are very similar to the roller blocks on a standard Mazero Rolling Road, but have been slightly modified so that they can drop in between the sleepers in a track panel to support the driven wheels. They work with all makes of proprietary track. The roller blocks must be positioned exactly under each driven axle of the loco, and as every loco is different, a bit of creative sleeper spacing will be necessary to position the roller blocks perfectly. The roller blocks are supported at the correct height by the plastic connecting spacer pieces in between the sleepers. Properly set up as in the illustration, it is possible to see your favourite loco's wheels turning inside a display case. The roller blocks are almost unnoticeable when the loco is placed on top. For displaying a radio controlled loco nothing else is required, but if two rail pick up operation is required, electricity must be taken from the controller to the roller blocks. Priced at £15.00.



Please see Peter Spoerer's advertisement in this issue for contact details

TENMILLE BRASS AXLEBOXES



Tenmille can now supply brass cast axleboxes ideal as replacements for whitemetal or plastic versions. These are also suitable for use on some four wheeled coaches. The patterns were created using CAD and 3D printing. They are available in packs of four at £6.90 per pack.

STEEL COUPLING HOOKS

Tenmille introduced steel, laser cut hooks as alternatives to the plastic version in their coupling hook sets some time ago. The hooks are now available separately in packs of two priced at £1.75 per pack

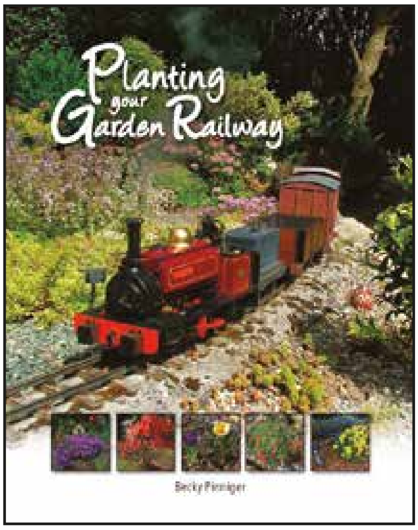
See the Tenmille advertisement in this issue for contact details

BOOK REVIEW

'PLANTING YOUR GARDEN RAILWAY' BY BECKY PINNIGER

reviewed by Steve Edwards (1023)

Whilst there have been books and articles over the years on how to construct a garden railway, I am not aware, until now, of one that addresses itself to the subject of the planting of our railways. It is difficult enough to get permission to build a railway in the garden when perhaps your wife has only seen a friend's efforts which is raised up on stilts resembling last year's Grand National.



Model railway fans are seldom gardeners – well here is your chance to see how a railway can enhance your garden.
For some reason Gauge Oners who are lucky enough to have their own track seldom make much, if any, effort to integrate their railway into the garden. I can think of exceptions such as Paul Abrams' beautiful garden railway in Kent. My own railway is not a thing of great beauty raised as it is on stilts with little attempt to disguise – my excuse is that it is sufficiently far away from the house not to matter. With G1 function is all – now contrast this with our friends in 16mm narrow gauge where we often see a railway that enhances the garden.

So it is no surprise that this book by Becky Pinniger is influenced by her husband Dave Pinniger who although a G1MRA member is better known in 16mm narrow gauge circles.

The book is beautifully presented with fabulous photography by Dave himself – which should give huge inspiration to those of us who would like to take our railway to the next stage. One of the most common reasons given for not planting up a garden railway is the concern that it only works for ground level lines which do not make for comfortable running. A quick flick through the book gives many examples of raised lines, and how they can benefit from planting.

But this is no coffee table book, it is very well written and highly readable. Becky Pinniger is too modest when she declares herself to be 'no horticultural expert'. Indeed the book would serve as a very useful first gardening book with lots of detail regarding plants – growing conditions, propagation and general care and maintenance.

The plant list towards the end of the book takes all the guess work out of planting choices – as it covers the conditions that the plant requires along with its ultimate size.

I would strongly recommend all G1MRA members to use this book to enhance their own railway.

I would be delighted to see more pictures appearing in the journal of attractive lines and would encourage Becky to write an article for the NL&J.

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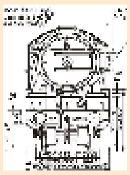

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Golden Age Models A4 'Dominion of Canada' and Pullmans - Other names available


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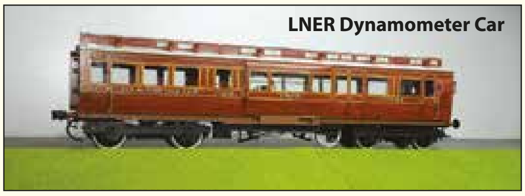
Sir Nigel Gresley in BR Blue


Mallard in BR Green


Pullman coaches


LNER Dynamometer Car


Triplet Restaurant Car set


Brighton Belle - all the sets


Coronation Twin

Photos : Tony Wright, Chris Nevard and Golden Age Models, including Gauge 1 samples and some OO and O Gauge photos for illustrative purposes

GOLDEN AGE MODELS LIMITED

Please contact for other names and full details.

www.goldenagemodels.net

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Tel: 01929 480210 (with answerphone) E-mail: quentin@goldenagemodels.net

Membership Secretary
Andrew Barnes, PO Box 1356 Buxton,
Norwich, NR10 5WR
Email: g1mra.membership@gmail.com

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(per annum)

Ordinary members: £20.00
Junior members : £10.00 (11-17 years)
Please pay by Standing Order if possible

Airmail supplement

Overseas members outside Europe may request their Newsletters by Airmail at a supplementary cost of £20.00 per annum.

Membership Coordinators

All Membership enquiries (including application forms) should be sent to the Membership Secretary, except for the countries below where members should use the applicable Membership Coordinator for all membership enquiries and payment of subscriptions.

Australia Tony Cotton, 13 Goldingham St, Taperoo 5017, South Australia
locowork@bigbutton.com.au

Canada David Morgan-Kirby, 10 Porter St, Stittsville, Ontario, K2S 1P8.
gaugeonelines@yahoo.com

France Garry Randoll, 3, Rue des Juifs, 80120 Dominois
garry.randoll249@orange.fr

Germany Joachim Wolf, Keltenweg 10, Bad Krozingen, D-79189
g1mra.d@googlemail.com

Netherlands Fred van der Lubbe, Nicolaas Beetslaan 6, 1985 HH Driehuis
fred.van.der.lubbe@planet.nl

New Zealand Geoff Hallam.
tirausue@clear.net.nz

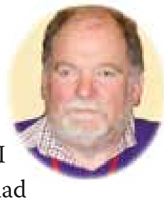
Switzerland Christian Schmutz, Hauptstrasse 72, Ch-4153 Reinach BL csc@csc-schmutz.ch

USA Ernie Noa, 791 County Farm Road Monticello IL 61856
ernienoa3@mchsi.com
217 649 6605

All other overseas members are requested to pay their subscriptions in Pounds Sterling by means of a Sterling cheque drawn on an UK Bank, direct to the UK Membership Secretary.

FROM THE SECRETARY

Remembering Peter Marshall



In 2013 our Association lost two long-standing members who I can say had both shown me particular kindness at various times over the years. I am referring to Ella Roberts and Tony Hall-Patch. However, these two losses were not the end of the sad story since I learnt at the end of October that my dear friend Peter Marshall had passed away quietly, but suddenly at home. He was 68.

Peter Marshall was one of the nicest people you could meet. He was kind, he was generous and he was a loyal friend through thick and thin.

I have known Peter since the Golden Jubilee celebrations at The Fosse in 1997 when I made the mistake of picking up an item on the Wagon & Carriage Works display to examine it. He instantly said "If you pick it up then you have to buy it". I was a bit taken aback but then saw the twinkle in the eye and knew immediately that here was someone who liked a bit of fun. And so it proved to be. Over the last 16 years we enjoyed many jokes and experiences together.

When Roy Scott and I took over the Aster distribution for the UK in 1998, Peter became the first and one of the most successful UK dealers. But we became friends too and travelled to the Diamondhead Steamups most years, sometimes with Peter O'Kane and sometimes with Geoff Calver. His home Gauge One track at Hoddesdon and his retail emporium at Sawbridgeworth were irresistible to anyone who liked model railways as long as they could put up with the almost ceaseless torrent of friendly abuse! It's hard to acknowledge we won't be seeing Peter again.

Peter's widow Vanessa has not decided how to manage the Wagon & Carriage Works business nor Peter's extensive collection of models. Perhaps members would kindly not pester Vanessa with enquiries for the time being? Thank you.

Andrew Pullen

THE TREASURER

Well, the AGM at Woking has been and gone, I really cannot believe how time flies. My wife says it is because I am becoming an elderly OAP and have little idea of time keeping, judging by the amount of time spent in the workshop and cold lunch when I fail to appear at the given time. 2014 is very nearly here and I think there is going to be a lot to look forward to: the North West Group have two events booked at the East Lancashire Railway with Withnell Junction, in February and then again in April. April is also the Yorkshire Group event at Bakewell, then we have the Spring Meeting in June, followed immediately by the Woodvale Rally again with Withnell Junction. The Manchester Model Engineering Exhibition, which for the last two years has started the season, is sadly no more due to the high cost of hiring the exhibition hall, which does leave a bit of a gap in the calendar.

As I write this, the Christmas tree is sparkling and presents are appearing, with little notes that say, no poking or touching, hopefully there may be an engine or two!!! Somehow I doubt it. I send my best wishes to all members and their families and hope that 2014 will be a good year.

The financial results at our year end, showed an improving situation, with funds around 60K, how we are going to use this money now to further the Premier Gauge I am not sure as the Committee plans have taken a bit of a knock. Whilst talking money, please will those members who have underpaid their subscription please send the additional £5 to the subscription officer and alter their standing order at their bank.

The one thing I really enjoy at the AGM is the members bring and buy table, I said at the AGM that what is rubbish to some is treasure to others, and I certainly purchased some 'Treasure'. Time now to open those treasures and see what sort of rubbish I have bought !!!!.

Regards, Peter Wood

MEMBERSHIP SECRETARY

Having said my farewells in NL&J 238 and wished my successor well, I regret that you have got me back again.

As many of you will know I have a full time job, with a six hour commute each day to and from my home in North Norfolk to my job in the City of London. In addition to that, I am an active director of the Bure Valley Railway in Norfolk. Against this background I was struggling to give the same level of time to the role of Membership Secretary as my predecessor had done. With this in mind, I offered my resignation to our Chairman with the view that a member with more time would be able to step into the role. Unfortunately, only one potential candidate came forward and when he understood the role and the work involved politely withdrew his interest.

Following a lengthy discussion with our Chairman in advance of the AGM, I have agreed to carry on as your Membership Secretary for another year, rather than leave the Association facing a quandary.

Please can I ask you to help me to support you, the membership? Please pay your subscriptions on time and for the correct amount. The best means to communicate with me is via the post or e-mail. The telephone is not good as I am out until late in the evening and out with other commitments at the weekend.

I shall endeavour to respond to queries as quickly and accurately as possible. I do not claim to be infallible, but please remember I am acting as a volunteer to support an Association I care about. Over the last couple of years I have received communications from many members, ranging from suggestions of how I should be doing the job, to open criticism of the way I undertake the role.

Please think before sending in such communications. I must give priority to attending to membership matters and do not have time for lengthy discourse over ways the membership should be better administered.

If however you are seriously interested in the role and think you could bring something to the Association, please do get in touch.

Now to Membership matters. At the AGM the membership stood at 2455. Of these, 1875 are in the UK the remainder overseas. Much has been said recently about encouraging younger members. I will leave you to define young, but at the time of writing the Association has 12 Junior members!

Turning to Subscriptions, despite appeals via the NL&J we still have 490 members who to date have only paid £15.00 for their 2013 membership rather than the new rate of £20 which applied from 1st January 2013. The only option left to us is to write to these members requesting payment of the shortfall. The cost of such a mailing will run into hundreds of pounds. If you think you may be one of these, please do pay the shortfall, or Article 9 will be applied.

On a positive note, we continue to see a steady growth in the Association membership.

Each year we see around 50 members leave the Association through the application of Article 9 for non-payment of their subs. In 2013 over 30 of these members rejoined as they had simply forgotten to renew and complained when they did not get their NL&J.

Subscriptions are now due. If you do not have a Standing Order for £20 in place, please use the form enclosed with this Newsletter to renew now.

Wishing you all a productive winter and I look forward to supporting you as your Membership Secretary over the coming year.

Best wishes, Andrew Barnes

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Charisma Creative

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Deadlines
Deadlines for contributions to be with the editor are 15 February, 15 May, 15 August and 15 November.

Contributions
The Newsletter and Journal contains contributions from G1MRA members. Members are asked to provide articles and pictures on any topic that they consider will be of interest to other members. The Editor reserves the right to edit or refuse contributions.

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Send copy by post or e-mail to the Editor at the address above. Copy preferably should be in any of the popular word processing formats or typed. Photographs are always welcome, and should be in digital format. Computer files may be sent on CD or by e-mail. Material will be returned after use only if requested.

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G1MRA DIARY

Please note that this diary contains national meetings, details of which are compiled by Peter Wood (see below). All submissions should be sent to him by the press deadlines indicated below. For Local Group meetings contact your organiser as listed on the Groups page.

January

17-19 **London Model Engineering Exhibition,** Alexandra Palace. Contact Jeff Brazier

February

15 **Stonehenge Group** - Bacon Butty Bash. Contact Dicky Boast, 01980 652921

23-24 **East Lancashire Railway Steam Gala** with Gauge One at Bury Transport Museum. North West Group with Withnell Junction. Contact Maurice Rushby.

March

1-2 **Homewood Model Railway Exhibition,** Tenterden, Kent. Invicta Portable Track. Contact Tony Whibley - 01580 762004 - Email: whibs.aj@tiscali.co.uk

April

12 **Gauge One North,** Bakewell, Derbyshire. Contact Alan Bullock or Peter Vincent or visit www.gauge1north.org.uk.

12 **The National Garden Railway Show** – East of England Showground. Peterborough.

26-27 **The Big Model Steam Train Show** at Bury Transport Museum. North West Group with Withnell Junction. Contact Maurice Rushby.

26-27 **Longjumeau Group, France.** 10am to 6pm in a large gymnasium. Open to the public. Free parking. Free entry. Four or five 45mm circuits - plenty of running capacity if you want to come along! Contact Hugh Hutchinson on h.hutchinson@free.fr

May

4 Richard Donovan, Burghill, Hereford, from 11am. Camping and Caravans in adjacent field. Picnics welcome. Contact Richard Donovan 01432-760881

June

14 G1MRA Spring Meeting, Shepshed High School, near Loughborough, LE12 9DA.

21 Get Together at Upton Snodsbury, Worcestershire, close to M5 and M40. From 2pm, buffet served at 5.15pm. Contact Graham Harding 01905-381335 in advance for catering purposes.

21-22 **Woodvale Rally,** Victoria Park, Southport. North West Gauge One with Withnell Junction. Contact Maurice Rushby.

28 **Stonehenge Group Summer Solstice Sausage Sizzle.** SP11 7HT. Contact Dicky Boast, 01980 652921

28-29 **Sinsheim, GERMANY: Spur 1 Treffen,** http://sinsheim.technik-museum.de/de/de/internationales-spur-1-treff en

July

5 Get Together, Adam Houghton, Staplehurst, Kent, from 12 noon. Picnics welcome.

12 Cliff Barker, Southend on Sea. G1 and G3. 10am – 6pm. Lunch available at £4 for advance orders please. Contact Cliff Barker.

16-20 **National Summer Steam up,** Sacramento, CA, USA. See www.summersteamup.com for details.

19-20 **Bekonscot GTG.** Contact Brian Newman-Smith.

October

25 2014 GIMRA AGM and Show. Contact Geoff Hammond.

November

1 **Padiham Model Railway Exhibition,** with North West Gauge One layout, Withnell Junction. Contact Tom Wallbank.

8-9 **Southport Model Railway Exhibition,** with North West Gauge One layout, Withnell Junction. Contact Tom Wallbank.

Please advise additions or changes to Peter Wood on 01704 576122 or by email: peterwood-gimra.nw@talktalk.net

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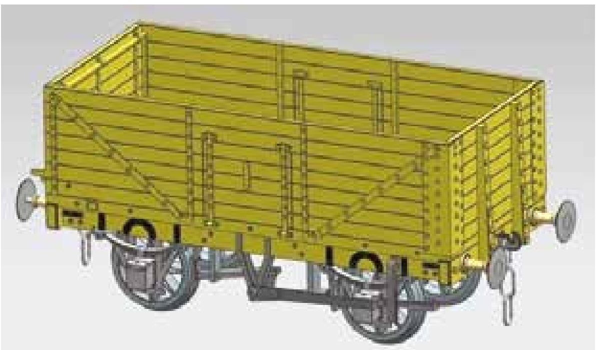

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